

Recommendation **ITU-T K.91 (01/2024)**

SERIES K: Protection against interference

**Guidance for assessment, evaluation and
monitoring of human exposure to radio
frequency electromagnetic fields**

Recommendation ITU-T K.91

Guidance for assessment, evaluation and monitoring of human exposure to radio frequency electromagnetic fields

Summary

There are many possible methods of exposure assessment and each of them has its own advantages and disadvantages. Recommendation ITU-T K.91 gives guidance on how to assess and monitor human exposure to radio frequency (RF) electromagnetic fields (EMFs) in areas with surrounding radiocommunication installations based on existing exposure and compliance standards in the 8.3 kHz to 300 GHz range. This includes procedures for evaluating exposure and how to show compliance with exposure limits with reference to existing standards.

Recommendation ITU-T K.91 is oriented to the examination of the area accessible to people in the real environment of currently operated services with many different sources of RF EMF, but also gives references to standards and Recommendations related to EMF compliance of products.

Recommendation ITU-T K.91 includes an electronic attachment containing an uncertainty calculator and the *Watt guard* modules.

History *

Edition	Recommendation	Approval	Study Group	Unique ID
1.0	ITU-T K.91	2012-05-29	5	11.1002/1000/11634
2.0	ITU-T K.91	2017-07-29	5	11.1002/1000/13276
3.0	ITU-T K.91	2018-01-13	5	11.1002/1000/13449
3.1	ITU-T K.91 (2018) Amd. 1	2018-09-21	5	11.1002/1000/13796
3.2	ITU-T K.91 (2018) Amd. 2	2018-09-21	5	11.1002/1000/13797
3.3	ITU-T K.91 (2018) Amd. 3	2019-05-22	5	11.1002/1000/13935
4.0	ITU-T K.91	2019-11-13	5	11.1002/1000/14072
5.0	ITU-T K.91	2020-06-29	5	11.1002/1000/14296
6.0	ITU-T K.91	2020-12-14	5	11.1002/1000/14570
7.0	ITU-T K.91	2022-01-13	5	11.1002/1000/14876
8.0	ITU-T K.91	2024-01-13	5	11.1002/1000/15777

Keywords

Exposure assessment, exposure evaluation, exposure monitoring, RF EMF exposure limits.

* To access the Recommendation, type the URL <https://handle.itu.int/> in the address field of your web browser, followed by the Recommendation's unique ID.

FOREWORD

The International Telecommunication Union (ITU) is the United Nations specialized agency in the field of telecommunications, information and communication technologies (ICTs). The ITU Telecommunication Standardization Sector (ITU-T) is a permanent organ of ITU. ITU-T is responsible for studying technical, operating and tariff questions and issuing Recommendations on them with a view to standardizing telecommunications on a worldwide basis.

The World Telecommunication Standardization Assembly (WTSA), which meets every four years, establishes the topics for study by the ITU-T study groups which, in turn, produce Recommendations on these topics.

The approval of ITU-T Recommendations is covered by the procedure laid down in WTSA Resolution 1.

In some areas of information technology which fall within ITU-T's purview, the necessary standards are prepared on a collaborative basis with ISO and IEC.

NOTE

In this Recommendation, the expression "Administration" is used for conciseness to indicate both a telecommunication administration and a recognized operating agency.

Compliance with this Recommendation is voluntary. However, the Recommendation may contain certain mandatory provisions (to ensure, e.g., interoperability or applicability) and compliance with the Recommendation is achieved when all of these mandatory provisions are met. The words "shall" or some other obligatory language such as "must" and the negative equivalents are used to express requirements. The use of such words does not suggest that compliance with the Recommendation is required of any party.

INTELLECTUAL PROPERTY RIGHTS

ITU draws attention to the possibility that the practice or implementation of this Recommendation may involve the use of a claimed Intellectual Property Right. ITU takes no position concerning the evidence, validity or applicability of claimed Intellectual Property Rights, whether asserted by ITU members or others outside of the Recommendation development process.

As of the date of approval of this Recommendation, ITU had not received notice of intellectual property, protected by patents/software copyrights, which may be required to implement this Recommendation. However, implementers are cautioned that this may not represent the latest information and are therefore strongly urged to consult the appropriate ITU-T databases available via the ITU-T website at <http://www.itu.int/ITU-T/ipr/>.

The author thanks the International Electrotechnical Commission (IEC) for permission to reproduce Information from its International Standards. All such extracts are copyright of IEC, Geneva, Switzerland. All rights reserved. Further information on the IEC is available from www.iec.ch. IEC has no responsibility for the placement and context in which the extracts and contents are reproduced by the author, nor is IEC in any way responsible for the other content or accuracy therein.

© ITU 2024

All rights reserved. No part of this publication may be reproduced, by any means whatsoever, without the prior written permission of ITU.

Table of Contents

		Page
1	Scope.....	1
2	References.....	1
3	Definitions	3
	3.1 Terms defined elsewhere	3
	3.2 Terms defined in this Recommendation.....	5
4	Abbreviations and acronyms	8
5	General guidance	9
	5.1 General public and occupational exposure.....	11
	5.2 Existing or planned transmitting station.....	11
	5.3 Collection of data concerning sources of radiation	11
	5.4 One or more sources of radiation, total exposure.....	12
	5.5 Field regions	13
	5.6 Basic restrictions and reference levels	14
	5.7 Exposure limits.....	14
	5.8 Compliance assessment	14
	5.9 Uncertainty evaluation.....	15
	5.10 Exposure assessment in areas around hospitals, schools, etc.....	15
6	General characteristics of typical sources of the radiation	15
	6.1 Amplitude modulation transmitting stations	16
	6.2 Shortwave transmitting station – main beam tilt.....	16
	6.3 Fixed amateur stations.....	17
	6.4 Fixed point-to-point systems	17
	6.5 Personal wireless communication devices	17
	6.6 Wireless power transfer systems	22
	6.7 VHF and UHF broadcasting transmitting stations	23
	6.8 2G, 3G and 4G mobile base stations	23
	6.9 Smart (adaptive) antennas	23
	6.10 Vehicle mounted antennas (such as police car).....	23
7	Exposure assessment for fixed stations	23
	7.1 Pre-analysis.....	23
	7.2 Measurements.....	24
	7.3 Calculations	30
	7.4 Comparison between measurement and calculation.....	35
	7.5 Monitoring RF EMF levels	36
8	Conclusions following the exposure assessment.....	37
9	Final report.....	37
10	Field levels around typical transmitting antennas	37

	Page
11 Conclusions.....	38
Appendix I – Exposure limits	39
I.1 Introduction	39
I.2 ICNIRP-2010 and ICNIRP-2020 guidelines in force.....	39
I.3 IEEE C95.1-2019	43
I.4 Simultaneous exposure to multiple sources	47
Appendix II – Time averaging	49
II.1 Analysis of time variations of measured electric fields from WCDMA mobile base stations.....	49
II.2 Analysis of time variations of measured electric fields from GSM mobile base stations.....	54
II.3 Results and discussion.....	54
II.4 Discussion about the number of sampling data and averaging time	57
Appendix III – Examples of RF EMF levels in areas accessible to the general public.....	59
Appendix IV – <i>Watt guard</i> software.....	63
Appendix V – Uncertainty calculator software.....	64
V.1 Introduction	64
V.2 Brief description of the software	64
V.3 Examples	66
Appendix VI – Examples for evaluating electromagnetic fields in general public environments with broadband radio signals	70
VI.1 Method for evaluating electromagnetic fields in general public environments with broadband signals	70
VI.2 Effect of RBWs on measured electromagnetic fields from mobile phone base stations.....	71
Appendix VII – Example of block diagram with possible activities during exposure assessment.....	74
VIII.1 Introduction	75
VIII.2 Brief description of the main features of "EMF exposure"	75
VIII.3 Brief description of the calculations done by "EMF exposure"	77
Bibliography.....	78

Electronic attachment containing an uncertainty calculator, ITU EMF guide and the Watt guard applications.

Recommendation ITU-T K.91

Guidance for assessment, evaluation and monitoring of human exposure to radio frequency electromagnetic fields

1 Scope

This Recommendation¹ gives guidance on how to assess and monitor human exposure to radio frequency (RF) electromagnetic fields (EMFs) in close proximity to wireless devices and in areas with surrounding telecommunication installations, such as base stations (BSs) as defined in [IEC 62232], radiocommunication installations based on existing exposure and compliance standards in the 8.3 kHz to 300 GHz range. This Recommendation presents and references in clear and simple ways, procedures for evaluating exposure and how to show compliance with exposure limits. Existing standards are product or service oriented. This Recommendation is oriented to the examination of the area accessible to people in the real environment of currently operated services with many different sources of RF EMFs, but also gives references to standards and Recommendations related to EMF compliance of products.

[b-ITU-T K-Sup.1] provides EMF information and education resources suitable for all communities, stakeholders and governments. It gives answers to questions commonly posed by the public on EMF and to related concerns. [b-ITU-T K-Sup.1] is also available as a mobile application that is available from <http://emfguide.itu.int/emfguide.html>. [b-ITU-T K-Sup.4] presents EMF considerations for smart sustainable cities. In [b-ITU-T K-Sup.9] discusses the main impact of 5G technology on human exposure to RF EMFs. [b-ITU-T K-Sup.13] provides information concerning RF-EMF exposure levels from mobile and portable devices during different conditions of use. In [b-ITU-T K-Sup.14] presents the impact of RF-EMF exposure limits stricter than the International Commission on Non-Ionizing Radiation Protection (ICNIRP) guidelines or Institute of Electrical and Electronics Engineers (IEEE) limits on 4G and 5G mobile network deployment. [b-ITU-T K-Sup.16] provides information concerning RF-EMF compliance assessments for 5G wireless networks. Information concerning "Electromagnetic field (EMF) strength inside underground railway trains" is included in [b-ITU-T K.Sup.19], and information concerning "Radiofrequency exposure evaluation around underground base stations" is included in [b-ITU-T K.Sup.20]. [ITU-T K.153] describes the procedure for establishing exclusion zones. [b-ITU-T K.Suppl.29] presents an analysis of EMF exposure around electric vehicles. [b-ITU-T K.Suppl.32] provides examples of exposure assessments to radio frequency electromagnetic fields.

2 References

The following ITU-T Recommendations and other references contain provisions which, through reference in this text, constitute provisions of this Recommendation. At the time of publication, the editions indicated were valid. All Recommendations and other references are subject to revision; users of this Recommendation are therefore encouraged to investigate the possibility of applying the most recent edition of the Recommendations and other references listed below. A list of the currently valid ITU-T Recommendations is regularly published. The reference to a document within this Recommendation does not give it, as a stand-alone document, the status of a Recommendation.

[ITU-T K.52] Recommendation ITU-T K.52 (2021), *Guidance on complying with limits for human exposure to electromagnetic fields*.

¹ This Recommendation includes an electronic attachment containing an uncertainty calculator, ITU EMF-guide and the *Watt guard* applications.

[ITU-T K.61]	Recommendation ITU-T K.61 (2018), <i>Guidance on measurement and numerical prediction of electromagnetic fields for compliance with human exposure limits for telecommunication installations.</i>
[ITU-T K.70]	Recommendation ITU-T K.70 (2020), <i>Mitigation techniques to limit human exposure to EMFs in the vicinity of radiocommunication stations.</i>
[ITU-T K.83]	Recommendation ITU-T K.83 (2024), <i>Monitoring of electromagnetic field levels.</i>
[ITU-T K.90]	Recommendation ITU-T K.90 (2018), <i>Evaluation techniques and working procedures for compliance with exposure limits of network operator personnel to power-frequency electromagnetic fields.</i>
[ITU-T K.100]	Recommendation ITU-T K.100 (2021), <i>Measurement of radio frequency electromagnetic fields to determine compliance with human exposure limits when a base station is put into service.</i>
[ITU-T K.113]	Recommendation ITU-T K.113 (2015), <i>Generation of radiofrequency electromagnetic field level maps.</i>
[ITU-T K.121]	Recommendation ITU-T K.121 (2016), <i>Guidance on the environmental management for compliance with radio frequency EMF limits for radiocommunication base stations.</i>
[ITU-T K.122]	Recommendation ITU-T K.122 (2016), <i>Exposure levels in close proximity of radiocommunication antennas.</i>
[ITU-T K.145]	Recommendation ITU-T K.145 (2020), <i>Assessment and management of compliance with radio frequency electromagnetic field exposure limits for workers at radiocommunication sites and facilities.</i>
[ITU-T K.153]	Recommendation ITU-T K.153 (2023), <i>Guidance on determining the compliance boundaries (the exclusion zones) of radio transmitter installations.</i>
[ITU-R BS.1195]	Recommendation ITU-R BS.1195 (2013), <i>Transmitting antenna characteristics at VHF and UHF.</i>
[ITU-R BS.1698]	Recommendation ITU-R BS.1698 (2023), <i>Evaluating fields from terrestrial broadcasting transmitting systems operating in any frequency band for assessing exposure to non-ionizing radiation.</i>
[IEC/IEEE 62209-1528]	IEC/IEEE 62209-1528:2020, <i>Measurement procedure for the assessment of specific absorption rate of human exposure to radio frequency fields from hand-held and body-worn wireless communication devices – Part 1528: Human models, instrumentation and procedures (Frequency range of 4 MHz to 10 GHz).</i>
[IEC 62232]	IEC 62232:2022, <i>Determination of RF field strength, power density and SAR in the vicinity of base stations for the purpose of evaluating human exposure.</i>
[IEC 62311]	IEC 62311:2019, <i>Assessment of electronic and electrical equipment related to human exposure restrictions for electromagnetic fields (0 Hz-300 GHz).</i>

[IEC/IEEE 62704-1]	IEC/IEEE 62704-1:2017, <i>Determining the peak spatial-average specific absorption rate (SAR) in the human body from wireless communications devices, 30 MHz to 6 GHz – Part 1: General requirements for using the finite difference time-domain (FDTD) method for SAR calculations.</i>
[IEC/IEEE 62704-2]	IEC/IEEE 62704-2:2017, <i>Determining the peak spatial-average specific absorption rate (SAR) in the human body from wireless communications devices, 30 MHz to 6 GHz – Part 2: Specific requirements for finite difference time domain (FDTD) modelling of exposure from vehicle mounted antennas.</i>
[IEC/IEEE 62704-3]	IEC/IEEE 62704-3:2017, <i>Determining the peak spatial-average specific absorption rate (SAR) in the human body from wireless communications devices, 30 MHz to 6 GHz – Part 3: Specific requirements for using the finite difference time domain (FDTD) method for SAR calculations of mobile phones.</i>
[IEC/IEEE 62704-4]	IEC/IEEE 62704-4:2020, <i>Determining the peak spatial-average specific absorption rate (SAR) in the human body from wireless communication devices, 30 MHz to 6 GHz – Part 4: General requirements for using the finite element method for SAR calculations.</i>

3 Definitions

3.1 Terms defined elsewhere

This Recommendation uses the following terms defined elsewhere:

3.1.1 antenna [ITU-T K.70]: Device that serves as a transducer between a guided wave (e.g., coaxial cable) and a free space wave, or vice versa. It can be used to emit or receive a radio signal. In this Recommendation the term antenna is used only for emitting antenna(s).

3.1.2 averaging time (T_{avg}) [ITU-T K.52]: The averaging time is the appropriate time period over which exposure is averaged for purposes of determining compliance with the limits.

3.1.3 basic restrictions [ITU-T K.70]: Restrictions on exposure to time-varying electric, magnetic and electromagnetic fields that are based directly on established health effects. Depending upon the frequency of the field, the physical quantities used to specify these restrictions are: current density (J), specific absorption rate (SAR) and power density (S).

3.1.4 compliance distance [ITU-T K.70]: Minimum distance from the antenna to the point of investigation where the field level is deemed to be compliant to the limits.

3.1.5 contact current [ITU-T K.52]: Contact current is the current flowing into the body by touching a conductive object in an electromagnetic field.

3.1.6 continuous exposure [ITU-T K.52]: Continuous exposure is defined as exposure for duration exceeding the corresponding averaging time. Exposure for less than the averaging time is called short-term exposure.

3.1.7 controlled/occupational exposure [ITU-T K.70]: Controlled/occupational exposure applies to situations where the persons are exposed as a consequence of their employment and in which those persons who are exposed have been made fully aware of the potential for exposure and can exercise control over their exposure. Controlled/occupational exposure also applies to the cases where the exposure is of transient nature as a result of incidental passage through a location where the exposure limits may be above the general population/uncontrolled environment limits, as long as the exposed

person has been made fully aware of the potential for exposure and can exercise control over his or her exposure by leaving the area or by some other appropriate means.

3.1.8 directivity [ITU-T K.70]: Is the ratio of the power radiated per unit solid angle over the average power radiated per unit solid angle.

3.1.9 equivalent isotropically radiated power (EIRP) [ITU-T K.70]: The EIRP is the product of the power supplied to the antenna and the maximum antenna gain relative to an isotropic antenna.

3.1.10 equivalent radiated power (ERP) [ITU-T K.70]: The ERP is the product of the power supplied to the antenna and the maximum antenna gain relative to a half-wave dipole.

3.1.11 exposure [ITU-T K.52]: Exposure occurs wherever a person is subjected to electric, magnetic or electromagnetic fields, or to contact currents other than those originating from physiological processes in the body or other natural phenomena.

3.1.12 exposure level [ITU-T K.52]: Exposure level is the value of the quantity used when a person is exposed to electromagnetic fields or contact currents.

3.1.13 exposure limits [ITU-T K.70]: Values of the basic restrictions or reference levels acknowledged, according to obligatory regulations, as the limits for the permissible maximum level of the human exposure to the electromagnetic fields.

3.1.14 exposure, non-uniform/partial body [ITU-T K.52]: Non-uniform or partial-body exposure levels result when fields are non-uniform over volumes comparable to the whole human body. This may occur due to highly directional sources, standing waves, scattered radiation or in the near field.

3.1.15 far-field region [ITU-T K.52]: That region of the field of an antenna where the angular field distribution is essentially independent of the distance from the antenna. In the far-field region, the field has predominantly plane-wave character, i.e., locally uniform distribution of electric field strength and magnetic field strength in planes transverse to the direction of propagation.

3.1.16 general population/uncontrolled exposure [ITU-T K.52]: General population/uncontrolled exposure applies to situations in which the general public may be exposed, or in which persons who are exposed as a consequence of their employment may not be made fully aware of the potential for exposure or cannot exercise control over their exposure.

3.1.17 general public [ITU-T K.52]: All non-workers (see definition of worker in clause 3.2.14) are defined as the general public.

NOTE – General public exposure – RF exposure of persons who have not received any form of RF safety awareness information or training. Typically, general public exposure occurs in uncontrolled environments and includes individuals of all ages and varying health status, including children, pregnant women, individuals with impaired thermoregulatory systems, individuals equipped with electronic medical devices and persons using medications that may result in poor thermoregulatory system performance [b-IEEE C95.7].

3.1.18 induced current [ITU-T K.52]: Induced current is the current induced inside the body as a result of direct exposure to electric, magnetic, or electromagnetic fields.

3.1.19 intentional emitter [ITU-T K.52]: Intentional emitter is a device that intentionally generates and emits electromagnetic energy by radiation or by induction.

3.1.20 intentional radiation [ITU-T K.70]: Electromagnetic fields radiated through the transmitting antenna even in directions which are not needed (for example to the back of the parabolic microwave antenna).

3.1.21 near-field region [ITU-T K.52]: The near-field region exists in proximity to an antenna or other radiating structure in which the electric and magnetic fields do not have a substantially plane-wave character but vary considerably from point to point. The near-field region is further subdivided into the reactive near-field region, which is closest to the radiating structure and that

contains most or nearly all of the stored energy, and the radiating near-field region where the radiation field predominates over the reactive field but lacks substantial plane-wave character and is complicated in structure.

NOTE – For many antennas, the outer boundary of the reactive near-field is taken to exist at a distance of one wavelength from the antenna surface.

3.1.22 power density, average (temporal) [ITU-T K.52]: The average power density is equal to the instantaneous power density integrated over a source repetition period.

NOTE – This averaging is not to be confused with the measurement averaging time.

3.1.23 power density, peak [ITU-T K.52]: The peak power density is the maximum instantaneous power density occurring when power is transmitted.

3.1.24 power density, plane-wave equivalent (S_{eq}) [ITU-T K.52]: The equivalent plane-wave power density is a commonly used term associated with any electromagnetic wave, equal in magnitude to the power flux-density of a plane wave having the same electric (E) or magnetic (H) field strength.

3.1.25 radio frequency (RF) [ITU-T K.70]: Any frequency at which electromagnetic radiation is useful for telecommunication.

NOTE – In this Recommendation, radiofrequency refers to the 9 kHz to 300 GHz range allocated by ITU-R Radio Regulations.

3.1.26 reference levels [ITU-T K.70]: Reference levels are provided for the purpose of comparison with exposure quantities in air. The reference levels are expressed as electric field strength (E), magnetic field strength (H) and power density (S) values. In this Recommendation, the reference levels are used for the exposure assessment.

3.1.27 specific absorption (SA) [ITU-T K.52]: Specific absorption is the quotient of the incremental energy (dW) absorbed by (dissipated in) an incremental mass (dm) contained in a volume element (dV) of a given density (ρ_m).

$$SA = \frac{dW}{dm} = \frac{1}{\rho_m} \frac{dW}{dV}$$

The specific absorption is expressed in units of joules per kilogram (J/kg).

3.1.28 unintentional emitter [ITU-T K.52]: An unintentional emitter is a device that intentionally generates electromagnetic energy for use within the device, or that sends electromagnetic energy by conduction to other equipment, but which is not intended to emit or radiate electromagnetic energy by radiation or induction.

3.1.29 unintentional radiation [ITU-T K.70]: Electromagnetic fields radiated unintentionally, for example through the transmitter enclosure or feeding line.

3.1.30 whole-body-exposure [b-IEEE C95.1]: The case in which the projected area of the entire body is exposed to the incident fields.

3.2 Terms defined in this Recommendation

This Recommendation defines the following terms:

3.2.1 antenna gain: The antenna gain $G_i(\theta, \phi)$ is the ratio of power radiated per unit solid angle multiplied by 4π to the total input power. The gain is frequently expressed in decibels with respect to an isotropic antenna (dBi). The formula defining the gain is:

$$G_i(\theta, \phi) = \eta \frac{4\pi}{P_{in}} \frac{dP_r}{d\Omega}$$

where:

θ, ϕ are angles in the polar coordinate system

η is the antenna efficiency due to dissipative losses

P_r is the radiated power in the (θ, ϕ) direction

P_{in} is the total input power

Ω is an elementary solid angle in the direction of observation.

NOTE 1 – In manufacturers' catalogues the antenna gain is understood as a maximum value of the antenna gain. Gain does not include losses arising from impedance and polarization mismatches. If an antenna is without dissipative loss, then its gain is equal to its directivity $D(\theta, \phi)$.

NOTE 2 – Based on the definition in [ITU-T K.70].

3.2.2 average (temporal) power (P_{avg}): The time-averaged rate of energy transfer defined by:

$$P_{avg} = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} P(t) dt$$

where:

$P(t)$ is the instantaneous power

t_1 and t_2 are the start and stop time of the exposure.

NOTE – Based on the definition in [ITU-T K.52].

3.2.3 body-mounted device, body-worn device: Portable device containing a wireless transmitter or transceiver that may be located close to a person's torso except the head during its intended use or operation of its radio functions (e.g., on a belt clip, holster, pouch or on a lanyard).

NOTE – Based on the definition in [IEC/IEEE 62209-1528].

3.2.4 electromagnetic field (EMF): A field determined by a set of four interrelated vector quantities that characterizes, together with the electric current density and the volumic electric charge, the electric and magnetic conditions of a material medium or of a vacuum.

3.2.5 hand-held device, mobile handset: A portable device containing a wireless transmitter or transceiver that can be located in a user's hand during its intended use or operation of its radio functions. A hand-held device for this Recommendation is a unit that is essentially not meant to be used close to the head or body but is held in the hand. A typical example of a hand-held device is a personal digital assistant (PDA) with an integrated RF module.

NOTE – Based on the definition in [IEC/IEEE 62209-1528].

3.2.6 point of investigation (POI): The location within the domain of investigation at which the value of the E -field, H -field or power density is evaluated.

3.2.7 power density (S): Power flux-density is the power per unit area normal to the direction of electromagnetic wave propagation, usually expressed in units of watts per square metre (W/m^2). In this Recommendation, this term is commonly used to refer to equivalent plane wave power density, see clause 3.1.26.

NOTE 1 – For plane waves, power flux-density, electric field strength (E), and magnetic field strength (H) are related by the intrinsic impedance of free space, $Z_0 \approx 377$ or $120\pi \Omega$. In particular,

$$S_{eq} = \frac{E^2}{Z_0} = Z_0 H^2 = EH$$

where E and H are expressed in units of V/m and A/m, respectively, and S in units of W/m^2 . Although many survey instruments indicate power density units, the actual quantities measured are E or H .

NOTE 2 – Based on the definition in [ITU-T K.52].

3.2.8 relative field pattern (radiation pattern), antenna pattern: The ratio of the absolute value of the field strength (arbitrarily taken to be the electric field) to the absolute value of the maximum field strength. It is related to the relative numeric gain (see clause 3.2.9) as follows:

$$f(\theta, \phi) = \sqrt{F(\theta, \phi)}$$

where $f(\theta, \phi)$ is the relative field pattern.

NOTE – Based on the definition in [ITU-T K.70].

3.2.9 relative numeric gain (normalized antenna gain), $F(\theta, \phi)$: The ratio of the antenna gain at each angle to the maximum antenna gain, which has a value ranging from 0 to 1.

NOTE – Based on the definition in [ITU-T K.70].

3.2.10 short-term exposure: The exposure for a duration less than the corresponding averaging time. Exposure for a duration exceeding the averaging time is called continuous exposure.

NOTE – Based on the definition in [ITU-T K.52].

3.2.11 specific absorption rate (SAR): The time derivative of the incremental energy (dW) absorbed by (dissipated in) an incremental mass (dm) contained in a volume element (dV) of a given mass density (ρ_m).

$$\text{SAR} = \frac{d}{dt} \frac{dW}{dm} = \frac{d}{dt} \left(\frac{1}{\rho_m} \frac{dW}{dV} \right)$$

SAR is expressed in units of watts per kilogram (W/kg).

SAR can be calculated by:

$$\begin{aligned} \text{SAR} &= \frac{\sigma E^2}{\rho_m} \\ \text{SAR} &= c \frac{dT}{dt} \\ \text{SAR} &= \frac{J^2}{\rho_m \sigma} \end{aligned}$$

where:

E is the r.m.s. value of the electric field strength in body tissue in V/m

σ is the conductivity of body tissue in S/m

ρ_m is the density of body tissue in kg/m³

c is the heat capacity of body tissue in J/kg °C

$\frac{dT}{dt}$ is the initial time derivative of temperature (at $t = 0$) in body tissue in °C/s

J is the value of the induced current density in the body tissue in A/m².

NOTE – Based on the definition in [ITU-T K.52].

3.2.12 transmitter: An electronic device used to intentionally generate radio frequency electromagnetic energy for the purpose of communication (in contrast to the definition for intentional emitter in clause 3.1.19). The transmitter output is connected via a feeding line to the transmitting antenna, which is the real source of intentional electromagnetic radiation.

NOTE – Based on the definition in [ITU-T K.70].

3.2.13 wavelength (λ): The wavelength of an electromagnetic wave is related to frequency (f) and propagation velocity (v) of an electromagnetic wave by the following expression:

$$\lambda = \frac{v}{f}$$

In free space, the propagation velocity is equal to the speed of light (c), which is approximately 3×10^8 m/s. In body tissue, the propagation velocity is reduced by the square root of the relative

dielectric constant, so that the wavelength in tissue is typically seven times shorter than that in free space.

NOTE – Based on the definition in [ITU-T K.52].

3.2.14 worker: Any person employed by an employer, including trainees and apprentices, but excluding domestic servants (see clause 3.1.6).

NOTE – Based on the definition in [ITU-T K.70].

4 Abbreviations and acronyms

This Recommendation uses the following abbreviations and acronyms:

AC	Alternating Current
AM	Amplitude Modulation
APC	Automatic Power Control
BS	Base Station
CDMA	Code Division Multiple Access
EIRP	Equivalent Isotropically Radiated Power
EMF	Electromagnetic Field
ERP	Equivalent Radiated Power
ERL	Exposure Reference Level
FDTD	Finite Difference Time Domain
FEM	Finite Element Method
FM	Frequency Modulation
FPP	Fixed Point-to-Point
GSM	Global System for Mobile communications
HRP	Horizontal Radiation Pattern
IF	Intermediate Frequency
IMT-2000	International Mobile Telecommunication-2000
LBS	Location-Based Service
LOS	Line of Sight
LW	Long Wave
MoM	Method of Moments
MW	Medium Wave
OFDM	Orthogonal Frequency-Division Multiplexing
PC	Personal Computer
PCS	Personnel Communication System
PDA	Personal Digital Assistant
POI	Point of Investigation
RBW	Resolution Bandwidth
RF	Radio Frequency

r.m.s.	root mean square
RSS	Root Sum Square
SA	Specific Absorption
SAM	Specific Anthropomorphic Mannequin
SAR	Specific Absorption Rate
SD	Standard Deviation
SW	Shortwave
S-DMB	Satellite-Digital Multimedia Broadcasting
TER	Total Exposure Rate
TRS	Trucked Radio System
TV	Television
UHF	Ultra-High Frequency
UMTS	Universal Mobile Telecommunication System
VHF	Very High Frequency
VRP	Vertical Radiation Pattern
WCDMA	Wideband Code Division Multiple Access
WiBro	Wireless Broadband Internet
Wi-Fi	Wireless Fidelity
WiMAX	Worldwide interoperability for Microwave Access

5 General guidance

This clause presents a general description of the procedure for assessing exposure to EMFs. The procedural steps are described in more detail in clauses 6 to 10. This procedure covers all possible steps, but in actual cases some steps will not be required. This procedure applies to exposure assessment in close proximity to wireless devices, and in areas around radiocommunication installations, i.e., around transmitting stations and BSs (in this Recommendation the term transmitting station is used). It does not cover exposure assessment for unintentional radiation from telecommunication equipment (e.g., radiation through the enclosure of the transmitter) or exposure to non-telecommunication equipment, such as industrial induction heating equipment or alternating current (AC) power systems. See [IEC 62232] for more general information about measurement or calculation methods for BSs. Information concerning implementation of [IEC 62232] is included in [b-IEC TR 62669].

It is recommended to first use the simplest method, even if other methods give higher accuracy. In many cases, the exposure level is far below the acceptable limit, and more sophisticated methods (more difficult to apply) are not necessary. The choice between measurement and calculation, as well as between different methods of exposure assessment, are discussed in this Recommendation.

Figure 5-1 presents the basic concept of the compliance check of human exposure to RF EMF.

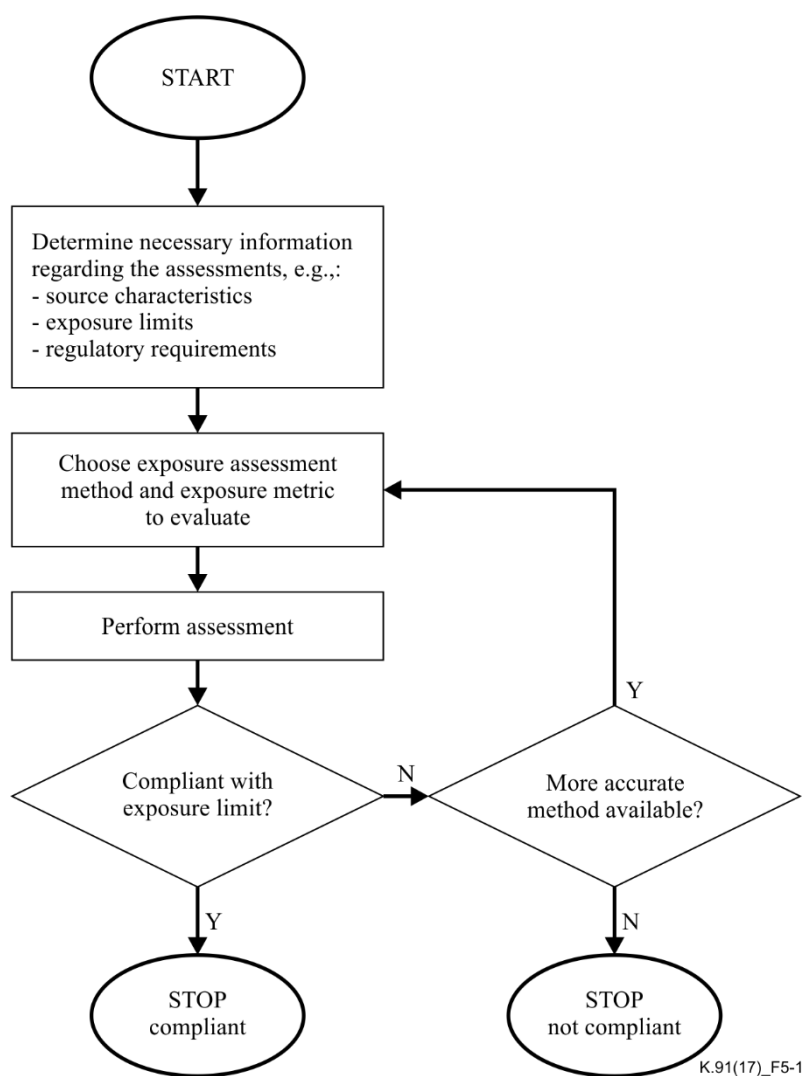


Figure 5-1 – General description of the exposure assessment procedure

Clauses 5.1 to 5.10 present a description of the possible activities that may be used during exposure assessment (see Appendix VII): the main problems; the advantages and disadvantages; the characteristics of the radiating sources that are used in radiocommunication; and exposure levels that may be expected in the areas around typical radiocommunication antennas.

In general, the exposure assessment is made by measurement or calculation. In some cases, it is more efficient to assess exposure to some sources of radiation by measurement (see [IEC 62232], [IEC/IEEE 62209-1528], [IEC 62209-3], [b-IEC TR 63170], [IEC TR 62905]) and for others by calculation (see [IEC 62232], [IEC 62704-1], [IEC 62704-2], [IEC 62704-3], [IEC 62704-4], [IEC TR 62905]). Also, in some cases, the exposure assessment for certain sources of radiation may be referenced against basic restrictions and for others against the reference levels. In any case, all sources of radiation should be considered once and all of them should be included in the total exposure evaluation. However, exposure due to mobile devices should not be combined with exposure due to sources that are in a fixed location, since there would not be any definite positional relationship between these two different types of sources and, more practically, the exposure levels from each different device type would often not be correlated in comparison with the basic restrictions during an assessment to determine compliance with the exposure limits.

5.1 General public and occupational exposure

In general, the RF EMF exposure assessment of the general public and workers (occupational exposure) use the same methods [ITU-T K.145]. However, there are some specific features that distinguish the two categories. The most important are:

- exposure limits for the general public are more conservative because workers have better knowledge about possible hazards and location of places with the highest exposure levels and they are exposed during working hours only;
- workers are more likely to approach closer than the general public to RF sources (transmitting antennas), so in more cases the area under examination is located in the reactive near-field region, which, as a consequence, requires more sophisticated methods for the exposure assessment by measurement or by calculation;
- workers in some cases are allowed to stay in areas with exposure levels above the limits, but for limited time periods or by using mitigation measures like protective clothes.

5.2 Existing or planned transmitting station

If the area under consideration is covered by emissions from existing RF sources only, then the exposure assessment can be performed either by measurement or by calculation. If new RF EMF sources are planned (even if there exist other operating radio systems) in the area under consideration, the possibility to check compliance is through the exposure assessment by calculation or by a combination of measurement and calculation.

[ITU-T K.121] presents guidance on the environmental management procedure for compliance with RF EMF limits for BSs. For the suggested procedure to perform when a BS is put into service, see [ITU-T K.100] and [IEC 62232].

5.3 Collection of data concerning sources of radiation

Collection of data concerning RF sources is helpful in many respects. If exposure assessment by measurement is planned, it is in principle possible to do without these data, but measurements without data concerning RF sources are more difficult and results of measurements less reliable. Data collection is required for exposure assessment by calculation. Generally, the more detailed descriptions of the RF sources are available – the more exact results of evaluation can be obtained. This Recommendation considers intentional radiation from transmitting antennas. So the description of these antennas is most important. However, much detailed information can be commercially sensitive and difficult to collect.

5.3.1 Data required for measurement

In general, measurements can be done without complete knowledge of the radiating sources if proper equipment that covers the full range of frequencies is available, knowing at least the range of frequencies to be measured. If measurements are made with wideband equipment (without frequency selection or shaped response), the results of such measurement will be conservative because they require the use of the limit value, which is most restrictive. In all cases of measurement, information concerning radiating sources is very helpful and makes measurements more accurate and reliable.

The following data are very helpful during measurements (for each radiating source):

- operating frequency – this allows use of a probe that has a band covering all operating frequencies;
- distance to the transmitting antenna – this allows the field region to be determined (for each operating frequency) and a proper measurement procedure chosen;
- maximum equivalent radiated power (ERP) – this allows estimation of the dynamic range required of the measurement equipment and the expected levels of measured values;

- whether the antennas are operating at the maximum transmitter power at the time of measurement;
- modulation characteristics – especially pulsed, intermittent or continuous operation.

Usually, this information can be obtained from the transmitting system documentation. Some data can be obtained during site inspection (e.g., distances to the transmitting antennas, operating frequencies based on the types and sizes of the transmitting antennas).

5.3.2 Data required for calculation

Exposure assessment by calculation in all cases requires information concerning radiating sources. In general, the more detailed the information, the more accurate are the results of calculations. These data are required for each operating (or planned) frequency. The required data put in order of growing level of accuracy of the exposure assessment by calculation are presented in the following.

The minimum data required for calculation, which leads to the most conservative approach, are:

- operating frequency;
- distance to the transmitting antenna;
- maximum equivalent isotropically radiated power (EIRP).

In this case, the point source model with isotropic antenna may be used for exposure assessment. The next step is:

- radiation patterns of the transmitting antenna.

These additional data allow use of the point source model with radiation patterns taken into account.

Many transmitting antennas are built as systems containing many identical radiating elements (panels in broadcasting or patches in mobile communication). The additional data allowing for more accurate results are:

- geometry of the transmitting antenna (spatial position of each panel or patch);
- radiation pattern of the individual panel or patch;
- feeding arrangement (amplitude or power and phase of each panel or patch feeding).

These data allow use of the synthetic model for the exposure assessment.

The data that allow for the most accurate calculations additionally contain:

- exact location of each metallic and dielectric part of the transmitting antenna;
- information concerning each excitation in the antenna system;
- or exact location of the metallic and dielectric parts of the antenna tower and the supporting structure in the vicinity of the transmitting antenna.

These data allow for the calculation using full-wave methods such as the method of moments (MoM) or finite difference time domain (FDTD).

For additional details, see [IEC 62232], [IEC TR 62669], [IEC 62704-1], [IEC 62704-2], [IEC 62704-3], [IEC 62704-4], [b-IEC PAS 63184], [ITU-R BS.1698], [ITU-R BS.1195], [b-ITU-R SpecMon], [b-ITU-R SM.2452], [ITU-T K.70] and [b-EN 50413].

5.4 One or more sources of radiation, total exposure

The area under consideration may be affected by one operating source (rare case) or by many sources of radiation, which is a typical case at present. If there is one source, the exposure assessment is relatively simple, and the measured or calculated levels require comparison with the appropriate exposure limits. If there is more than one source, then the total exposure should be considered.

If the assessment is made by measurement, all operating frequencies have to be covered by the measuring equipment. In the assessment by calculation, the exposure to each radiating source (operating frequency) requires separate evaluation. In both cases, the total exposure to all sources should be evaluated. However, exposure due to mobile devices should not be combined with exposure due to transmitters that are in a fixed location since there would not be any definite positional relationship between these two different types of sources. Thus, the total exposure from all sources in a single mobile device should be considered in a mobile device exposure assessment, and separately the total exposure from all sources in a multiple source environment should be considered in a fixed-source exposure assessment.

The exposure assessment in the multiple source environment, according to existing standards, requires the evaluation of the total exposure (in some standards called the total exposure ratio). All operating frequencies must be considered in a weighted sum, where each individual source is pro-rated according to the limit applicable to its frequency. According to most standards [ITU-T K.52], [IEC 62232], any sources with values less than 5% of the relevant limit do not need to be included, as the methodology is inherently conservative.

In general, the total exposure for induced current density and electrical stimulation effects (relevant up to 10 MHz) and for electric and magnetic field components for the thermal effect (relevant above 100 kHz) should be compared with the limits (see Appendix I).

Outside the reactive near-field region, the compliance for the electric field components is sufficient. For typical radiocommunication antennas, exposure limits are more restrictive for the thermal effect than for electro-stimulation.

Clause I.4 gives guidance for combining exposure from multiple sources.

Additional details can be found in [b-ICNIRP 1998], [b-ICNIRP 2020], [b-IEEE C95.1], [IEC/IEEE 62209-1528], [IEC 62232], [ITU-T K.70], [ITU-T K.52], [b-EN 50413] and [b-IEC TR 62630].

5.5 Field regions

Borders of the field regions generally depend on the distance to, and design of, the transmitting antenna. In the reactive near-field region, located closest to the transmitting antenna, the electric and magnetic fields have a phase and amplitude relationship that is strongly dependent on location. There is a considerable amount of reactive energy, where the phase between the electric and magnetic fields changes. This reactive energy does not propagate.

In the far field, the electric and magnetic fields are in phase and orthogonal to each other. The energy propagates away from the antenna and drops with the square of the distance.

The radiative near field is a transition region between the reactive near field and the far field.

For fixed stations (e.g., base stations), usually, the general public has access to areas located in the far-field region, or in some cases, in the radiating near-field region. In the far-field region, the EMF has a simple structure similar to a plane wave and calculations or measurements are comparatively easy.

The distance to the beginning of the far-field region is the most important factor. The usually assumed value is $2D^2/\lambda$, where D is the maximum size of the antenna. The distance is rather long and some POIs can be located at a shorter distance, which implies more complex calculations. According to Annex B of [IEC 62232] for antennas with the maximum size $D > 2.5\lambda$, and radiating elements in a linear configuration (this is true for the typical BS panels and also for the faces of the broadcasting antennas) this distance is described by the formula:

$$d = \frac{2D^2}{\lambda} \sin^2 \beta - \frac{\lambda}{32} + \frac{D}{2} |\cos \beta| \quad (5-2)$$

where β is the angle between the main axis of the antenna and line from the antenna centre to the POI. Equation (5-2) gives a substantially smaller distance of the beginning of the far-field region for directions outside the main beam ($\beta = 0$) of the antenna, which extends the area of applicability of the methods appropriate for the far-field region. In Annex B of [IEC 62232], there is also a simplified formula valid for the same types of antennas giving distance $d = 0.6D^2/\lambda$.

For additional details, see [IEC 62232] and [ITU-T K.61].

5.6 Basic restrictions and reference levels

Basic restrictions on exposure to RF EMFs are quantities most closely related to established adverse biological effects. They include quantities to be determined in body tissue such as specific absorption rate (SAR) inside the body. If it is difficult to calculate or measure these values, reference levels for human exposure to external electric and magnetic fields and power density are derived from the basic restrictions using the realistic worst-case assumption about exposure. If the reference limits are met, then the basic restrictions will also be met; if reference levels are exceeded, that does not necessarily mean that the basic restrictions are exceeded (see Figure 5-1). This approach means that the demand for the compliance with reference levels is conservative.

For additional details, see [IEC 62232] and [ITU-T K.52].

5.7 Exposure limits

Exposure limits are acknowledged, according to obligatory regulations, as the limits for the permissible maximum level of human exposure to EMFs. Exposure limits based on established scientific data and currently available knowledge are given in internationally recognized documents: [b-ICNIRP 1998], [b-ICNIRP 2020] and [b-IEEE C95.1] (see Appendix I). In many cases, the exposure limits are also given by national authorities and sometimes they are prepared in different ways based on different assumptions. The national regulations are obligatory, and they should be respected during exposure assessment. In most countries, the ICNIRP or IEEE exposure limits are valid. In countries that do not have their own regulations, the ICNIRP (recommended by the World Health Organization (WHO)) or IEEE exposure limits are highly recommended.

For exposure limits in various countries, see [b-WHO Internet].

For additional details, see Appendix I, [b-ICNIRP 2020], [b-ICNIRP 1998], and [b-IEEE C95.1]. Some problems with very restrictive exposure limits are presented in [b-ITU-T K-Sup.14].

5.8 Compliance assessment

If only one radiating source is considered, the results of calculation or measurement can be directly compared with the exposure limit for the compliance assessment.

If there is simultaneous exposure to multiple sources, the total exposure ratio should be less than 1.0 to comply with the regulations. In two frequency ranges: below 10 MHz (for the induced current density and electrical stimulation effect); and above 100 kHz (in which the thermal effect is dominating) – the total exposures are calculated in different ways and have to be compared separately with each limit; both values should be less than 1.0.

Usually, reference levels are used for compliance assessment. However, if the reference levels are exceeded, this does not necessarily mean that the basic restrictions are violated. In this case, the basic restrictions (which are less restrictive) should be evaluated and compared with the exposure limits. Many national regulations allow for the use of the reference levels only. Because of this, the exposure limits in these countries are more restrictive.

For additional details, see: [IEC 62232], [ITU-T K.52], [ITU-R BS.1698], [b-ITU-R SpecMon], [b-ITU-R SM.2452], and [b-IEC TR 62630].

5.9 Uncertainty evaluation

When performing calculation or measurement, the expanded uncertainty of the results requires specification with a 95% confidence level. The combined uncertainty is comprised of several uncertainty components. Each uncertainty component (the standard uncertainty $u(x_i)$) shall be registered with their names, the probability distribution, the sensitivity coefficients and the values).

Before combining, each uncertainty component shall be expressed in terms of the measured quantity and as a standard uncertainty having a NORMAL distribution. If any of the standard uncertainty components is not expressed in terms of the measured quantity, then it should be converted using the appropriate sensitivity coefficient, c_i , and the equation $u_i(y) = c_i u(x_i)$. If it does not have a Normal probability distribution, or if it uses a confidence level other than 1σ (68%), then it should be appropriately scaled, as described in [JCGM 100:2008]. All uncertainty contributions shall be appropriately combined, as described in [JCGM 100:2008].

The combined standard uncertainty $u_c(y)$ shall then be evaluated by taking the square root of the sum of squares of the uncertainty contributions $u_i(y)$ for independent or uncorrelated input quantities.

$$u_c(y) = \sqrt{\sum_{i=1}^m u_i^2(y)} \quad (5-3)$$

(see [IEC 62232] and [b-JCGM 100]).

The combined expanded uncertainty, having a 95% confidence level, is determined by multiplying the combined standard uncertainty by two:

$$U_c(y) = 2 \cdot u_c(y)$$

It is good practice to maintain the expanded uncertainty below a threshold. Some standards have requirements for the maximum uncertainty.

For fixed stations (e.g., base stations) compliance assessment [IEC 62232], the target expanded uncertainty for *in situ* field measurements should be ≤ 4 dB, which is considered industry best practice. The expanded uncertainty for the RF exposure evaluation used for *in situ* field measurements should not exceed 6 dB. For mobile devices, [IEC/IEEE 62209-1528] requires that the expanded uncertainty is below 30%.

Additional information and examples of evaluation can be found in [b-JCGM 100], [IEC 62232] and [ITU-T K.61].

5.10 Exposure assessment in areas around hospitals, schools, etc.

With respect to human exposure, currently there is no technical requirements for any special consideration for locating BSs close to areas such as hospitals and schools due to the fact that existing exposure guidelines incorporate in the exposure limits safety margins that are applicable to all locations.

NOTE – Using the mobile telephone within areas of good reception also decreases exposure as it allows the telephone to transmit at reduced power [b-WHO 2014].

6 General characteristics of typical sources of the radiation

Information concerning mobile BSs, wireless access systems, radiocommunication systems, broadcasting transmitting stations (television (TV), frequency modulation (FM), amplitude modulation (AM)) is presented in Appendix II of [ITU-T K.70]. Important information for some specific systems is presented in clauses 6.1 to 6.9, which may be useful for their exposure evaluation. Table 6-1 lists the frequency and wavelength ranges of typical radiocommunication services. Information concerning additional services for BSs can be found in [IEC 62232].

Table 6-1 – Examples of frequency and wavelength ranges of broadcasting and mobile downlink systems

Service	Frequency range (MHz)	Wavelength range (m)
LW broadcasting	0.148 5-0.255	2 018.8-1 175.7
MW broadcasting	0.526 5-1.606 5	569.4-186.6
SW broadcasting	3.95-26.1	75.9-11.5
FM broadcasting	87.5-108	3.43-2.78
VHF broadcasting	174-230	1.72-1.30
UHF broadcasting	470-790	0.64-0.38
Mobile 700	694-790	0.43-0.38
Mobile 800	790-860	0.38-0.35
Mobile 850	869-894	0.35-0.34
Mobile 900	935-960	0.32-0.31
T-DAB	1 452–1 492	0.20-0.21
Mobile 1 800	1 805-1 880	0.17-0.16
Mobile 1 900	1 930-1 990	0.16-0.15
Mobile 2 100	2 110-2 170	0.14-0.14
Wi-Fi	2 400-2 483	0.12-0.12
Mobile 2 600	2 620-2 690	0.11-0.11
WiMAX and Mobile 3 500	3 400-3 800	0.09-0.08
Wi-Fi	5 170-5 725	0.06-0.05
Mobile 26 000	24 250-29 500	0.012-0.010
VHF: Very high frequency UHF: Ultra-high frequency SW: Shortwave Wi-Fi: Wireless fidelity WiMAX: Worldwide interoperability for microwave access		

6.1 Amplitude modulation transmitting stations

AM broadcasting uses very long electromagnetic waves, see Table 6-1.

The EMF has a very complex structure in close proximity to the transmitting antenna – in the reactive near-field whose outer border is equal to one wavelength. Table 6-1 shows that the reactive near-field is largest (in descending order) in LW, MW and SW transmitting antennas. In these cases, the areas accessible to people (including the general public) are frequently located in the reactive near-field. Since the magnetic component is not proportional to the electric component in this region, both the electric and magnetic components should be measured. For computation in this case, the MoM is especially efficient.

For additional information concerning the measurement and the calculation of RF EMF around LW, MW and SW antennas, see [ITU-R BS.1698].

6.2 Shortwave transmitting station – main beam tilt

SW transmitting stations also have their own specificity, i.e., the shape of the vertical radiation pattern (VRP). The typical transmitting antenna in broadcasting or mobile BSs has a VRP with the

main beam tilted to the ground direction. This beam tilt is rather small for the broadcasting antennas (usually from 0° to 1°) and bigger for the mobile BS antennas (usually from 0° to 15°). In contrast, SW transmitting antennas usually have a VRP pointed upwards (from -7° to -40°) in order to reach the ionosphere. In this case, the radiation levels are highest at these high elevation angels.

For additional information concerning the measurement and the calculation of EMF around SW antennas, see [ITU-R BS.1698].

6.3 Fixed amateur stations

Measuring the EMFs of fixed amateur stations is generally a very complex task. In the case of amateur transmitting equipment, it is particularly important that the propagation conditions of the EMF in the near-field region of the transmitting antenna are taken into account, as such equipment can use large wavelengths and operate in densely populated housing areas. Such a complex environment in the near-field region can require time-consuming and expensive near-field measurement.

The *Watt guard* calculation tool enables a simple and fast assessment of fixed amateur stations. The software module runs on a wide variety of operating systems and is published together with this Recommendation. For additional information, see Appendix IV.

6.4 Fixed point-to-point systems

As opposed to other antennas, microwave radio relay antennas have a main beam with a very narrow radiation pattern and very low level side lobes in all other directions. Because of this, the RF EMF levels are higher in the region close to the main beam direction and decrease quickly outside it. Thus, these antennas may be considered as radiating in the point-to-point direction only. Additionally, if there is any large obstacle in the propagation path between a pair of radio relay antennas, then the communication link is broken, and this is acted on promptly. Thus, the probability that any object is present in the propagation path, including people, is very low.

RF EMF exposure has to be assessed only in close proximity to such antennas. The evaluation of the total exposure, which has to be performed in the environment where many transmitting antennas are operating simultaneously, may omit the radiation from the radio relay antennas, except in close vicinity to them.

For additional detailed information concerning the measurement and the calculation of RF EMF around radio relay antennas, see [ITU-R BS.1698] and [b-ITU-R SM.2452].

6.5 Personal wireless communication devices

Personal wireless communication devices, such as hand-held devices (mobile phones, two-way radios, tablets), laptops, desktop computers and body-mounted wireless devices require either SAR, incident power density, or absorbed power density evaluation.

For SAR calculation, FDTD (see [IEC/IEEE 62704-1] and [IEC/IEEE 62704-3]) and FEM (see [IEC/IEEE 62704-4]) can be used to model the handset and the human body that is close to the handset.

For SAR measurements, procedures for SAR evaluation of personal wireless communication devices are covered in clauses 6.5.1 and 7.2.8.

Incident power density evaluation could be performed for portable wireless devices in accordance with [b-IEC TR 63170].

Absorbed power density evaluation standards are currently under development within IEC.

6.5.1 SAR compliance procedures for personal wireless communication devices used against the head

This clause concerns the compliance assessment of RF EMF transmitting devices intended to be used with the radiating part of the device in close proximity to the human head and positioned against the

ear, such as mobile phones, cordless phones, and handsets operating in the 4 MHz to 10 GHz range. The metrics used to quantify the RF EMF exposure is SAR. The measurement of SAR requires specific instrumentation as well as highly advanced techniques and requirements. This is why, in line with the agreement between ITU-T and IEC, the user of this Recommendation shall refer to the latest version of [IEC/IEEE 62209-1528] to assess SAR compliance with the exposure limits. The compliance assessment shall be achieved by applying measurement requirements and measurement procedures specified in [IEC/IEEE 62209-1528]. The objective of [IEC/IEEE 62209-1528] is to specify the measurement method for demonstration of compliance with the SAR for the previously mentioned devices. In order to give some general idea about the content of [IEC/IEEE 62209-1528], a very brief description is provided in the following paragraphs.

[IEC/IEEE 62209-1528] has two normative references [IEC 62209-3] and [ISO/IEC Guide 98-3:2008]. It defines terms, definitions, symbols and abbreviations in clauses 3 and 4. It has a quick start guide in clause 5. Technical information related to general requirements, such as the phantom shell, tissue simulating liquid, specifications of the SAR measurement equipment, scanning system specifications and device holder specifications are explained in clause 6 of [IEC/IEEE 62209-1528]. The measurement-related requirements and information, including the measurement procedure, are explained in clause 7 of [IEC/IEEE 62209-1528]. Figure 6-1, taken directly from [IEC/IEEE 62209-1528], explains the procedures that shall be performed for each test condition. The head phantom used in the measurements for wireless devices held to the ear is the specific anthropomorphic mannequin (SAM), which is described in Annex G of [IEC/IEEE 62209-1528]. As an example case, the phantom and handset device is shown in Figure 6-2 and a head SAR measurement system is shown in Figure 6-3. Application-specific phantoms, such as wrist phantoms for smart watches and face-down phantoms for virtual-reality headsets, are described in Annex K of [IEC/IEEE 62209-1528].

One of the main issues in measurements is uncertainty evaluation. Clause 8 of [IEC/IEEE 62209-1528] is devoted to an extensive uncertainty estimation, where all components contributing to the uncertainty are identified and calculated. There are templates for the uncertainty evaluation for the handset SAR test, measurement uncertainty evaluation for system validation and measurement repeatability evaluation for system check. Clause 8 of [IEC/IEEE 62209-1528] specifies the measurement report.

In certain cases, local or national regulatory agencies or standards bodies may recommend national or regional measurement practices based on [IEC/IEEE 62209-1528] or other procedures. The SAR measurement procedures defined in [IEC 62209-3] have not yet been adopted by national regulatory agencies, and the IEC is working on improving this standard.

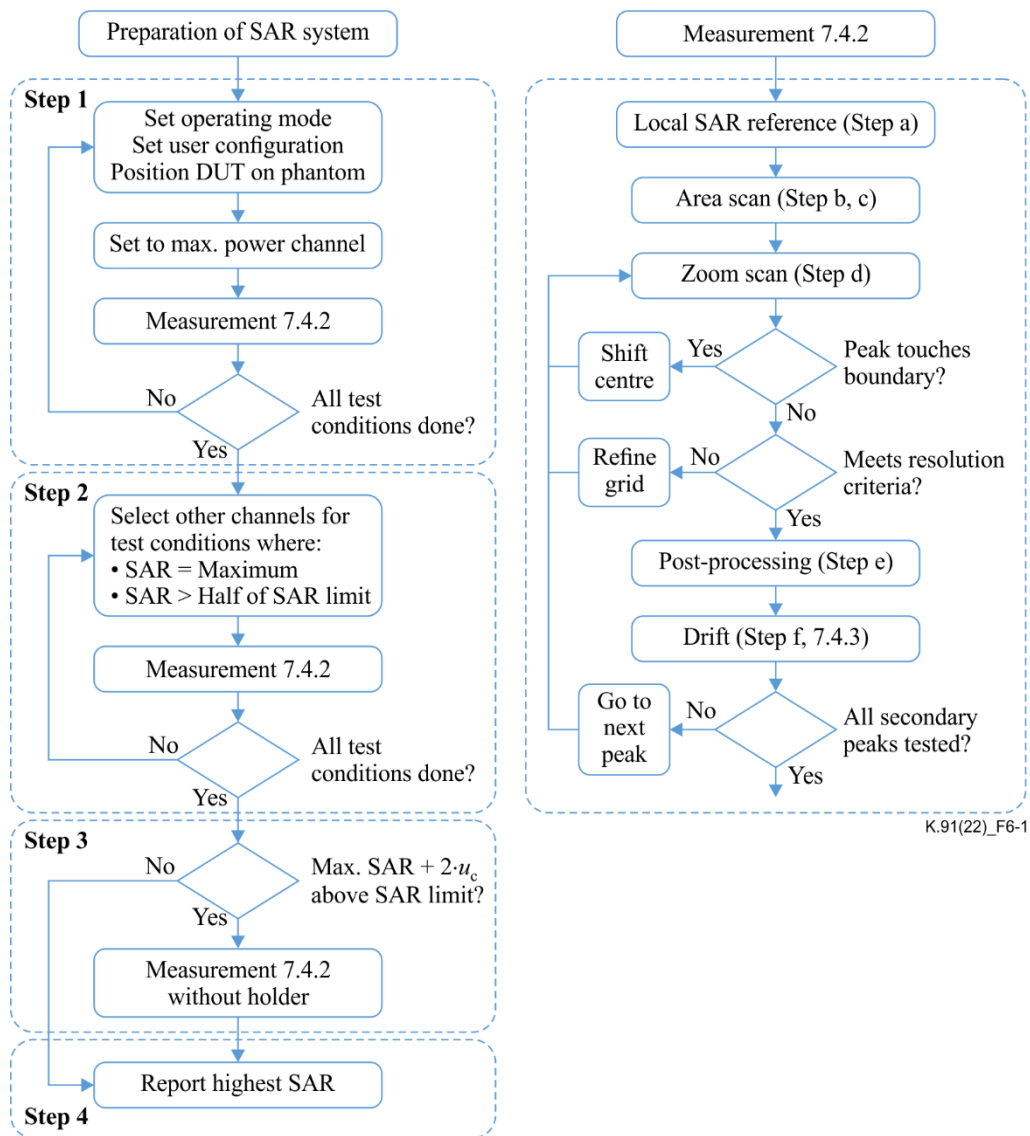


Figure 6-1 – Block diagram of the tests to be performed for the assessment of SAR of a wireless communication device*

* Reproduced with permission from IEC/IEEE 62209-1528 ed 1.0 "Copyright © 2020 IEC Geneva, Switzerland. www.iec.ch".

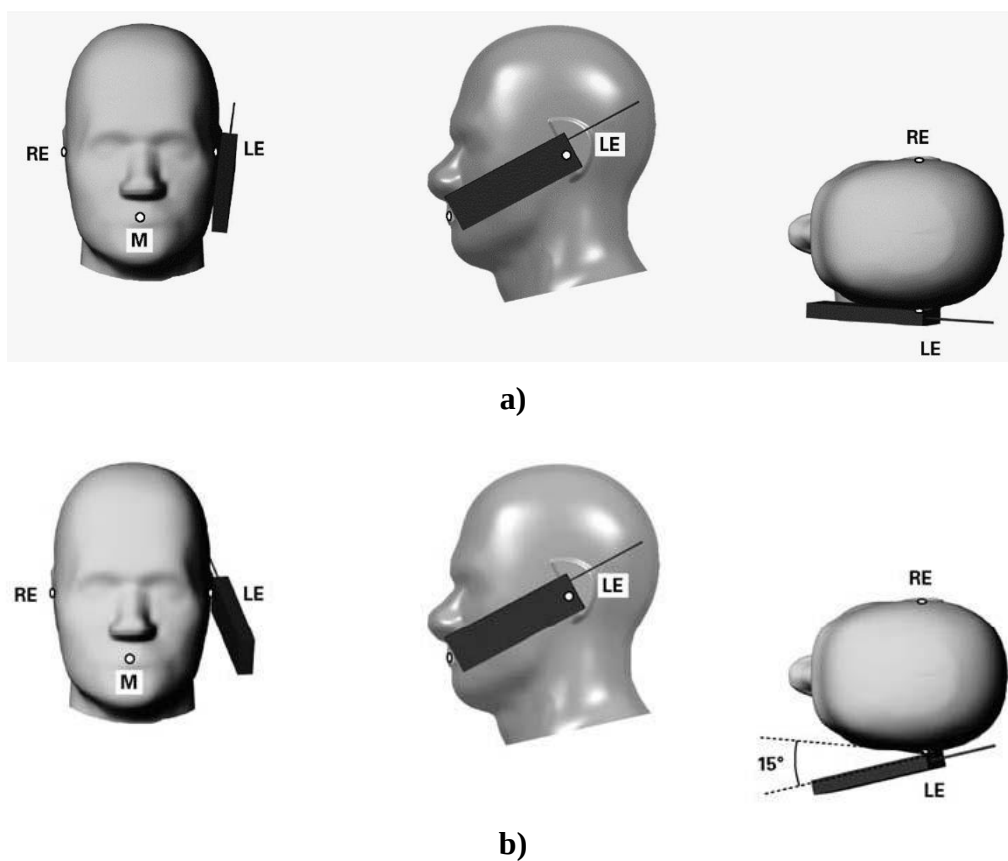


Figure 6-2 – a) Cheek position and b) tilt position of the wireless communication device used next to the ear on the left side of SAM phantom



Figure 6-3 – A SAR measurement system where phantom, robot with calibrated probe, and a post-processing unit is shown

All mobile handsets on the market have to comply with regulations concerning the limits of human exposure to EMFs. The safety factors are taken into account for the worst-case scenarios and required margins for human exposure limits.

6.5.2 SAR compliance of personal wireless communication devices used in close proximity to the human body

This clause considers compliance assessment of RF EMF transmitting devices intended to be used with the radiating part of the device in close proximity to the human body (excluding holding the device to the ear in the 4 MHz to 10 GHz range). The user of this Recommendation shall refer to the latest version of [IEC/IEEE 62209-1528] in order to assess SAR compliance with exposure limits. The compliance assessment shall be achieved by applying measurement requirements and measurement procedures as described in [IEC/IEEE 62209-1528]. The objective of [IEC/IEEE 62209-1528] is to specify the measurement method for demonstration of compliance with the SAR for the previously mentioned device types. [IEC/IEEE 62209-1528] is applicable to wireless communication devices capable of transmitting RF EMF intended to be used at a position near the human body with the radiating part(s) of the device at distances up to and including 200 mm from a human body, i.e., when held in the hand or in front of the face, mounted on the body, combined with other transmitting or non-transmitting devices or accessories (e.g., belt-clip, camera or Bluetooth add-on), or embedded in garments. [IEC/IEEE 62209-1528] may also be used to measure simultaneous exposures from multiple radio sources (within the same product/device) used in close proximity to the human body. Test distance and positioning procedures are provided for the following general categories of device types: desktop device; front-of-face device; limb-worn device; clothing-integrated device; large device relative to phantom surface area; devices used with body-worn accessories; devices used next to or against the body without an accessory; devices with hinged or swivel antennas; hand-held device, not at the head or torso; generic device. The types of devices considered include, but are not limited to, mobile telephones, cordless microphones, and radio transmitters in personal, desktop and laptop computers, for multiband operations using single or multiple antennas, including push-to-talk devices. [IEC/IEEE 62209-1528] gives guidelines for a reproducible and conservative measurement methodology for determining the compliance of wireless devices with the SAR limits. [IEC/IEEE 62209-1528] also does not apply for exposures to transmitting or non-transmitting implanted medical devices. [IEC/IEEE 62209-1528] does not apply for exposure from devices at distances greater than 200 mm away from the human body.

The main difference between the two parts of IEC 62209 is the phantom shape for body SAR and head SAR assessment, which is a flat phantom (as shown in Figure 6-4) rather than the SAM phantom. Most of the text in the two parts, such as tissue simulating liquid and uncertainty evaluation, is essentially the same.

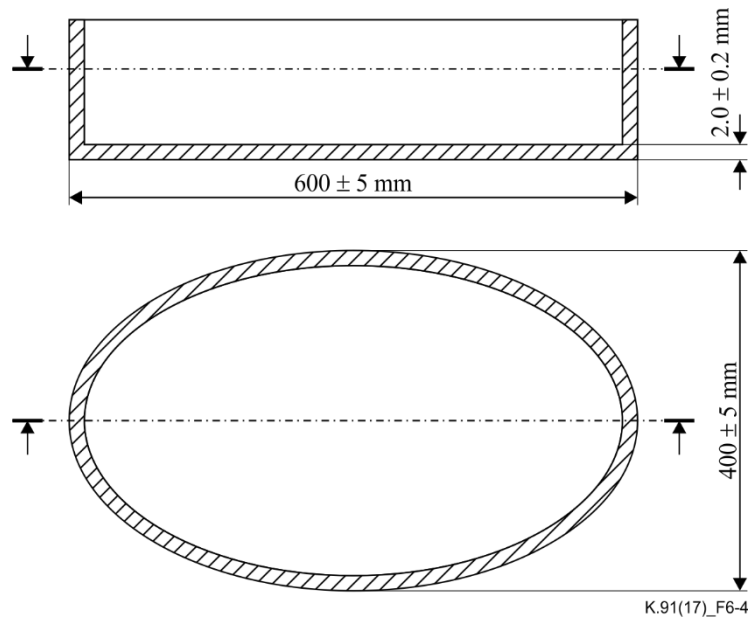


Figure 6-4 – Flat phantom for SAR measurement of wireless communication devices used in close proximity of the torso

6.5.3 Compliance of the wireless communication device with hands-free kit

All wireless communication devices with or without hands-free accessories must comply with exposure limits in all modes of operation.

NOTE – [b-WHO 2014] indicates that a person keeping the mobile phone 30 to 40 cm away from the body will have a much lower exposure to RF fields than someone holding the handset against their head. This can be achieved by using the phone in loudspeaker mode or by using a hands free device such as wireless [b-IEEE 802.15.1] or wired (as separate accessory or built-in) headsets.

6.5.4 Power density compliance of personal wireless communication devices

For portable wireless devices operating at frequencies between 6 GHz and 300 GHz, the basic restriction is specified in terms of power density. The assessment of peak spatial-average power density is specified in [b-IEC TR 63170]. In summary, this document provides detailed information on the following:

- laboratory requirements
- field probe and measurement instrumentation
- description of the scanning system and device holder
- field reconstruction algorithms
- tests to be performed
- measurement procedure
- evaluation surfaces
- combining power density and SAR for simultaneous transmission
- uncertainty estimation
- calibration of measurement equipment
- system check and system validation

6.6 Wireless power transfer systems

Wireless power transfer (dynamic and static charging) systems are used to provide power to devices without the use of cables. Electric vehicles and mobile phones are examples of devices that are

charged using a wireless power transfer system. These systems use electric fields, magnetic fields, or electromagnetic fields to transfer power. [b-IEC PAS 63184] provides procedures to evaluate human exposure to electromagnetic fields from wireless power transfer systems.

6.7 VHF and UHF broadcasting transmitting stations

The antennas for these systems are usually built as a set of panels (up to 64) consisting of dipoles and screens and mounted on the antenna tower. For additional information, see [ITU-R BS.1195]. Consideration of real cases shows that FM high-power transmitting antennas generally contribute the most to exposure levels. In most cases, the synthetic model (see clause 7.3.1.2) is appropriate for exposure assessment by calculation. In many cases of exposure assessment by measurement, frequency-selective measurements and post-processing are required.

6.8 2G, 3G and 4G mobile base stations

The transmitting antenna of a mobile base station is typically a sector antenna in which one panel is serving a 120° sector. In many cases, the POIs are located in the far-field region and the point source model (see clause 7.3.1.1) is sufficient for exposure assessment by calculation. Broadband measurement is also sufficient in most cases, especially when close to the antennas.

6.9 Smart (adaptive) antennas

Smart antennas produce a number of narrow beams directed to individual users to reduce interference and optimize communication. Smart antennas are controlled in such a way as to optimize radiation patterns in time depending on the distribution of the users in the served area. Because the maximum exposure level caused by the smart antennas is usually low, the point source model with an isotropic radiation pattern can be considered as the most conservative approach (see Equation B.2 in [ITU-T K.70] or clause F.10 of [IEC 62232]). If more exact (less conservative) assessment is required, then the procedure described in clause F.10 of [IEC 62232] is recommended. Smart antennas are planned for wide use in 5G mobile systems as described in [b-ITU-T K-Sup.9] and [b-ITU-T K-Sup.16].

6.10 Vehicle mounted antennas (such as police car)

A source of intentional RF EMF is antennas usually located on the body of vehicles. These use a vertical dipole(s)/monopole so they have an omnidirectional horizontal radiation pattern (HRP). A VRP strongly depends on the antenna size, configuration and location. Exposure assessment should be conducted inside and outside of the vehicle. Assessment inside the car should take into account the screening effect of the body of the vehicle. The assessment method may be measurement or calculation. Methods for calculating SAR from vehicle-mounted antennas are detailed in [IEC/IEEE 62704].

7 Exposure assessment for fixed stations

In general, the exposure assessment can be done by measurement or calculation. Both methods give similar levels of accuracy and have advantages and disadvantages.

7.1 Pre-analysis

The goal of the pre-analysis is to choose the best and most convenient method for proper exposure assessment. The pre-analysis may contain:

- a collection of the data concerning radiating sources in the area under consideration;
- an evaluation of the field regions for each radiating source.

Based on the national, accessible data and resources, the pre-analysis may include:

- a decision whether measurements or calculations will be used;

- an estimation of the expected field levels (by calculation and using the most simple formulas);
- an evaluation of the directions and points with the expected highest levels of exposure.

Detailed pre-analysis makes exposure assessment easier and more accurate. It also saves time and costs.

7.2 Measurements

It is recommended that first the simplest method be used (i.e., broadband RF EMF measurement). If the measured exposure level is not in compliance with the reference level, then the frequency-selective RF EMF measurement should be used to get more accurate results. If there is still no compliance with the exposure limit, then the most sophisticated method based on the SAR measurement against basic restrictions may be used.

According to ICNIRP and IEEE guidance, the measuring equipment should respond to, and indicate the root mean square (r.m.s.) value; however, national regulations may request other values.

When using a spectrum analyser to measure the RF field power density, the resolution bandwidth (RBW) has quite a large effect on the measurement results, particularly for broadband signals such as wideband code division multiple access (WCDMA) and orthogonal frequency-division multiplexing (OFDM). Therefore, it should be carefully selected to sufficiently cover the occupied bandwidth of the target radio source for an accurate field power density measurement (see Appendix VII).

In the reactive near-field region, the measurements are complex. Measurement of both the electric and magnetic components of the EMF is required. Also, all three orthogonal components should be measured. However, this region is very close to the transmitting antenna and such measurements are only required in special cases. In practice, these cases are for occupational exposure only.

In the far-field region, and in part of the radiating near-field region, the measurements are not so complex. Measurements only of the orthogonal electric field components of the EMF are sufficient in this region.

Both broadband and narrowband measurements are possible in this case.

For additional details, see [IEC 62232], [ITU-T K.61], [ITU-R BS.1698], [b-ITU-R SpecMon] and [b-ITU-R SM.2452].

7.2.1 Selection of the points of investigation

When evaluating the RF EMF exposure in the neighbourhood of any radiocommunication station, the points where the measurements will be performed must be carefully selected. Ideally, the effective EIRP, the antenna height, the azimuth, the downtilt and the radiation pattern should be known to cautiously select the POIs. However, mainly for broadcast antennas, knowledge of the exact radiation pattern is not easily available. These antennas may be designed for specific scenarios and may contain sets of elements (e.g., panels) with individually designed feeding arrangements.

If the available data are not sufficient, the POI may be selected by careful visual inspection and some calculations taking notes of nearby buildings, and considering their heights and distances to the source.

A single dipole antenna approach (see Equation 7-1) can be considered as a worst-case scenario for almost all situations. This means that if there is not enough information about the antenna radiation pattern or no software tools (or enough time) to perform the estimation of the RF EMF exposure in the region of interest, then it is possible to consider a worst-case scenario. If the RF EMF exposure is compliant in this scenario, it will be in the real conditions as well. The power density, taking into account the reflection coefficient equal to 1 ($\rho = 1$), is given by Equation 7-2.

$$F(\theta, \phi) \approx \cos^2 \theta \quad (7-1)$$

$$S(R, \theta, \phi) = (1 + \rho)^2 \frac{\text{EIRP}}{4\pi R^2} F(\theta, \phi) = \frac{\text{EIRP}}{\pi R^2} \cos^2 \theta \quad (7-2)$$

The envelope curves for the frequency ranges of 10-400 MHz and 400-2000 MHz can be traced using Equations 7-3 and 7-4, respectively. These equations are valid for the following conditions: general public exposure equals the frequency range limit ($S = 2 \text{ W/m}^2$ for 10-400 MHz and $S = f/200 \text{ W/m}^2$ for 400-2000 MHz) and the reflection coefficient equals 1 ($\rho = 1$).

$$h = H - \sqrt{-x^2 + x \sqrt{\frac{\text{EIRP}}{2\pi}}}, \text{ for } 0 \leq x \leq \sqrt{\frac{\text{EIRP}}{2\pi}} \quad (7-3)$$

$$h = H - \sqrt{-x^2 + x \sqrt{\frac{200 \times \text{EIRP}}{f \times \pi}}}, \text{ for } 0 \leq x \leq \sqrt{\frac{200 \times \text{EIRP}}{f \times \pi}} \quad (7-4)$$

where:

h is the envelope curve height, in metres

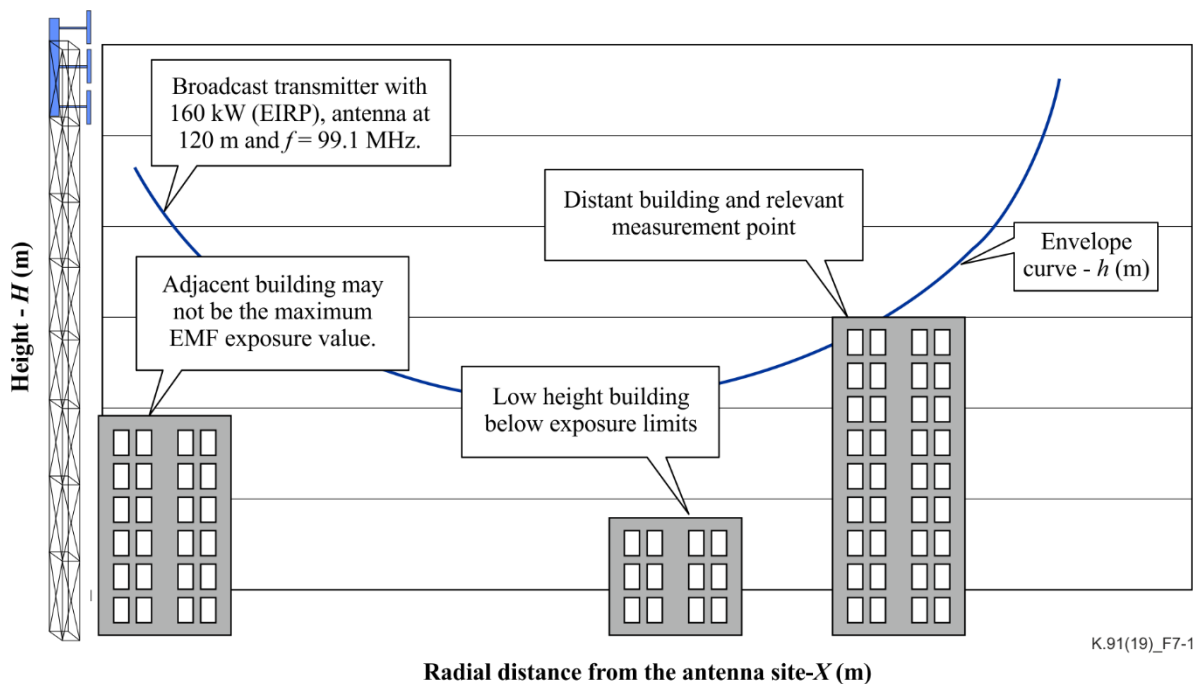
H is the antenna height, in metres

x is the distance from the base of the antenna to the base of the building, in metres

f is the frequency, in megahertz

EIRP is the equivalent isotropically radiated power in the direction of the largest antenna gain, in watts.

In Figure 7-1 a typical scenario of an FM broadcasting station is shown, where technician staff would choose an adjacent building to measure the RF EMF exposure. However, the top of a distant building may be exposed to higher E -field (or H -field) strengths and should be chosen for the RF EMF exposure assessment. It can be guaranteed that the real E -field on the top of the two closer buildings is compliant. The same is not guaranteed for the distant building, as long as the real radiating pattern is unknown.



NOTE – The distant building is a place with potentially higher exposure levels.

Figure 7-1 – A big city with an FM broadcasting station located downtown

7.2.2 Spatial averaging

Multi-path reflections can create non-uniform field distributions. Therefore, in some regulations to assess the whole-body human exposure, an averaging process is required. The field values are then determined at N points (3, 6, 9, 20 or other) as described in [IEC 62232] and [ITU-T K.61]. Three points, at heights 1.1 m, 1.5 m and 1.7 m above the terrain level are basically recommended, but if higher accuracy is required, the number can be increased in accordance with national or regional standards and regulations (see [ITU-T K.61]).

7.2.3 Time averaging for measurement procedure

Signals used in radiocommunication exhibit a variation with time due to the traffic or modulation (mobile communication systems) or due to the picture content (analogue TV systems) or non-continuous nature of the signal (radar systems) and for many other reasons (e.g., maintenance time). On the other hand, some signals have very small time variation (e.g., FM signal).

[b-ICNIRP 1998] specifies the use of a 6 min averaging time for the reference levels for frequencies between 100 kHz and 10 GHz, or $68/f^{1.05}$ min for the basic restrictions for frequencies above 10 GHz. [b-ICNIRP 2020] specifies for frequencies from 100 kHz to 300 GHz the use of a 30 min averaging time for whole body exposures and 6 min for local exposure. The ICNIRP averaging times are based on the thermal time constants of heating phenomenon of body tissues, not the time variation of the radiocommunication signals.

In the case of frequency-selective measurement, when spatial averaging is applied and there are many radiating sources, measurement procedures at each POI may take a long time due to the 6 min averaging time.

A comparison of the measurements for a 6 min and a 1 min averaging time shows that the difference in results is on a level lower than 1 dB. Figure 7-2a shows the results of the RF EMF measurement for seven mobile base stations and for the averaging time of 10, 20, 30, 40, 50, 60, 120, 180 and 360 s. In Figure 7-2b a standard deviation (SD) is shown for these measurements. It can be seen that the difference between results of measurement using different averaging times is less than 0.4 dB and is small in comparison with the day-long variation (about 3 dB) and week-long variation (up to 10 dB).

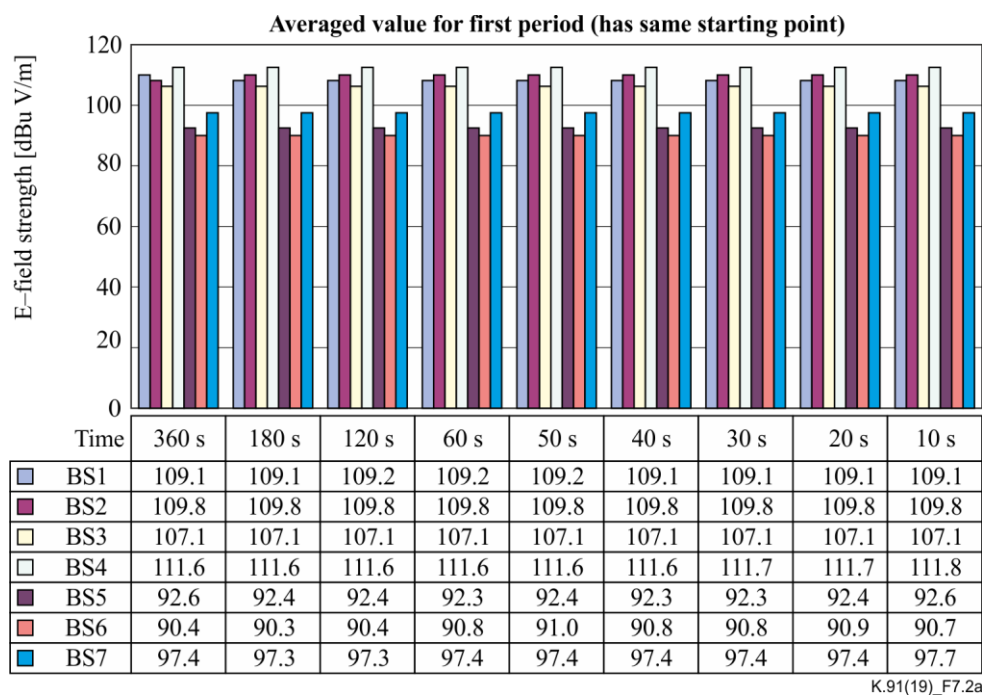


Figure 7-2a – Averaged value

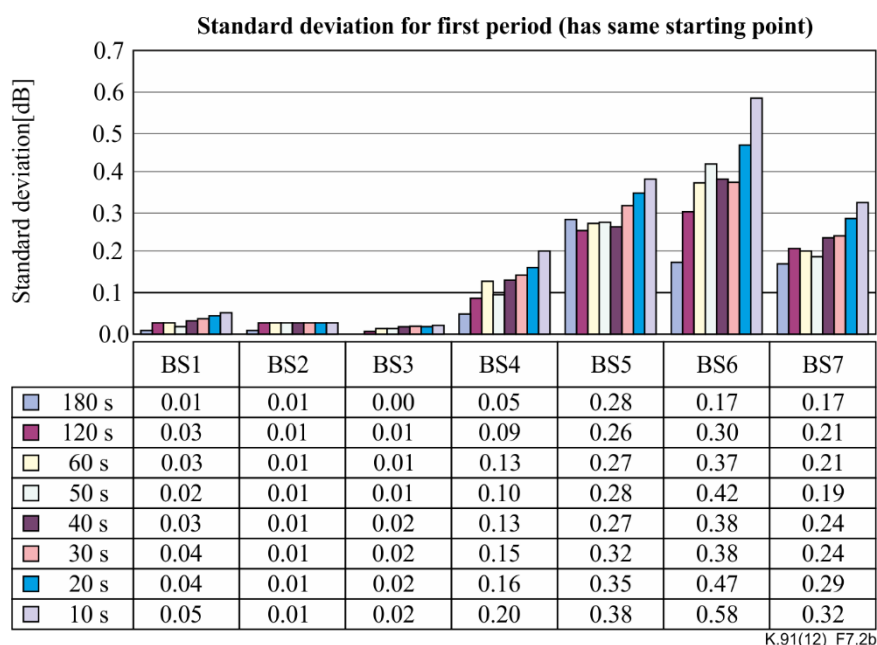


Figure 7-2b – Standard deviation

For RF EMF measurements of typical radiocommunication systems, such as code division multiple access (CDMA), it looks permissible to make measurements with a 1 min averaging time if the total exposure is far (3 dB) below the exposure limit. In addition, a shorter averaging time is also available if a larger uncertainty is allowed or if a conservatively large margin to the exposure limit due to the averaging is introduced. In such a case, sufficient rationale for the selection of the value should be provided. If the total exposure is closer to the exposure limit, then the 6 min averaging time should be used. This simplification is not valid for signals of a non-continuous nature (radar systems).

[IEC 62232] recommends time averaging according to the relevant exposure limits at the time of day that would ensure the highest average RF field strength conditions (i.e., under maximum traffic conditions). Long-term measurement (over a day or a week if possible) may be useful to determine when the highest average RF field strength occurs, if this information is unknown from appropriate time averaging, to determine compliance with the relevant exposure limits.

The number of sampling data that is taken in the averaging procedure is also important. It has as much effect on the averaged results as averaging time. This effect is described in Appendix II.

7.2.4 Broadband measurement

Broadband measurement of the RF EMF in the far-field region is the simplest measurement method for compliance assessment. In this method, the sum of the RF EMF contributions (electric or magnetic field depending on probe type) from all radio sources within a wide frequency range is measured. Problems may occur if radio sources operating outside the frequency range of the probe are present. On the market, there are electric field probes available that cover these frequency bands, e.g., from 100 kHz to 6 GHz or 3 MHz to 18 GHz. These frequency bands cover the operating frequencies of almost all most widely used radiocommunication services. Outside the band of the first probe are many fixed point-to-point (FPP) systems (e.g., 11 GHz, 13 GHz, 23 GHz). However, FPP systems radiate very narrow beams, thus they can usually be neglected in the exposure assessment as systems giving very small contributions in the areas accessible to people. Outside the band of the second probe are the AM broadcasting systems (LW, MW and SW) and some FPP systems (e.g., 23 GHz band). Other broadband field probes cover wireless applications, including mmWave 5G RF bands, up to 40, 60 and 90 GHz.

For broadband measurement, the exposure ratio corresponding to the most restrictive reference level at the considered frequency range has to be taken, unless the measurement device incorporates a

weighting function consistent with the exposure limits. This means that the broadband measurement usually gives conservative results.

Broadband measurements are simpler and less time consuming than those that are frequency selective, but tend to be less sensitive for measurement of low-level exposures. Even though they are less accurate, however, they can be applied in many cases. If the exposure level is not compliant with the limit, then frequency-selective measurements are required.

For additional information, see [IEC 62232].

7.2.5 Frequency-selective measurement

Frequency-selective measurement can provide information on the field strength and the spectral characteristics of the RF EMFs measured. These measurements allow for the use of different exposure limits in different frequency ranges, but require post-processing. These measurements give information about frequencies of measured sources and their individual contributions to the total exposure.

The results of exposure assessment using frequency-selective measurements are more accurate in comparison with the results of the broadband measurements. In Figure 7-3, an example of the results of frequency-selective measurement is presented. Table 7-1 presents another example of the numerical results of frequency-selective measurements (the combined electric field strength is presented for each frequency band).

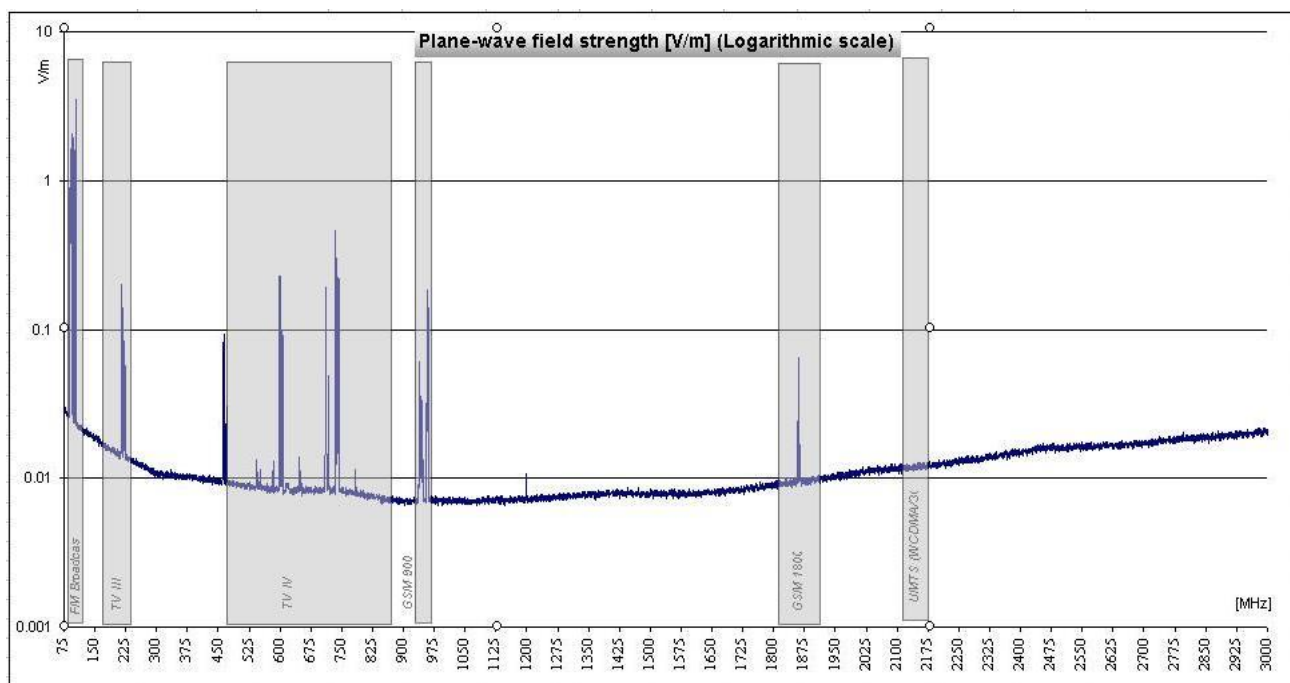


Figure 7-3 – Example of the results of frequency-selective measurement

Table 7-1 – Example of the numerical results of frequency-selective measurement

Measurement point	Band	Frequency (MHz)	Reference level (V/m)	Height (m)	Measured value (V/m)	Exposure ratio
1-2	FM broadcast	87.5-108	28	1.1	0.550 6	0.000 386 7
				1.5	0.667 3	0.000 568 0
				1.7	0.833 0	0.000 885 1
				Average		0.000 613 2
	VHF TV	174-230	28	1.1	0.072 7	0.000 006 7
				1.5	0.071 2	0.000 006 5
				1.7	0.070 0	0.000 006 3
				Average		0.000 006 5
	UHF TV	470-862	30	1.1	0.159 8	0.000 028 4
				1.5	0.160 3	0.000 028 6
				1.7	0.162 1	0.000 029 2
				Average		0.000 028 7
	GSM 900	925-960	42	1.1	0.155 4	0.000 013 7
				1.5	0.143 6	0.000 011 7
				1.7	0.185 0	0.000 019 4
				Average		0.000 014 9
	GSM 1800	1805-1910	58	1.1	0.062 8	0.000 001 2
				1.5	0.078 3	0.000 001 8
				1.7	0.110 7	0.000 003 6
				Average		0.000 002 2
	Universal mobile tele-communication system (UMTS) (WCDMA/3G)	2110-2170	61	1.1	0.101 8	0.000 002 8
				1.5	0.105 9	0.000 003 0
				1.7	0.083 9	0.000 001 9
				Average		0.000 002 6
						Sum

The results in Figure 7-3 and in Table 7-1 correspond to two different POIs. For comparison with the exposure limits, post-processing is required in both cases. This is done in Table 7-1 and the results show that the total exposure ratio of this POI is compliant. The reference levels are from [b-ICNIRP 1998].

For additional information, see [IEC 62232].

7.2.6 Near-field measurement

In the near-field region, the electric (E) and magnetic (H) field components are not proportionally related to each other and depend on the RF EMF source properties. Therefore, both the electric and magnetic field components shall be measured. The total exposure for the combined E -field component and the combined H -field component shall be compared with exposure limits independently.

7.2.7 Post-processing

Post-processing is not required if only one RF EMF source is measured. If many RF EMF sources are in operation, then post-processing consisting in the evaluation of the total exposure is usually required.

If a broadband measurement is performed and the band of the probe covers the whole range of operating frequencies, then post-processing is not required. If a broadband measurement needs more than one probe to cover all operating emissions, then post-processing is also required.

Frequency-selective measurement usually requires post-processing to evaluate the total exposure, taking into account the measured values and the respective exposure limits for each operating frequency. Some measurement instruments allow for the evaluation of the combined E using the most restrictive exposure limit in the considered frequency band. In this case, Equation 7-5 is used:

$$E = \sqrt{\sum_{i=1}^n E_i^2} < E_{\text{lim}} \quad (7-5)$$

where E is the total electric field, due to all emissions.

In Equation 7-5 for the electric field, the r.m.s. values should be used, but in practice, the root sum square (RSS) values are measured [IEC 62311]. This is a typical approach not only for wideband, but also frequency-selective equipment. This result is then compared with the value of the strongest limit for the frequency band covering all emissions. This is very convenient, but the evaluation is conservative. If such results show no compliance, then appropriate frequency-selective measurements should be done together with post-processing to reassess compliance.

7.2.8 SAR measurement

If compliance with reference levels cannot be demonstrated, the basic restrictions may instead be used for compliance verification. For mobile communication frequency bands, SAR is usually the exposure metric to consider. Compliance assessments based on SAR are in general required for RF sources used close to the body (see [IEC 62232] and [IEC/IEEE 62209-1528]). SAR measurements are mainly conducted by specialized laboratories, since the measurements are complex and require special equipment. SAR measurement methodologies make use of phantoms simulating the human head or body that have been developed to provide conservative results with respect to exposure in real humans.

7.3 Calculations

Depending on the distance to the radiating source, different methods of calculation should be used to ensure sufficient accuracy of results. In general, if the relative distance (in relation to a wavelength) is shorter, then more sophisticated and technically difficult methods have to be applied. Calculations are relatively simple in the far-field region.

For additional details, see clauses 7.3.1 to 7.3.3, as well as [IEC 62232], [ITU-T K.52], [ITU-T K.70], [ITU-R BS.1698] [b-ITU-R SpecMon] and [b-ITU-R SM.2452].

7.3.1 Methods of calculation in different field regions

There are many methods of calculation with varying degrees of accuracy and complexity. The use of the simplest appropriate method for the exposure assessment is advised. The choice is highly dependent on the field region in which POIs are located in relation to the radiating source. Clauses 7.3.1.1 to 7.3.1.5 present the most frequently used methods, starting from the simplest.

7.3.1.1 Point-source model

The point-source model is a simple, but very effective, model that may be used to calculate exposure levels to be used in comparison with the reference levels (see [IEC 62232], [ITU-T K.52],

[ITU-T K.61] and [ITU-T K.70]). It is assumed that the transmitting antenna is represented by a single point source, situated in the antenna electric centre and having a radiation pattern of the transmitting antenna considered. In this case, the equivalent plane-wave power density S_{eq} at the distance R to the antenna is:

$$S_{eq} = \frac{P \cdot G_i}{4\pi R^2} F(\theta, \phi) \quad (7-6)$$

where:

S_{eq} is the equivalent plane-wave power density (W/m²) in a given direction

R is the distance (m) from the radiation source

P is the average power (W) emitted when the transmitter operates at maximum emission settings (all channels transmitting at their respective maximum power setting) and supplied to the radiation source (transmitting antenna)

G_i is the maximum gain of the transmitting antenna, relative to an isotropic radiator

$F(\theta, \phi)$ is the antenna numeric gain (normalized gain), in which θ is the elevation angle, ϕ is the azimuth angle.

The accuracy of this model depends on the field region and on the antenna gain and is fully acceptable in the far-field region. The boundaries of the field regions are defined according to slightly different criteria that can be found in [IEC 62232], [ITU-R BS.1698], [ITU-T K.52] and [ITU-T K.61]. This model is implemented in the software EMF-estimator (Appendix I of [ITU-T K.70]) and in the "EMF exposure" mobile app (see Appendix VIII).

The simplest version of this method is the use of the isotropic radiation pattern with maximum ERP. In this case, the formula for the equivalent plane-wave power density S_{eq} at the distance R to the antenna has simplest form:

$$S_{eq} = \frac{P \cdot G_i}{4\pi R^2} \quad (7-7)$$

Equation 7-7 is very useful for a first approximation of the expected field level, especially during pre-analysis. In actual cases, Equation 7-7 can give a very high estimate. However, if calculations based on Equation 7-7 show compliance, then no further assessment is needed.

7.3.1.2 Synthetic model

In the synthetic model ([IEC 62232] or [b-EN 50413]), each antenna is considered to be a set of elementary sources that have identical parameters. Such an approach is natural, for example, for broadcast antennas that usually contain sets (up to 64) of identical panels operating as one transmitting antenna (see [ITU-R BS.1195]). In the case of global system for mobile communications (GSM) panels (or similar antennas), they can be divided into "patches" (usually containing one dipole with a screen) and each of them considered as a separate radiating source. This model can be employed for distances beyond the near-field distance, which is calculated with respect to the maximum size of an elementary radiating source (patch in GSM, panel in broadcasting) and not the size of the whole system. In this case, the electric field strength E in the POI is as follows:

$$E = \sum_n \frac{\sqrt{30P_n G_n}}{r_n} \exp \left[j \left(\gamma_n + \frac{2\pi r_n}{\lambda} \right) \right] \quad (7-8)$$

where:

P_n is the average power (W) delivered to the input of the n th radiating panel or patch

r_n is the distance between POI and the centre of the n th radiating panel or patch

γ_n is the relative phase of voltage delivered to the n th radiating panel or patch

G_n is the gain of the n -th radiating panel or patch towards the POI relative to an isotropic antenna

λ is the wavelength.

The synthetic model leads to very accurate results, but the accuracy is lower than in full-wave modelling, because the coupling between radiating sources is neglected. In many cases (e.g., for all broadcasting antennas or mobile BS panels), this assumption is well fulfilled.

A disadvantage of this model is that exact information concerning the feeding arrangement (P_n and γ_n) of the antenna system containing many radiating sources is required.

7.3.1.3 Full-wave methods

The highest accuracy of the calculation of reference levels is achieved by numerical modelling using one of the full-wave methods based on solving Maxwell's equations in the frequency or time domain (see [IEC 62232], [ITU-R BS.1698] and [ITU-T K.61]). It includes the MoM and FDTD. Such methods of calculation may be used for any source region. They use detailed-segmented models of systems where the more detailed-segmented model leads to higher accuracy of the evaluated field distribution.

A typical transmitting station or BS consists of many transmitting antennas. The area required for consideration is usually big and inhomogeneous. In such a case, the MoM is more convenient than other methods.

The application of numerical modelling, as described in the first paragraph, requires appropriate software, experience in electromagnetics, and very detailed input data. There are many commercial packages available. In some cases, there are limitations concerning the number of unknowns (the number of segments) that may be used in calculations. The accuracy of the results of calculations strongly depends on the exactness and the range of accessible data concerning a transmitting antenna, which includes antenna geometry and its feeding arrangement. Many antennas (e.g., broadcasting antenna systems, cellular panels) contain a huge number of active radiating elements (up to 256) that are fed with different amplitudes and phases. Without such information, the calculation is impossible or may be simplified for a general assessment only. In many cases (e.g., for GSM panels, which also consist of many dipoles), it is very difficult to obtain the required data; it is the manufacturer's proprietary information.

Procedures such as the MoM also allow all other objects that have an influence on the radiation level to be taken into account as secondary sources causing reflections (e.g., facings made of metallic elements, antenna towers and antenna supporting structures, ground).

In general, full-wave methods provide a good opportunity to take into account almost all substantial factors influencing radiation, but they are only useful in rather simple cases or for the reactive near-field region in which other methods are not sufficiently accurate. This is partly so because it is very difficult to collect all the data needed. Additionally, sophisticated software and experience in electromagnetics are required, together with significant computer resources.

7.3.1.4 SAR calculations

For cases when compliance with reference levels cannot be demonstrated, basic restrictions may instead be used for compliance verification. For mobile communication frequency bands, SAR is usually the exposure metric to consider. Compliance assessments based on SAR are in general required for RF sources used close to the body (see [IEC 62232]). Numerical models of the device under test and the head or body phantom are required for the SAR calculation. SAR calculations are complex and require experience in numerical modelling. There are several software packages available and the most frequently used method is FDTD, but other methods may also be used.

7.3.1.5 Influence of the reflections

Calculations in many cases are done for free space conditions. A reflection from the ground or from other structures, such as buildings or fencing (especially metallic objects), may lead to an increase in

the value of the field level. In the conservative approach, reflections from the ground can be taken into account as in Appendix II of [ITU-T K.52]. In typical cases, where an observation point is located up to a few metres above the terrain level, the electric field strength is multiplied by a factor of 1.6, which means that the corresponding power density is multiplied by a factor of 2.56.

Reflections from other structures are discussed in detail in [IEC 62232], together with information about factors that can be used in many situations. It should be noted that in a complex environment – with many reflections present – the highest multiplication factor should be used. In practice, the maximum value of the multiplication factor is 2 for the electric field strength, which corresponds to 4 for the power density.

7.3.2 Modelling of the transmitting antennas

The real source of intentional RF EMF is the transmitting antenna – not the transmitter itself, because the transmitting antenna is the radiating source that determines the RF EMF distribution in the vicinity of a transmitting station. From the viewpoint of protection against exposure, the most important parameters of the transmitting antenna (or radiating source) are operating frequency, EIRP and radiation pattern, which also determine the antenna directivity and the antenna gain. The antenna radiation pattern requires modelling.

The transmitting antenna is represented by the 3D radiation pattern $f(\theta, \phi)$ (see [ITU-R BS.1195]). In numerical modelling (e.g., MoM, FDTD) the radiation pattern (or directivity) is determined by the numerical solution that depends on the geometry and feeding arrangement of the transmitting antenna. The radiation pattern $f(\theta, \phi)$ is a function of the elevation and azimuth angles. The best accuracy can be achieved if the exact 3D radiation pattern $f(\theta, \phi)$ is known and is directly used in the calculation. However, in practice the telecommunication operator knows only the two cross-sections of the radiation pattern: in a horizontal plane (called the HRP):

$$H(\phi) = f(\theta, \phi)|_{\theta=\theta_{\max}} \quad (7-9)$$

where:

$H(\phi)$ is the HRP

θ is the elevation angle

ϕ is the azimuth angle

$\theta_{\max}$ is the elevation angle at which the maximum radiation occurs.

and in a vertical plane (called VRP):

$$V(\theta) = f(\theta, \phi)|_{\phi=\phi_{\max}} \quad (7-10)$$

where:

$V(\theta)$ is the VRP

$\phi_{\max}$ is the elevation angle at which the maximum radiation occurs.

The actual values of the radiation pattern for any elevation and azimuth angles are usually approximated by the relationship (see [ITU-R BS.1195]):

$$f(\theta, \phi) = H(\phi) \cdot V(\theta) \quad (7-11)$$

There are many models of transmitting antennas with different degrees of complexity and accuracy that may be used. The most important are the following.

- Antenna with isotropic radiation pattern and with maximum EIRP. This model is the simplest and may be used to model any transmitting antenna; however, it gives most conservative results with the highest overestimation of the level of exposure.
- Antenna as a point source with HRP and VRP. This model is sufficient for the far-field region; radiation patterns may be delivered by the manufacturer or evaluated using specialist

antenna software (the MoM may be used for this purpose). If the radiation patterns are evaluated, data concerning the transmitting antenna configuration (including feeding arrangement) are required.

- Antenna as a point source with radiation pattern as a two-dimensional function $f(\theta, \phi)$. In general, the applicability is as in the previous entries; however, results of exposure assessment will be more accurate, because there is no need for interpolation of the radiation pattern for the direction outside HRP and VRP.
- Antenna model in the full-wave software (like MoM or FDTD). This model is most accurate, with applicability even in the reactive near-field region; it also allows for the SAR evaluation (FDTD, MoM/finite element method (FEM)) if compliance with basic restrictions is needed. This model requires very detailed data concerning the configuration of each metal or dielectric part of the antenna (also antenna tower if necessary) and concerning the feeding arrangement.

The radiation pattern is a function of frequency and for a wideband antenna system varies substantially across the antenna band. This means that for wideband transmitting antennas the radiation patterns have to be evaluated for each operating frequency (e.g., FM, VHF, UHF broadcasting); for other radiocommunication systems one radiation pattern is sufficient for all operating channels (mobile communication – in each band).

7.3.3 Radiated power ERP/EIRP evaluation

The ERP is a product of the power supplied to an antenna and the maximum antenna gain relative to the half-wave dipole. If the antenna gain is taken relative to the isotropic antenna then EIRP is evaluated. The following relations hold between ERP and EIRP:

$$\text{EIRP [dBd]} = \text{ERP [dBi]} + 2.15 \quad \text{dB} \quad (7-12)$$

$$\text{EIRP [W/W]} = 1.64 \text{ ERP [W/W]} \quad (7-13)$$

where 2.15 dB is the gain of the half-wave dipole relative to the isotropic antenna.

The EIRP has to include not only the directivity of the antenna and its dissipative losses, but also the attenuation of the feeding system. The main contribution to the attenuation of the feeding system is due to the feeder (usually coaxial cable), but also to the connectors and in some cases to the combiners and additional connecting lines. A combiner is used to combine many transmitters into one feeder and in consequence to one antenna system. This is a typical situation in broadcasting where, for example, many FM transmitters feed one antenna system. Additional losses may also be present in mobile communication when combining many channels into one transmitting antenna panel or by a diplexer giving separation of the transmitting and receiving frequency bands. Thus, the EIRP should be calculated using the formula:

$$\text{EIRP [dBm]} = P [\text{dBm}] + G_i [\text{dBi}] - L \quad [\text{dB}] \quad (7-14)$$

where:

P is the average transmitter power in dBm (relative to 1 mW)

G_i is the antenna gain (including internal losses) relative to an isotropic antenna

L represents the transmission losses in dB (including feeder losses and additional losses).

For example, if the GSM900 antenna (with antenna gain $G_i = 17$ dBi) is fed by 30 m of the foam coaxial cable of diameter 7/8" [22.2 mm] (attenuation about 6.9 dB/100 m on 925 MHz) then the feeder losses equal 2.07 dB. Additional losses for the connectors may be assumed to be 0.3 dB, so the total losses equal 2.37 dB. If the GSM transmitter has 30 W output power then the EIRP = 871.2 W. If the losses are neglected, then EIRP = 1 503.6 W. It can be seen that if losses are neglected, then there is substantial overestimation of the EIRP and in consequence the level of

exposure. Typical levels of the losses are in the range from 1 to 3 dB, but in some cases may reach 6 dB or more.

In the case of human exposure evaluation, the average transmitter power P should be used. For many services (e.g., all digital, FM), the average power is equal to the nominal (rated) transmitter power. However, there are some services, like analogue TV or LW, MW and SW transmitters, in which the average power is different, when a conversion factor should be used.

For additional information, see [ITU-T K.70] and [ITU-R BS.1698].

7.4 Comparison between measurement and calculation

Measurements require measuring equipment and experienced staff. Measurements referring to basic restrictions are done on phantoms of the human body (especially head) and are very complex and expensive. They are usually performed by specialized laboratories. Measurements of the reference levels are much easier and cheaper, and may be wideband or frequency selective. More accurate are the frequency-selective measurements, but measuring equipment is more expensive and the measurements are more time consuming.

The main advantages of measurements as the method for exposure assessment are:

- it takes into account all the radiating sources present in a given region, with their real parameters;
- it takes into account the actual environment (e.g., reflections, antenna supporting hardware, obstacles);
- it takes into account of simultaneous exposure in a realistic way (the phase differences of different waves);
- it can be done with little knowledge about radiating sources (only a preliminary identification of the occupied spectrum is required);
- good quality measurement equipment is available on the market;
- a live demonstration of the measurement to interested people is possible.

The main disadvantages are:

- measurement is not possible when a radiating source (new or additional) is at the design stage, unless a temporary transmitter can be installed;
- in many cases (e.g., mobile communication), the exposure level depends on the time of measurement (peak or off peak-hours), so the measurement may not reflect the maximum exposure and the results can be underestimated;
- the influence of personnel and equipment on the local distribution of the EMF will increase uncertainty and should be minimized;
- it is difficult to confirm and check that all the sources in a given environment are operating at maximum EIRP;
- in a multiple sources environment, post-processing is required because of different limits for different operating frequencies;
- there is no probe covering the whole required frequency range, so usually a set of probes is needed for measurements;
- out-of-band emissions may have an influence on the results of measurements in the considered frequency band (this should be eliminated during the post-processing).

Calculation requires specialized software, experienced staff and data concerning the transmitting systems (antennas). There are many commercial packages for RF EMF evaluation, based on various models offering different level of accuracy, and requiring different experience and input data. Generally, the more sophisticated the software, the bigger are the requirements concerning input data

(parameters of the transmitting antennas) and qualifications of staff, but the results of the evaluation are more accurate.

The calculations of quantities used for basic restrictions (e.g., SAR) or RF EMF exposure levels (to compare with the reference levels) are possible. The calculations of SAR are more complex and time consuming and require the numerical model of the human body.

The advantages of calculations as the method for exposure assessment are:

- the method allows planned radiating sources (not yet in operation) to be considered;
- it allows maximum possible exposure conditions (EIRPs) to be applied that lead to the evaluation of the maximum possible exposure level;
- a choice among many calculation methods of varying levels of complexity and accuracy;
- calculations with high levels of overestimation are easy to perform and are sufficient in cases where exposure levels are far below the limits;
- costs of calculation are lower than those of measurement;
- the simulation of mitigation techniques, if required – [ITU-T K.70] provides an analysis of the influence of the variation of transmitting equipment parameters on exposure levels;
- calculations for areas with no access (e.g., safety reasons, no right of access) are possible;
- the data on a much denser grid (than in measurement) can be obtained and a visualization of the results can be easily prepared for the whole considered area (e.g., horizontal or vertical cross-sections).

The disadvantages of calculation are:

- for accurate evaluation, a detailed description of the radiating antennas (e.g., geometry, feeding arrangement, data concerning panels) is necessary;
- the effect of reflections from various objects is difficult to model;
- the method requires knowledge of the software and experience in using it (understanding of the background theory is also recommended);
- it requires at least basic knowledge of the transmitting antennas theory.

There are many methods of measurement and calculation, of varying levels of complexity and accuracy. The uncertainty of calculation and measurement depends on the method used. In general, the best approach is to start with the simplest method (of measurement or calculation) with the least accuracy and, preferably, high overestimation. If exposure levels appear to be close to exposure limits, then a more sophisticated (complex) method that offers higher accuracy, but that also needs more precise input data, better tools and more experienced staff, should be used.

[IEC 62232] provides guidance on verification of the software and considers uncertainties in some detail.

7.5 Monitoring RF EMF levels

As an effort to increase information about RF EMF exposure levels, a long-term RF EMF measurement system may be a good solution. Such a system monitors the exposure levels of the general public in specific locations. The results of continuous measurement of the exposure level in the whole spectrum used in radiocommunication are registered by the automatic system in order to show exposure as a function of time. The results of measurements may be presented for the general public by a user-friendly tool accessible via the Internet (see [ITU-T K.113]). Such a system is more transparent and may be reassuring to citizens worried whether RF EMF levels are under control.

For additional information, see [ITU-T K.83], [b-ITU-T EMF LA], [b-ITU-R SM.2452] and in [b-ITU-R SpecMon].

8 Conclusions following the exposure assessment

If all areas accessible to people are compliant with limits, then the exposure assessment can be finished, and no further action is required.

If there are areas accessible to people that are not compliant with the limits, then immediate action is required, and two solutions are possible:

- 1) areas not compliant with the limits should be properly marked or barricaded to prevent the presence of the general public or to limit the time workers are present [ITU-T K.145];
- 2) some mitigation techniques should be applied to decrease the exposure levels and to achieve compliance with the limits in all the areas accessible to people.

The simplest method of achieving compliance with the limits is to decrease a transmitter (transmitters) output power (it is easy to apply, but it decreases the coverage of the radiocommunication service) or adjust the location of the antennas (e.g., BS antennas can be mounted above an accessible rooftop). Many other methods of mitigation are possible that do not lead to degradation of the service quality.

For additional information, see [ITU-T K.70] or in [b-IEEE C95.7].

9 Final report

Requirements concerning the form of final reports may vary in different countries; however, some of them are common. The content of the final report should at least contain:

- description of the transmitting station (or area) that is the subject of the report;
- description of the RF EMF sources in the area under consideration;
- description of the methods of exposure assessment that are used;
- information concerning measurement equipment or calculation tools;
- information about factors that may influence results of measurements or calculations (e.g., humidity, type of terrain, possible reflecting objects);
- results of the exposure assessment;
- the final conclusions – the most important is the statement about whether the area considered accessible to people is compliant with regulations;
- information concerning authors of the report (affiliation, names).

It is advisable to use tables and charts (especially for the presentation of results of exposure assessment) in order to make information as clear as possible, even for inexperienced readers.

For additional information, see [IEC 62232].

10 Field levels around typical transmitting antennas

In Appendix III, some examples of RF EMF levels measured in areas accessible to the general public are presented. Note that in general the exposure levels are much lower than exposure limits. This is because it is most important for the telecom operator to achieve distribution of the RF EMF in the coverage area that is as uniform as possible. This means that the radiation patterns of the antenna systems are designed in such a way that the highest radiation levels are pointed into longer distances and the lower levels into shorter distances. In consequence, the highest part of the energy is radiated above the terrain close to the antenna tower.

For information concerning exposure levels that may be expected in the areas in close proximity to the transmitting antennas, see [ITU-T K.122]. Such areas are usually not accessible to the general public, but the presence of maintenance staff may be temporarily required there.

In the case of mobile handsets, the antenna radiation pattern is close to omnidirectional, because the BS in which the connection is established can be in any direction. It is not reasonable to reduce the radiation level in the direction of the human head, because it may be the same direction as to the BS. However, in mobile handsets, the radiation level is set as low as possible for reliable communication (automatic power control (APC)) and it is strictly controlled before the device is put on the market.

In general, digital radiocommunication systems are more efficient and use lower radiated power. This means that, in general, transition from analogue to digital systems leads to lower levels of human exposure.

11 Conclusions

There are many possible methods of exposure assessment and each of them has its own advantages and disadvantages. This Recommendation gives guidance on the method to be chosen, depending on needs and circumstances.

For additional information, see [IEC 62232], [ITU-T K.52] and [ITU-T K.100].

Appendix I

Exposure limits

(This appendix does not form an integral part of this Recommendation.)

This appendix builds on [b-ICNIRP 2010] and [b-ICNIRP-2020], IEEE 95.1-2019 and the ITU-D Question 7/2 October 2021 [b-ITU-D Q 7/2].

I.1 Introduction

The International Commission on Non-Ionizing Radiation Protection (ICNIRP) has revised its 1998 ICNIRP Guidelines for Limiting Exposure to Time-varying Electric, Magnetic and Electromagnetic Fields (up to 300 GHz) [b-ICNIRP 1998]. The final ICNIRP Guidelines were published in March 2020 [b-ICNIRP 2020]. The ICNIRP Guidelines for limiting exposure are as follows: ICNIRP (1998) [b-ICNIRP 1998] up to 300 GHz, ICNIRP (2010) [b-ICNIRP 2010] 1 Hz – 100 kHz, [b-ICNIRP 2020] 100 kHz to 300 GHz. The current limits below 100 kHz are the ones published in [b-ICNIRP 2010]. With the publication of the 2020 RF Guidelines, the 1998 Guidelines have become obsolete. On 4 October 2019, the Institute of Electrical and Electronics Engineers (IEEE) published [b-IEEE C95.1], "IEEE Standard for Safety Levels with Respect to Human Exposure to Radio Frequency Electromagnetic Fields, 0 Hz to 300 GHz" [b-IEEE 95.1-2019], an update to [IEEE C95.1-2005] standard.

Administrations are encouraged to follow the guidelines set by the science-based ICNIRP and IEEE expert groups, or limits set by their own experts. It is strongly recommended to adopt harmonized international standards and EMF exposure limits. It should be emphasized that [b-IEEE C95.1] and ICNIRP 2020 Guidelines are largely harmonized.

The [b-ICNIRP-1998] and [b-ICNIRP 2020], and [b-IEEE C95.1] separate persons in unrestricted environments (general public) and persons permitted in restricted environments (occupational). The basic restrictions and the reference levels for power density of [b-ICNIRP 2020] and the IEEE Standard for whole-body exposure to continuous fields above 30 MHz are identical:

- SAR equals 2 W/kg for general public and 10 W/kg for occupational.
- Exposure reference levels equals at:
 - 400 to 2 000 MHz $f_M/200 \text{ W/m}^2$ for general public and $f_M/40 \text{ W/m}^2$ for occupational
 - 2 000 to 300 000 MHz 10 W/m² for general public and 50 W/m² for occupational.

I.2 ICNIRP-2010 and ICNIRP-2020 guidelines in force

I.2.1 Elaborating the tables and figures of [b-ICNIRP 2020]

In Tables I.1 to Table I.4 (taken from Tables 1, 2, 5 and 6 of [b-ICNIRP 2020]) and the following figures (not taken from the Guidelines), underlined text² indicates the significant parameters. Comparisons with [b-ICNIRP 2010] (for frequencies lower than 100 kHz) are inserted.

² The underlined text in this clause does not appear as underlined in the original tables.

Table I.1 – Quantities and corresponding SI units used in the [b-ICNIRP 2020] Guidelines (ICNIRP Table 1)

Quantity	Symbol*	Unit
Absorbed energy density	U_{ab}	joule per square meter ($J m^{-2}$)
Incident energy density	U_{inc}	joule per square meter ($J m^{-2}$)
Plane-wave equivalent incident energy density	U_{eq}	joule per square meter ($J m^{-2}$)
Absorbed power density	S_{ab}	watt per square meter ($W m^{-2}$)
Incident power density	S_{inc}	watt per square meter ($W m^{-2}$)
Plane-wave equivalent incident power density	S_{eq}	watt per square meter ($W m^{-2}$)
Induced electric field strength	E_{ind}	volt per meter ($V m^{-1}$)
Incident electric field strength	E_{inc}	volt per meter ($V m^{-1}$)
Incident magnetic field strength	H_{inc}	ampere per meter ($A m^{-1}$)
Specific energy absorption	SA	joule per kilogram ($J kg^{-1}$)
Specific energy absorption rate	SAR	watt per kilogram ($W kg^{-1}$)
Electric current	I	ampere (A)
Frequency	f	hertz (Hz)
Time	t	second (s)
* <i>Italicized</i> symbols represent variables; quantities are described in scalar (not vector) form because direction is not used to derive the basic restrictions or reference levels.		

The following table (ICNIRP Table 2) refers to EMF exposure from 100 kHz to 300 GHz.

Table I.2 – Basic restrictions, for averaging intervals ≥ 6 min (ICNIRP Table 2)

Exposure scenario	Frequency range	Whole-body average SAR ($W kg^{-1}$)	Local Head/Torso SAR ($W kg^{-1}$)	Local Limb SAR ($W kg^{-1}$)	Local S_{ab} ($W m^{-2}$)
Occupational	100 kHz to 6 GHz	0.4	10	20	NA
	> 6 to 300 GHz	0.4	NA	NA	100
General public	100 kHz to 6 GHz	0.08	2	4	NA
	> 6 to 300 GHz	0.08	NA	NA	20

Note:

- 1) "NA" signifies "not applicable" and does not need to be taken into account when determining compliance.
- 2) Whole-body average SAR is to be averaged over 30 min.
- 3) Local SAR and S_{ab} exposures are to be averaged over 6 min.
- 4) Local SAR is to be averaged over a 10-g cubic mass.

- 5) Local Sab is to be averaged over a square 4-cm² surface area of the body. Above 30 GHz, an additional constraint is imposed, such that exposure averaged over a square 1-cm² surface area of the body is restricted to two times that of the 4-cm² restriction.

The following two tables (Tables 5 and 6 of [b-ICNIRP 2020]) detail reference levels for exposure, to 'electromagnetic fields from 100 kHz to 300 GHz (unperturbed rms values)'.

Table I.3 – Reference levels, averaged over 30 minutes and the whole body (ICNIRP Table 5)

Exposure scenario	Frequency range	Incident E-field strength; E_{inc} (V m ⁻¹)	Incident H-field strength; H_{inc} (A m ⁻¹)	Incident power density; S_{inc} (W m ⁻²)
Occupational	0.1 – 30 MHz	$660/f_M^{0.7}$	$4.9/f_M$	NA
	> 30 – 400 MHz	61	0.16	10
	> 400 – 2000 MHz	$3f_M^{0.5}$	$0.008f_M^{0.5}$	$f_M/40$
	<u>> 2 – 300 GHz</u>	NA	NA	<u>50</u>
General Public	0.1 – 30 MHz	$300/f_M^{0.7}$	$2.2/f_M$	NA
	> 30 – 400 MHz	27.7	0.073	2
	> 400 – 2000 MHz	$1.375f_M^{0.5}$	$0.0037f_M^{0.5}$	$f_M/200$
	<u>> 2 – 300 GHz</u>	NA	NA	<u>10</u>

Notes (from [b-ICNIRP 2020]):

- 1) 'NA' signifies 'not applicable' and does not need to be taken into account when determining compliance.
- 2) f_M is frequency in MHz.
- 3) S_{inc} , E_{inc} and H_{inc} are to be averaged over 30 minutes over the whole-body space. Temporal and spatial averaging of each of E_{inc} and H_{inc} must be conducted by averaging over the relevant square values (see Eqn. 8 in Appendix A for details).
- 4) For frequencies of 100 kHz to 30 MHz, regardless of the far-field/near-field zone distinctions, compliance is demonstrated if neither E_{inc} or H_{inc} exceeds the above reference level values.
- 5) For frequencies of > 30 MHz to 2 GHz: (a) within the far-field zone: compliance is demonstrated if either S_{inc} , E_{inc} or H_{inc} does not exceed the above reference level values (only one is required); S_{eq} may be substituted for S_{inc} ; (b) within the radiative near-field zone, compliance is demonstrated if either S_{inc} , or both E_{inc} and H_{inc} do not exceed the above reference level values; and (c) within the reactive near-field zone: compliance is demonstrated if both E_{inc} and H_{inc} do not exceed the above reference level values; S_{inc} cannot be used to demonstrate compliance, and so basic restrictions must be assessed.
- 6) For frequencies of > 2 GHz to 300 GHz: (a) within the far-field zone: compliance is demonstrated if S_{inc} does not exceed the above reference level values; S_{eq} may be substituted for S_{inc} ; (b) within the radiative near-field zone, compliance is demonstrated if S_{inc} does not exceed the above reference level values; and (c) within the reactive near-field zone, reference levels cannot be used to determine compliance, and so basic restrictions must be assessed.

Table I.4 – Reference levels for local exposure, averaged over 6 minutes (ICNIRP Table 6)

Exposure scenario	Frequency range	Incident E-field strength; E_{inc} (V m ⁻¹)	Incident H-field strength; H_{inc} (A m ⁻¹)	Incident power density; S_{inc} (W m ⁻²)
Occupational	0.1 – 30 MHz	$1504/f_M^{0.7}$	$10.8/f_M$	NA
	> 30 – 400 MHz	<u>139</u>	<u>0.36</u>	<u>50</u>
	> 400 – 2000 MHz	$10.58f_M^{0.43}$	$0.0274f_M^{0.43}$	$0.29f_M^{0.86}$
	> 2 – 6 GHz	NA	NA	<u>200</u>
	> 6 – < 300 GHz	NA	NA	$275/f_G^{0.177}$
	300 GHz	NA	NA	<u>100</u>
General Public	0.1 – 30 MHz	$671/f_M^{0.7}$	$4.9/f_M$	NA
	> 30 – 400 MHz	<u>62</u>	<u>0.163</u>	<u>10</u>
	> 400 – 2000 MHz	$4.72f_M^{0.43}$	$0.0123f_M^{0.43}$	$0.058f_M^{0.86}$
	> 2 – 6 GHz	NA	NA	<u>40</u>
	> 6 – 300 GHz	NA	NA	$55/f_G^{0.177}$
	300 GHz	NA	NA	<u>20</u>

Notes (from [b-ICNIRP 2020]):

- 1) 'NA' signifies 'not applicable' and does not need to be taken into account when determining compliance.
- 2) f_M is frequency in MHz; f_G is frequency in GHz.
- 3) S_{inc} , E_{inc} and H_{inc} are to be averaged over 6 minutes, and where spatial averaging is specified in Notes 6-7, over the relevant projected body space. Temporal and spatial averaging of each of E_{inc} and H_{inc} must be conducted by averaging over the relevant square values (see Eqn. 8 in Appendix A for details).
- 4) For frequencies of 100 kHz to 30 MHz, regardless of the far-field/near-field zone distinctions, compliance is demonstrated if neither peak spatial E_{inc} or peak spatial H_{inc} , over the projected whole-body space, exceeds the above reference level values.
- 5) For frequencies of ≥ 30 MHz to 6 GHz: (a) within the far-field zone, compliance is demonstrated if one of peak spatial S_{inc} , E_{inc} or H_{inc} , over the projected whole-body space, does not exceed the above reference level values (only one is required); S_{eq} may be substituted for S_{inc} ; (b) within the radiative near-field zone, compliance is demonstrated if either peak spatial S_{inc} , or both peak spatial E_{inc} and H_{inc} , over the projected whole-body space, does not exceed the above reference level values; and (c) within the reactive near-field zone: compliance is demonstrated if both E_{inc} and H_{inc} do not exceed the above reference level values; S_{inc} cannot be used to demonstrate compliance; for frequencies > 2 GHz, reference levels cannot be used to determine compliance, and so basic restrictions must be assessed.
- 6) For frequencies of ≥ 6 GHz to 300 GHz: (a) within the far-field zone, compliance is demonstrated if S_{inc} , averaged over a square 4-cm² projected body surface space, does not exceed the above reference level values; S_{eq} may be substituted for S_{inc} ; (b) within the radiative near-field zone, compliance is demonstrated if S_{inc} , averaged over a square 4-cm² projected body surface space, does not exceed the above reference level values; and (c) within the reactive near-field zone, reference levels cannot be used to determine compliance, and so basic restrictions must be assessed.
- 7) For frequencies of ≥ 30 GHz to 300 GHz, exposure averaged over a square 1-cm² projected body surface space must not exceed twice that of the square 4-cm² restrictions.

ICNIRP (2020) Table 8 states (pay attention to the bold text) "*reference levels for **local exposure** to EMFs from 100 kHz to 10 MHz (unperturbed rms values), for **peak values**, the occupational limit is 170 V/m and the general public is 83 V/m*".

The following figure, [b-Mazar 2016] revised 2021 Chapter 9, Figure 9.6, depicts the differences between the ICNIRP (2020) field-strength and power-density exposure levels of occupational and general-public exposure, averaged over 30 minutes and the whole body. The power-density ratio of 5 in ICNIRP (2020) Table 5 (e.g., at 30 – 400MHz, Watts ratio 50/10) results in a V/m ratio $61.0/27.7 = 2.2 \approx \sqrt{5}$.

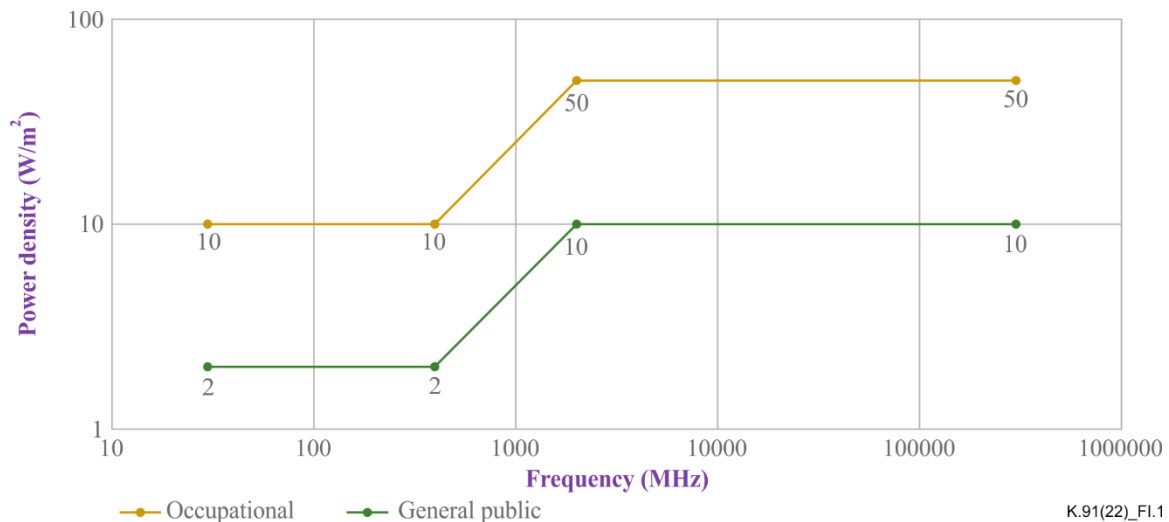


Figure I.1 – ICNIRP (2020) Table 5, power density for occupational versus general public exposures 30 MHz – 300 GHz

I.3 IEEE C95.1-2019

This clause elaborates the tables and figures of the 2019 version of the C95.1-2019 standard [b-IEEE 95.1] (including C95.1-2019/Cor 1-2019).

I.3.1 Reference levels: safety factors applying 100 kHz – 6 GHz; thermal effects

- Whole body averaged (WBA)
Behavioural effects in animals over many frequencies, threshold at 4 W/kg, before dividing by:
10x – 0.4 W/kg for upper tier (restricted environments)
50x – 0.08 W/kg for lower tier (unrestricted environments – general public)
- Localized exposure (averaged in 10 g)
Cataract observed in rabbits, threshold at 100 W/kg, before dividing by:
10x – 10 W/kg for upper tier
50x – 2 W/kg for lower tier
- SAR is averaged over 30 minutes for WBA exposure and 6 minutes for local exposure
- Epithelial power density through body surface is averaged over 6 minutes.

I.3.2 Dosimetric reference limits and exposure reference level

This clause is based on IEEE (2019), Tables 5 to 8, Figures 3 and 4.

The following two tables specify dosimetric reference limits (DRLs) below and above 6 GHz. No continuity at 6 GHz.

Table I.5 – Dosimetric reference limits, DRLs, 100 kHz to 6 GHz (Table 5 of [b-IEEE C95.1])*

Conditions	Persons in unrestricted environments, SAR (W/kg)^a	Persons in restricted environments, SAR (W/kg)^a
Whole-body exposure	0.08	0.4
Local exposure ^b (head and torso)	<u>2</u>	<u>10</u>
Local exposure ^b (limbs and pinnae)	4	20

^a SAR is averaged over 30 minutes for whole-body exposure and 6 minutes for local exposure.

^b Averaged over any 10 g of tissue (defined as a tissue volume in the shape of a cube). The averaging volume of 10 g of tissue would be represented as a 10 cm³ cube (approximately 2.15 cm per side).

Table I.6 – DRLs, 6 GHz to 300 GHz (Table 6 of [b-IEEE C95.1])*

Conditions	Epithelial power density (W/m²)^{a,b,c}	
	Persons in unrestricted environments	Persons permitted in restricted environments
Body surface	20	100

^a Epithelial power density through body surface is averaged over 6 minutes.

^b Averaged over any 4 cm² of body surface at frequencies between 6 GHz and 300 GHz (defined as area in the shape of a square at surface of the body).

^c Small exposed areas above 30 GHz: If the exposed area on the body surface is small (< 1 cm² as defined by –3 dB contours relative to the peak exposure), the epithelial power density is allowed to exceed the DRL values of Table 6 by a factor of 2, with an averaging area of 1 cm² (defined as area in the shape of a square at the body surface).

The following two tables do not provide the electric and magnetic field strengths above 400 MHz. The following table details exposure reference level (ERLs) for whole-body exposure of persons in unrestricted environments, averaging time 30 minutes.

* Adapted and reprinted with permission from IEEE. Copyright IEEE 2019. All rights reserved.

Table I.7 – ERLs, 100 kHz to 300 GHz (Table 7 of [b-IEEE C95.1])*

Frequency range (MHz)	Electric field Strength (E) ^{a,b,c} (V/m)	Magnetic field strength (H) ^{a,b,c} (A/m)	Power density (S) ^{a,b,c} (W/m ²)	
0.1 to 1.34	614	$16.3/f_M$	S_E	S_H
			1 000	$100\,000/f_M^2$
1.34 to 30	$823.8/f_M$		$1\,800/f_M^2$	
30 to 100	27.5	$158.3/f_M^{1.668}$	2	$9\,400\,000/f_M^{3.336}$
100 to 400		0.0729	2	
400 to 2000			$f_M/200$	
2000 to 300 000			10	

NOTE – S_E and S_H are plane-wave equivalent power density values, based on electric or magnetic field strength respectively, and are commonly used as a convenient comparison with ERLs at higher frequencies and are sometimes displayed on commonly used instruments.

^a For exposures that are uniform over the dimensions of the body, such as certain far-field plane-wave exposures, the exposure field strengths and power densities are compared with the ERLs in [IEEE 95.1] Table 7. For more typical non-uniform exposures, the mean values of the exposure fields, as obtained by spatially averaging the plane-wave-equivalent power densities or the squares of the field strengths, are compared with the ERLs in Table 7.

^b f_M is the frequency in MHz.

^c The E , H , and S values are those rms values unperturbed by the presence of the body.

The following figure depicts Figure 3 of [b-IEEE C95.1] – Graphical representations of the ERLs in Table 7 of IEEE standard, electric and magnetic fields, and plane-wave-equivalent power density – Persons in **unrestricted** environments.

NOTE – At frequencies below 30 MHz, the wavelength is longer than 10 metres. There is no resonance with our body (shorter than 2 metres). We are not an obstacle to the signal, and only a small part of the RF energy enters our body. This note does not appear in the [IEEE 95.1] standard.

* Adapted and reprinted with permission from IEEE. Copyright IEEE 2019. All rights reserved.

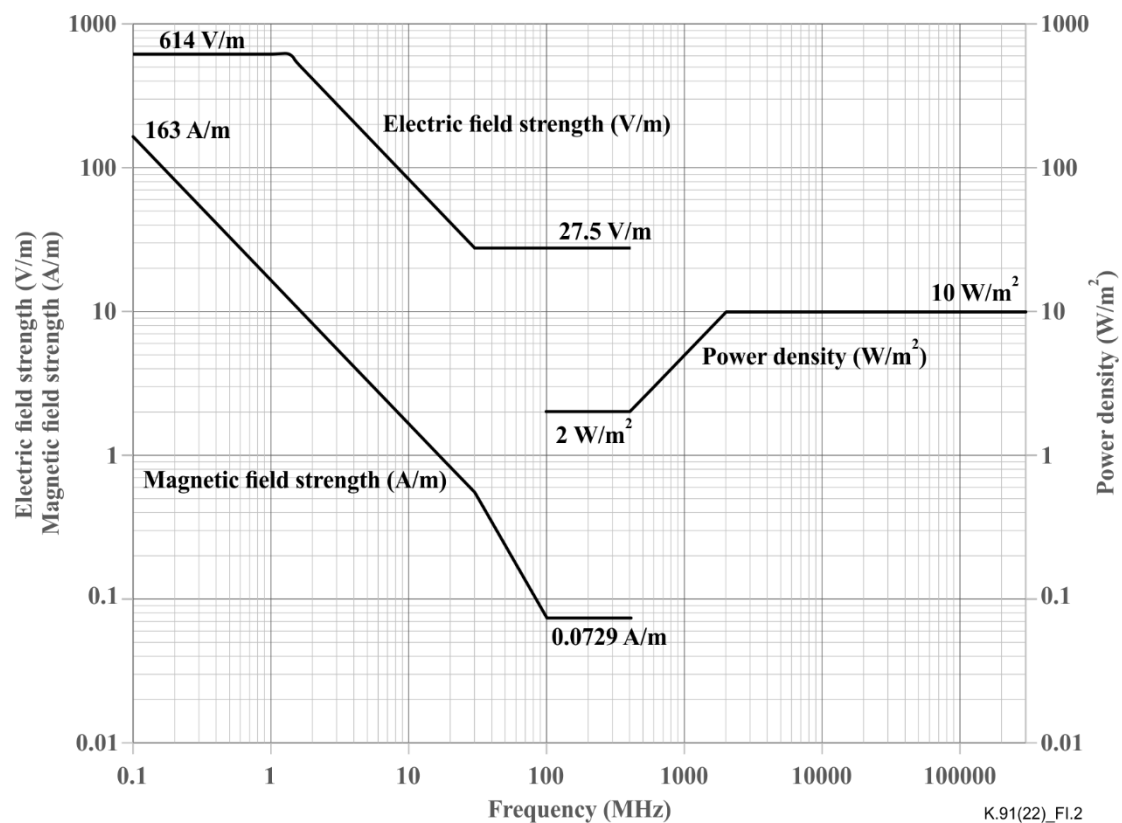


Figure I.2 – EMFs and power density in unrestricted environments
(Figure 3 of [b-IEEE C95.1])

The following table details Table 8 of [b-IEEE C95.1]: ERLs for whole-body exposure of persons permitted in restricted environments (100 kHz to 300 GHz), averaging 30 minutes.

Table I.8 – ERLs in restricted environments, 100 kHz to 300 GHz
(Table 8) of [b-IEEE C95.1] *

Frequency range (MHz)	Electric field Strength (E) ^{a,b,c} (V/m)	Magnetic field strength (H) ^{a,b,c} (A/m)	Power density (S) ^{a,b,c} (W/m ²)	
0.1 to 1.0	1 842	16.3/ f_M	S_E	S_H
			9 000	100 000 f_M^2
1.0 to 30	1 842/ f_M		$9\,000 / f_M^2$	
30 to 100	61.4		10	
100 to 400		0.163	10	
400 to 2000			$f_M/40$	
2000 to 300 000			50	

NOTE – S_E and S_H are plane-wave equivalent power density values, based on electric or magnetic field strength respectively, and are commonly used as a convenient comparison with ERLs at higher frequencies and are sometimes displayed on commonly used instruments.

^a For exposures that are uniform over the dimensions of the body, such as certain far-field plane-wave exposures, the exposure field strengths and power densities are compared with the ERLs in [IEEE 95.1]

* Adapted and reprinted with permission from IEEE. Copyright IEEE 2019. All rights reserved.

Table 8. For more typical non-uniform exposures, the mean values of the exposure fields, as obtained by spatially averaging the plane-wave-equivalent power densities or the squares of the field strengths, are compared with the ERLs in Table 8.

^b f_M is the frequency in MHz.

^c The E , H , and S values are those rms values unperturbed by the presence of the body.

The following figure depicts Figure 4 of [b-IEEE C95.1]: Graphical representations of the ERLs in IEEE (Table 8) for electric and magnetic fields and plane-wave-equivalent power density – Persons permitted in **restricted** environments.

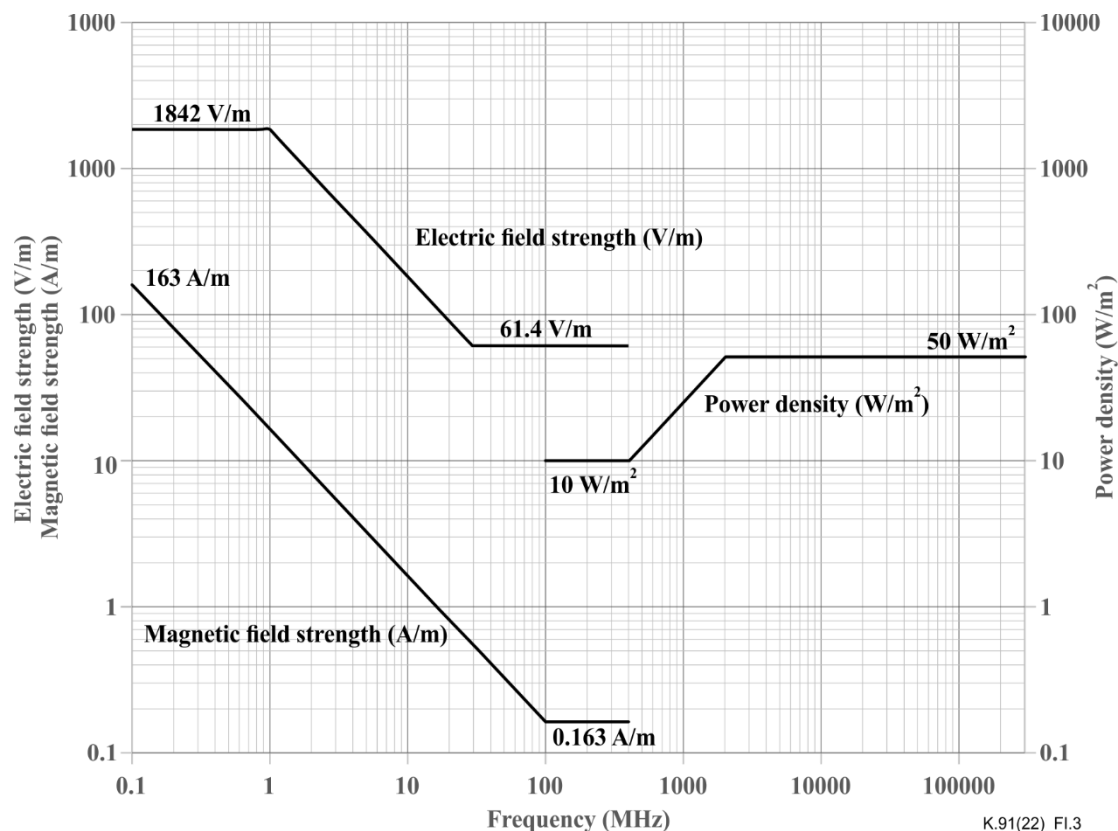


Figure I.3 – EMFs and power density in **restricted** environments
(Figure 4 of [b-IEEE C95.1])*

I.4 Simultaneous exposure to multiple sources

The exposure assessment in the multiple sources environment, according to the existing standards, requires the calculation of the cumulative exposure (in some standards called also the total exposure ratio). All the operating frequencies must be considered in a weighted sum, where each individual source is pre-rated according to the limit applicable to its frequency. For compliance with the regulations, the total exposure should be less than 1.

I.4.1 Basic restrictions

In the case of simultaneous exposure to fields of different frequencies, the following criteria should be satisfied in terms of the basic restrictions for intervals ≥ 6 min; see 'Basic restrictions for intervals 6 min', [b-ICNIRP 2020] p. 498. For thermal effects, relevant from 100 kHz, specific energy absorption rates and power densities should be added according to [b-ICNIRP 2020] p. 498 equation (1):

* Adapted and reprinted with permission from IEEE. Copyright IEEE 2019. All rights reserved.

$$\sum_{i=100 \text{ kHz}}^{300 \text{ GHz}} \frac{SAR_i}{SAR_{BR}} \leq 1$$

where SAR_i and SAR_{BR} are the whole-body average SAR levels at frequency i and the whole-body average SAR basic restrictions given in Table 2 ICNIRP (2020) respectively. SAR_{BR} is the SAR basic restriction given in Table 2 of [b-ICNIRP 2020], see Table I.2.

For simultaneous exposure to fields at different frequencies, the compliance with the exposure limits is evaluated using the equation below. All conditions for the appropriate frequency ranges are to be satisfied, see ICNIRP (2020) p. 498 Equation 4:

$$\sum_{i=100 \text{ kHz}}^{30 \text{ MHz}} \left\{ \left(\frac{E_{inc,i}}{E_{inc,RL,i}} \right)^2 + \left(\frac{H_{inc,i}}{H_{inc,RL,i}} \right)^2 \right\} + \sum_{i>30 \text{ MHz}}^{2 \text{ GHz}} \text{MAX} \left\{ \left(\frac{E_{inc,i}}{E_{inc,RL,i}} \right)^2, \left(\frac{H_{inc,i}}{H_{inc,RL,i}} \right)^2, \left(\frac{S_{inc,i}}{S_{inc,RL,i}} \right) \right\} + \sum_{i>2 \text{ GHz}}^{300 \text{ GHz}} \left(\frac{S_{inc,i}}{S_{inc,RL,i}} \right) \leq 1$$

where, $E_{inc,i}$ and $E_{inc,RL,i}$ are the local incident electric field strength and local incident electric field strength reference level given in Table 6, see Table I.4, at frequency i , respectively; $H_{inc,i}$ and $H_{inc,RL,i}$ are the local incident magnetic field strength and local incident magnetic field strength reference level given in Table 6, at frequency i , respectively; $S_{inc,i}$ and $S_{inc,RL,i}$ are the local incident power density and local incident power density reference level given in Table 6, at frequency i , respectively; inside the body above 6 GHz, S_{inc} terms are to be treated as zero; equation (4) must be satisfied for every position in the human body.

Appendix II

Time averaging

(This appendix does not form an integral part of this Recommendation.)

II.1 Analysis of time variations of measured electric fields from WCDMA mobile base stations

This appendix presents the results of investigation of the effect of measurement of the time-variation on measured E -fields. The measurement has been done around WCDMA mobile base stations in Japan. Figure II.1 shows the SD for different measurement times. The results indicate very small variation in SDs. The maximum and minimum SDs are 1.7 and 1.5 dB, respectively, obtained for measurement times of 10 and 20 s. Table II.1 lists time-averaged E -fields for each measurement time. The values are normalized to the average value obtained for a 6 min measurement time. In Table II.1, the numbers of measured data are also listed. It can be seen that the variation of the average E -field is very small for measurement times of up to 6 min. However, there is a sudden increase in the average value for 1 h and 24 h, even though their SDs are almost the same as those obtained for measurements of up to 6 min.

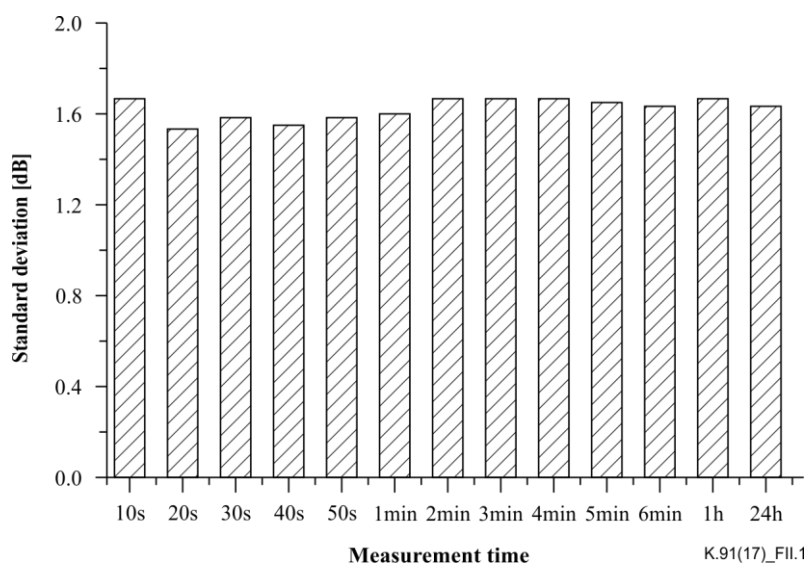


Figure II.1 – Standard deviation as a function of measurement time

Table II.1 – Time-averaged E -fields normalized to the value of the 6 min measurement time

Time	10 s	20 s	30 s	40 s	50 s	1 min	2 min	3 min	4 min	5 min	6 min	1 h	24 h
Number of measurements	62	115	172	230	280	334	654	976	1 299	1 660	1 940	174	1 429
Average E -field (dB)	−0.4	−0.3	−0.2	−0.2	−0.3	−0.4	−0.3	−0.1	−0.1	−0.0	0	−4.0	−4.2
NOTE – The number of measured data for each measurement time is also listed.													

Very small variations of 0.1 and 0.4 dB in SD and time-averaged E -field, respectively, have been obtained for measurement times of up to 6 min.

In the Republic of Korea, the effect of time reduction has been considered by the analyses of the RF EMF around CDMA BSs in 18 periods of time from 180 to 10 s with a 10 s interval (180, 170, 160, ..., 30, 20, 10 s).

Table II.2 shows the variation of time-averaged electric field strengths in one point for several time periods ranging from over 10 to 360 s of the seven target BSs. These values were calculated using data obtained during the first time-periods that had the same starting time.

**Table II.2 – Comparative data of electric field strength for different averaging times
(unit: dB μ V/m)**

Base station No.	Time (s)				
	360	180	60	40	10
1	109.14	109.14	109.16	109.14	109.08
2	109.80	109.79	109.79	109.79	109.80
3	107.07	107.07	107.09	107.07	107.11
4	111.59	111.55	111.60	111.63	111.81
5	92.56	92.36	92.34	92.33	92.56
6	90.41	90.28	90.76	90.84	90.74
7	97.38	97.26	97.39	97.40	97.67

In Table II.2 each element corresponds to the calculated value using the data obtained during the first period (1–10 s) among 36 periods (1–10, 11–20, 21–30, ..., 341–350, 351–360 s). Similarly, in the case of the 40 s column, it is that of the first period (1–40 s) among nine periods (1–40, 41–80, 81–120, 121–160, 161–200, 201–240, 241–280, 281–320, 321–360 s). These average values seem to be nearly the same. As described previously, since the reference levels can be averaged over any 6 min or 360 s, this comparison is appropriate.

In Figure II.2, the SD between the 360 s average value and different time periods is presented. These were estimated by the several average values calculated during successive time periods, which have no overlapping time period. For example, SD values at 60 s were estimated by the six values averaged during 1~60, 61~120, 121~180, 181~240, 241~300, and 301~360 s. The maximum SD among all BSs occurred in the case of BS No. 6, and its value was 0.58 dB at 10 s. As shown in Figure II.2, over 60 s no BSs have an SD exceeding 0.4 dB; the shortest period of 60 s was, therefore, selected as the suitable averaging time for human exposure measurement.

A dataset of measured *E*-fields of a WCDMA BS signal *in situ* from 10 s to 1 week is presented in the remainder of this clause.

Measurements configurations

Measurements were carried out at 2000 MHz band for *in situ* measurement. A Narda selective radiation meter (SRM-3000) system and field nose (ARC) were located on the rooftop of a building (Figures II.3 and II.4). The measured BS was in an urban area of Tokyo. The distance between the measurement location and the BS was 750 m. The height from the floor to the receiving antenna was set at 85 cm, due to restrictions on positioning the measurement devices at the site. The RBW was set at 100 kHz. During the measurement from 10 s to 6 min, the time intervals of the measurement were set at 10 s and 20 s for the SRM-3000 and field nose, respectively. For measurement times longer than 6 min, 12 min time-intervals were employed because of limitations on the data storage.

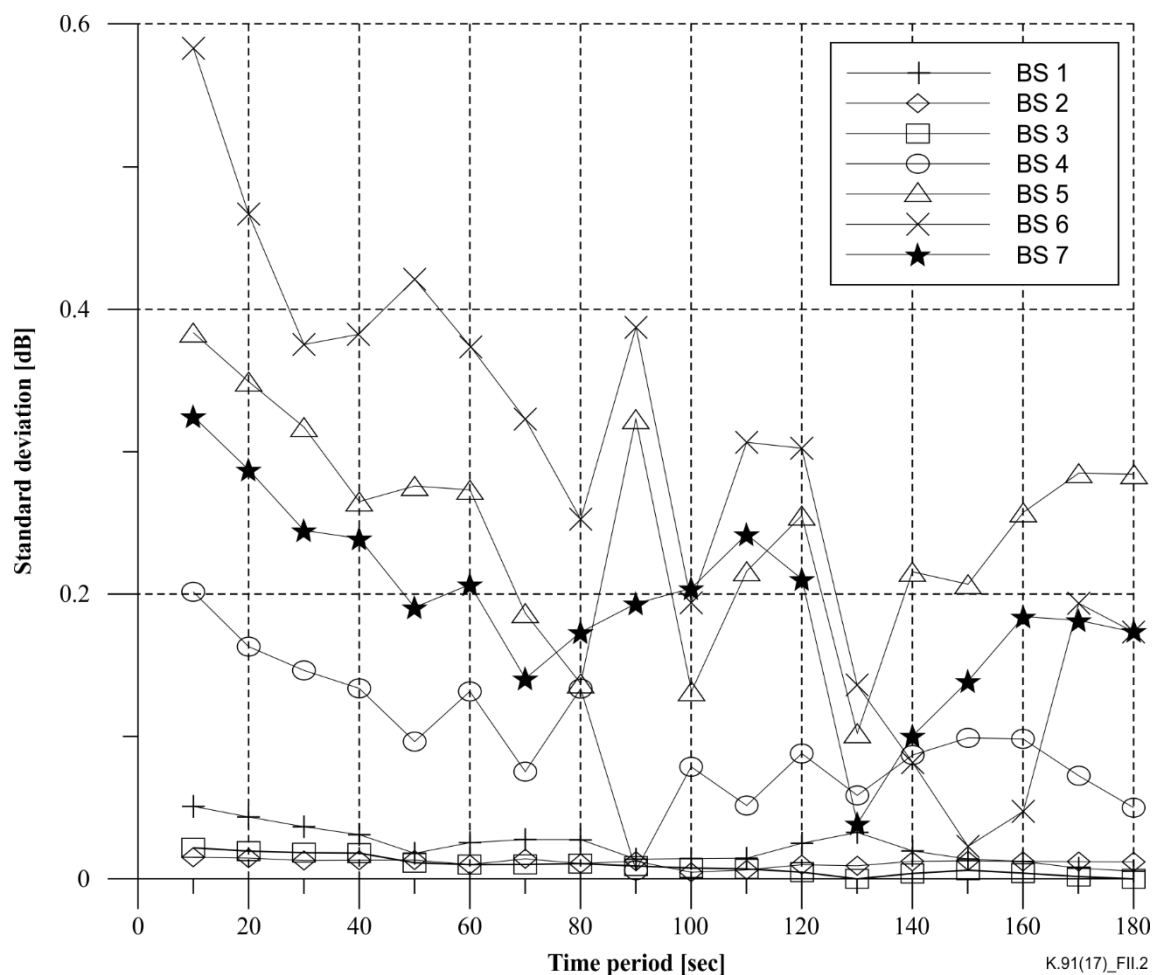


Figure II.2 – Standard deviation for different averaging times compared with that of 360 s



Figure II.3 – Base station

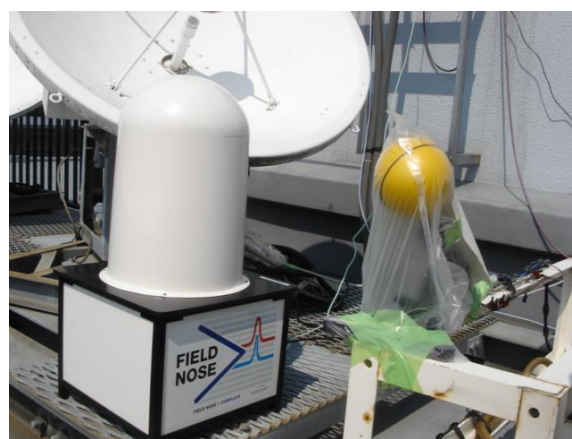


Figure II.4 – Field nose (left) and SRM-3000 (right)

Results and discussion

Figure II.5 shows the SD for different measurement times during a day. The results show that the maximum SDs of field nose and SRM-3000 are 2.1 and 2.6 dB, respectively, obtained for a measurement time of 1 h. It can be seen that the SDs are similar for measurement times of up to 6 min

for both systems. Though there is a sudden increase in the SDs for 1 h and 24 h, the SDs are still smaller than 3 dB.

Figure II.6 shows the SDs in each measurement day, along with the average over an entire week. In this case, the maximum SDs of field nose and SRM-3000 are 1.9 and 2.9 dB, respectively, and it is almost comparable to the SD of the entire week. Figure II.7 shows time-averaged *E*-fields for each measurement time. The values are normalized to the average value obtained for a 6 min measurement time. It can be seen that the average *E*-field from 6 min measurements are within the SD of 1 day measurement for both measurement systems, and the value of shorter averaging time tends to be larger than the week-averaged *E*-field. One reason for this is that the measurements were started during the daytime when traffic was heaviest, therefore the 1 day average tends to be smaller because it includes the quietest time of early morning.

Figure II.8 shows the scatter plot of a 1 week measurement with the SRM-3000. A larger variation (max-min) up to 18 dB is shown.

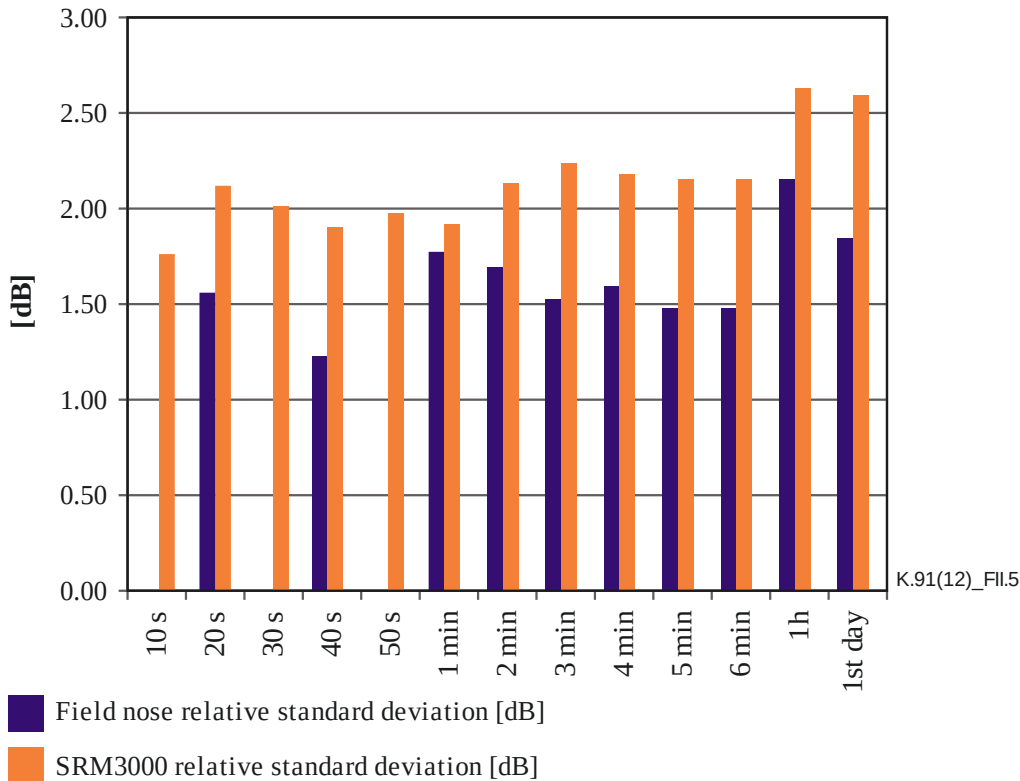


Figure II.5 – Standard deviation as a function of measurement time during a day

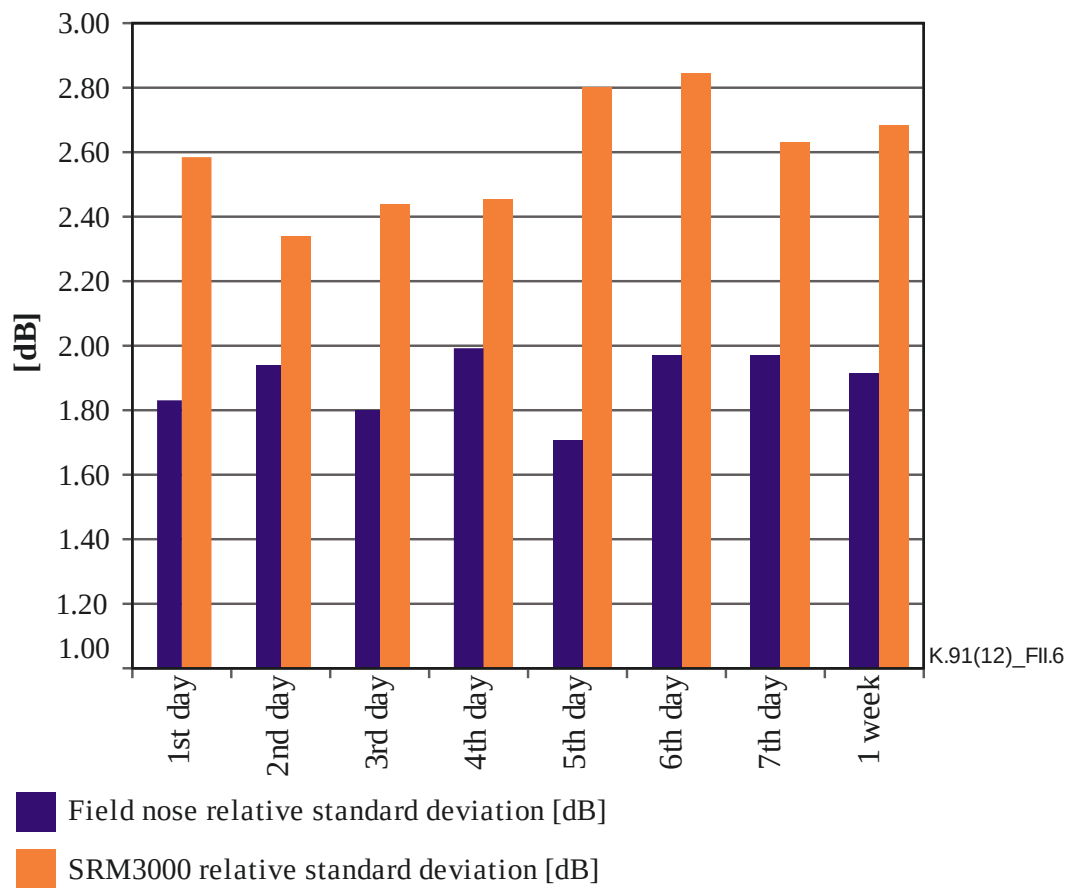


Figure II.6 – Standard deviation as a function of measurement time over 1 week

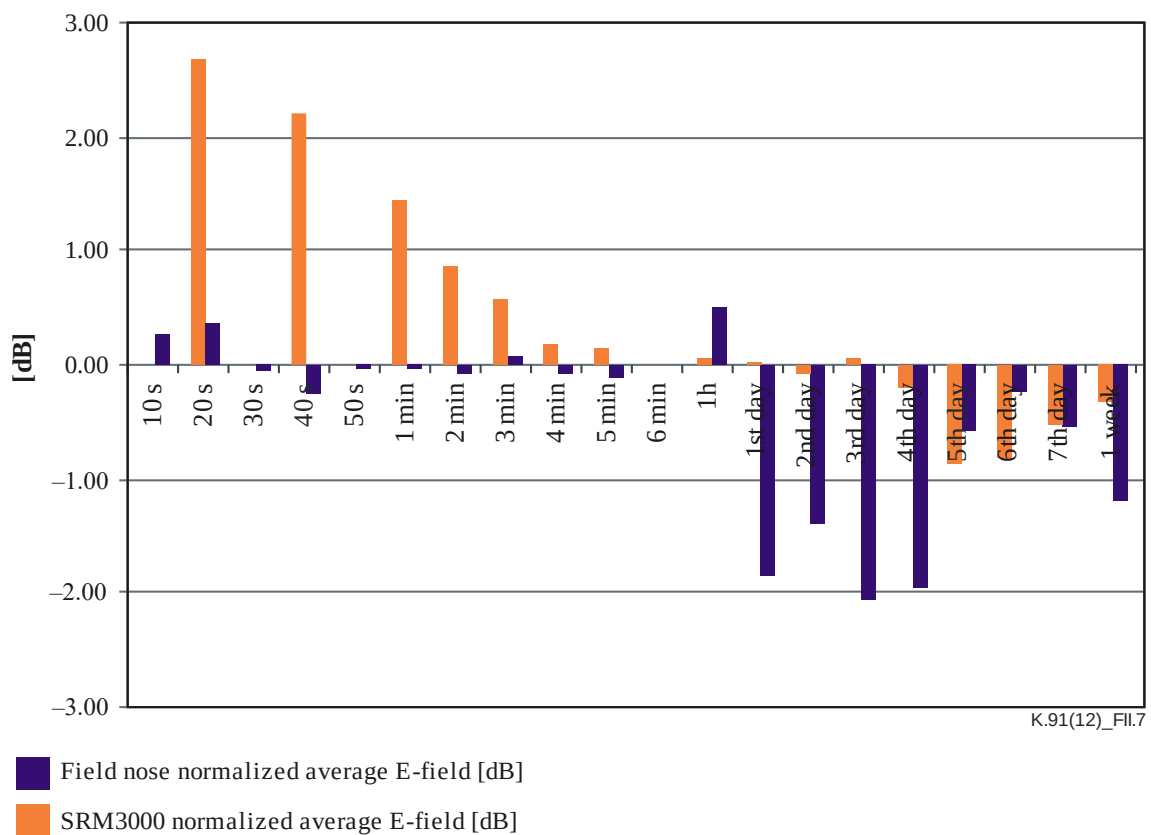
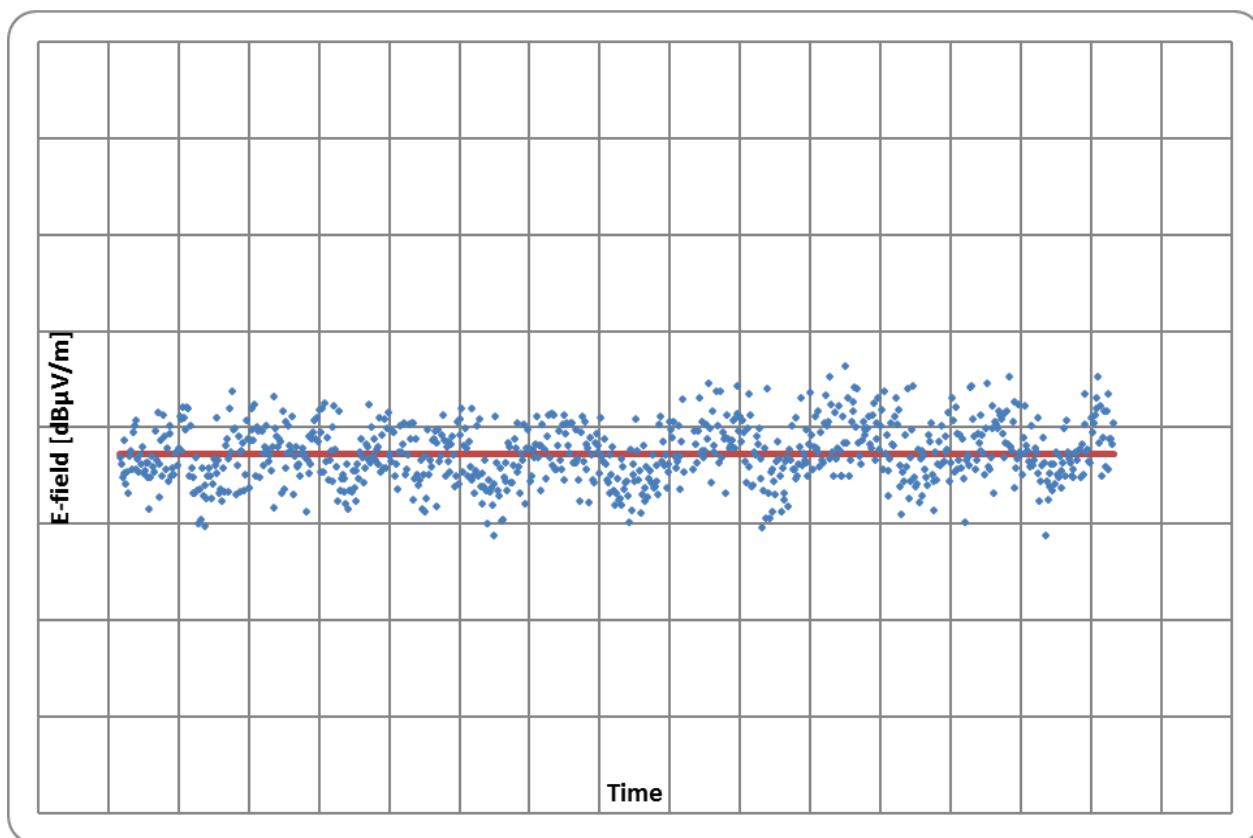


Figure II.7 – Time-averaged *E*-fields normalized to the value of the 6 min measurement time



NOTE – Red line represents the averaged value.

Figure II.8 – Measured *E*-field strength over 1 week with the SRM-3000

II.2 Analysis of time variations of measured electric fields from GSM mobile base stations

The effect of RBW and measurement times done in the United Republic of Tanzania for GSM signals are presented in clause II.3. A Narda SRM-3006 with isotropic probe was used in the measurements.

Measurement configuration

Measurements were done at 1 800 MHz band for *in situ* measurement of two GSM mobile phone base stations, referred to here as BS1 and BS2. An SRM-3006 meter was located on the second-floor balcony of a building, as shown in Figure II.9. The distances between the measurement location and BS1 and BS2 were 630 and 750 m, respectively, with a clear line of sight (LOS). Figure II.10 shows BS1. The two BSs belong to two different service providers. Each service provider has a downlink frequency band of 12.5 MHz. All measurements were taken with the isotropic probe fixed at a height of 1.5 m from the floor. Total fields were obtained by integrating *E*-fields in the downlink frequency band. The result type of the Narda equipment was set to actual.

II.3 Results and discussion

II.3.1 RBW

In this clause, the effect of RBW on the SD and level of measured signal for both BS1 and BS2 were investigated. A measurement time of 6 min and time interval of 18 s were used.

Figure II.11 shows the SD as a function of RBW. For BS1, SD increases up to the RBW of 100 kHz and then starts to decrease. The minimum SD is obtained with an RBW of 30 kHz. The maximum difference is 0.2 dB. A different trend is observed for the case of BS2, with maximum and minimum SDs obtained at RBWs of 200 and 50 kHz, respectively. The maximum variation is 0.5 dB.

The effect of RBW on the measured E -field is plotted in Figure II.12. The measured values from the BSs were normalized to the E -field of 200 kHz RBW, which is the channel bandwidth of the GSM signal. The maximum variations (max–min) obtained are 1 and 1.7 dB, respectively, for BS1 and BS2. The maximum E -fields were recorded at the frequency of maximum SD for both BSs. The smaller RBW (30 kHz) may be used for accuracy of measured results. Measurement from BS1 was found to be in agreement, because the minimum SD was obtained at an RBW of 30 kHz. This was not the case for BS2. The simulation may not be able to take into account other external factors such as reflections.

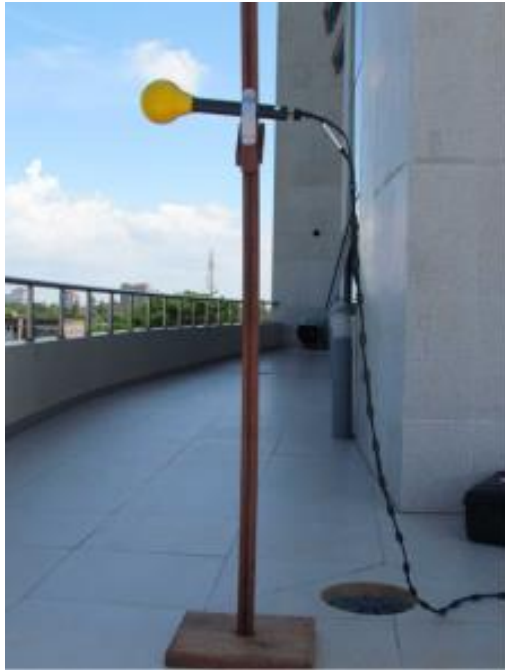


Figure II.9 – Narda probe fixed on a wooden stand



Figure II.10 – BS1

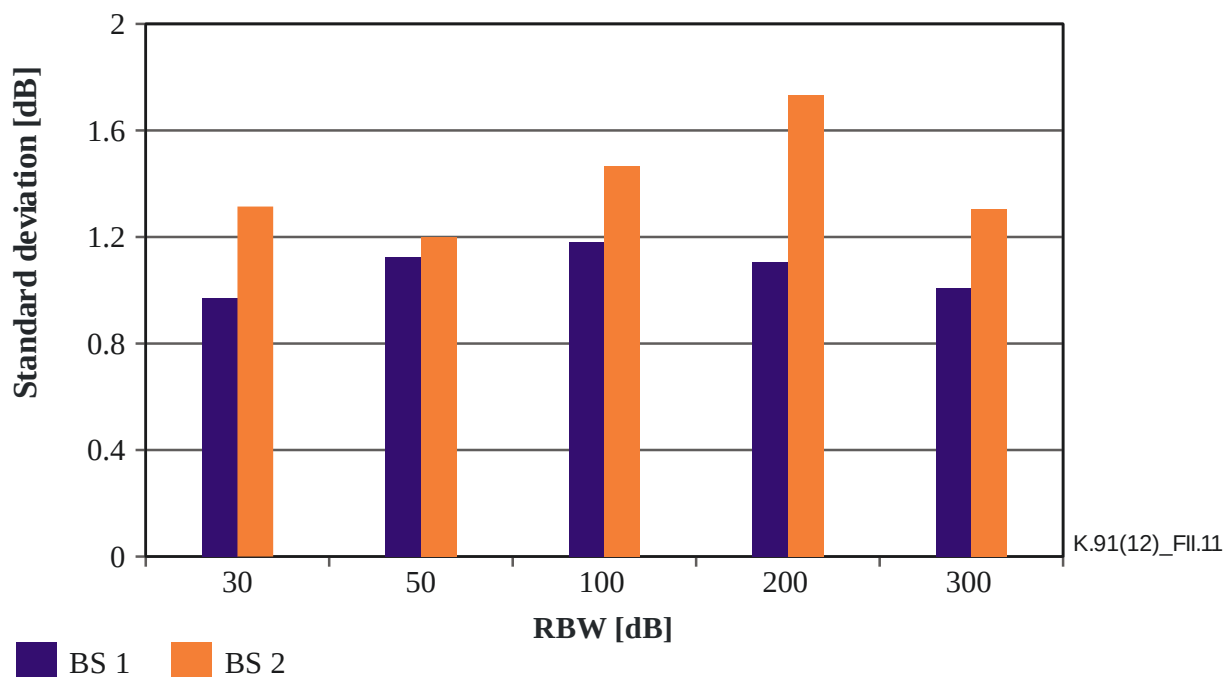


Figure II.11 – Standard deviation as a function of RBW for BS1 and BS2

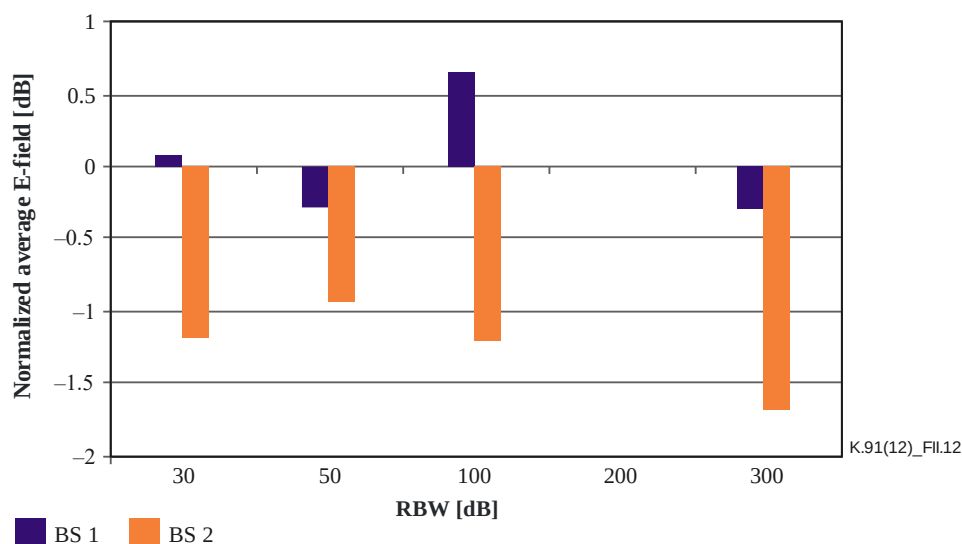


Figure II.12 – Average E -fields normalized to the value of the 200 kHz RBW

II.3.2 Measurement time

In this investigation, measurements of up to 1 h were carried out. For measurements from 12 s to 6 min, the time interval of the measurement was set at 6 s; 1 min time-intervals were employed for 1 h measurements.

Figure II.13 shows the SD for different measurement times for two different BSs. The results show that maximum SDs of 1.7 and 5.1 dB, respectively, were obtained for measurement times of up to 1 h. The SDs obtained are in the same range with the values obtained for the WCDMA system, and larger than the SD of the CDMA signal. The recorded SDs are of similar values for measurement times of up to 4 min (except for the 12 s value of BS2) for both BSs. A sudden increase in the SDs for 5 min, 6 min and 1 h was observed. A SD exceeding 3 dB was obtained for the signal from BS2.

The time-averaged E -fields of the two BSs are shown in Figures II.14 and II.15. The values are normalized to the average value obtained for a 6 min measurement time. The maximum average E -fields variations (max–min) are 0.9 and 1.9 dB for BS1 and BS2, respectively.

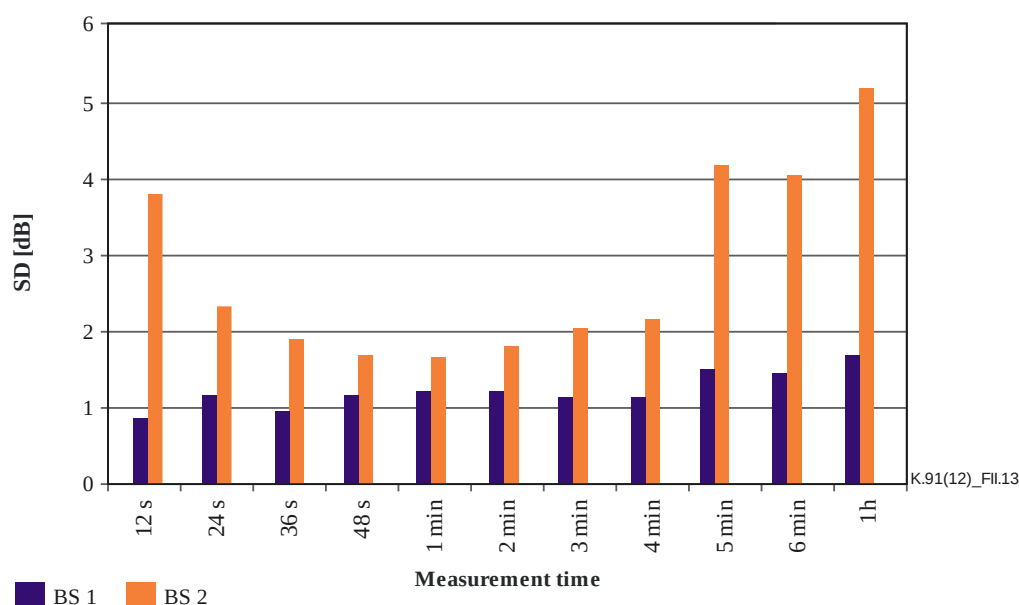


Figure II.13 – Standard deviation as a function of measurement time for BS1 and BS2

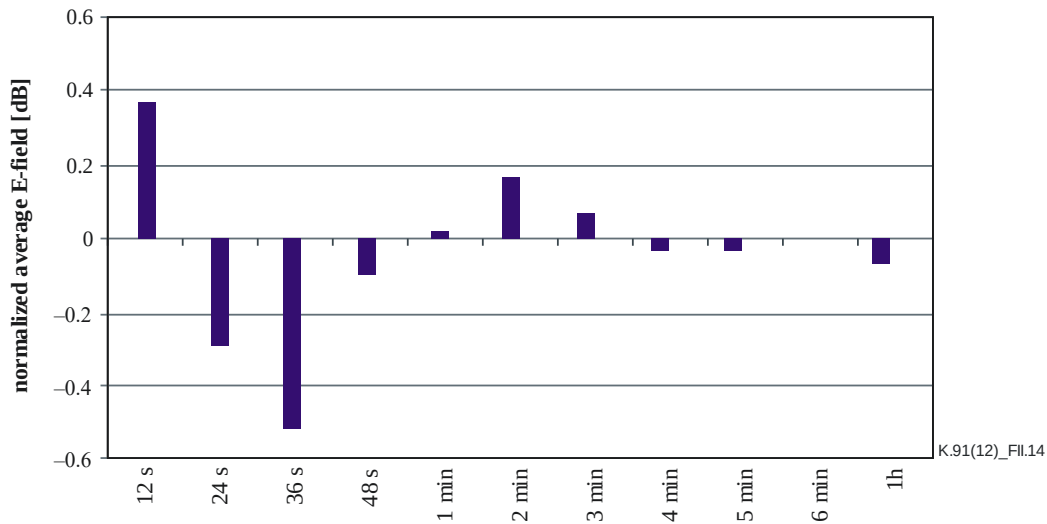


Figure II.14 – Time-averaged E -fields of BS1 normalized to the value of the 6 min measurement time

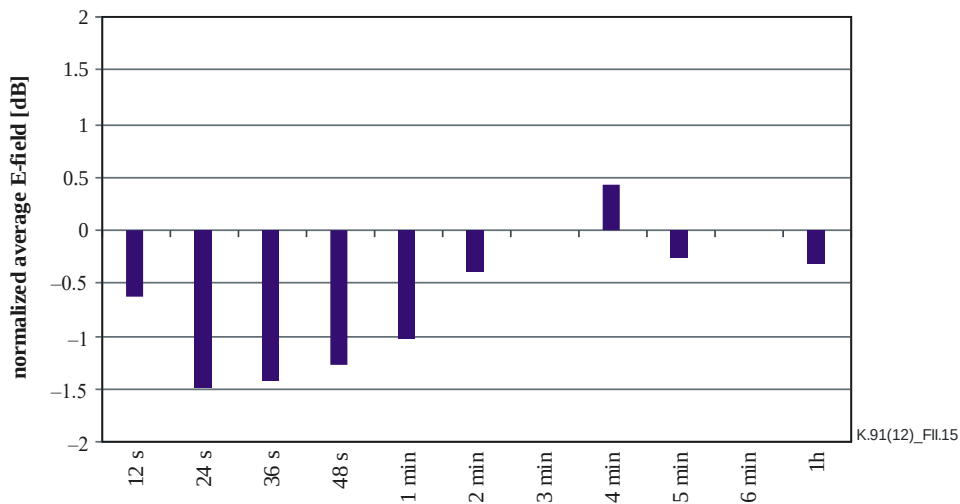


Figure II.15 – Time-averaged E -fields of BS2 normalized to the value of the 6 min measurement time

II.4 Discussion about the number of sampling data and averaging time

The measurement data of WCDMA BS measurement have been processed. The measurements were carried out with a maximum duration of 1 h. The minimum sampling interval was 10 s, due to the restriction by specification of the field measurement system SRM-3000 (Narda). A running average of each measurement time was considered.

Figure II.16 shows the variation of the relative SD of the running average at each measurement time during the measurement of two different averaging times with two different sampling intervals: 1 min averaging with a 10 s interval (scheme 1); 6 min averaging with a 10 s interval (scheme 2); and 6 min averaging with a 1 min interval (scheme 3). The numbers of data for each averaging time are 7, 37 and 7, respectively.

Figure II.16 shows that the shorter the averaging time, the larger the deviation becomes. However, except the quite large fluctuation around 20 min, the maximum fluctuations of schemes 1 and 3 are approximately the same level of 10 dB. This implies that this fluctuation is caused by the same number of the sampling data. Therefore, care must be taken when comparing averaged results. In this measurement, the SD of a whole 1 h averaging time was -4.5 dB. Consequently, these data show that

a shorter averaging time does not always show larger SD, and it depends on the traffic of the signal at the measured time.

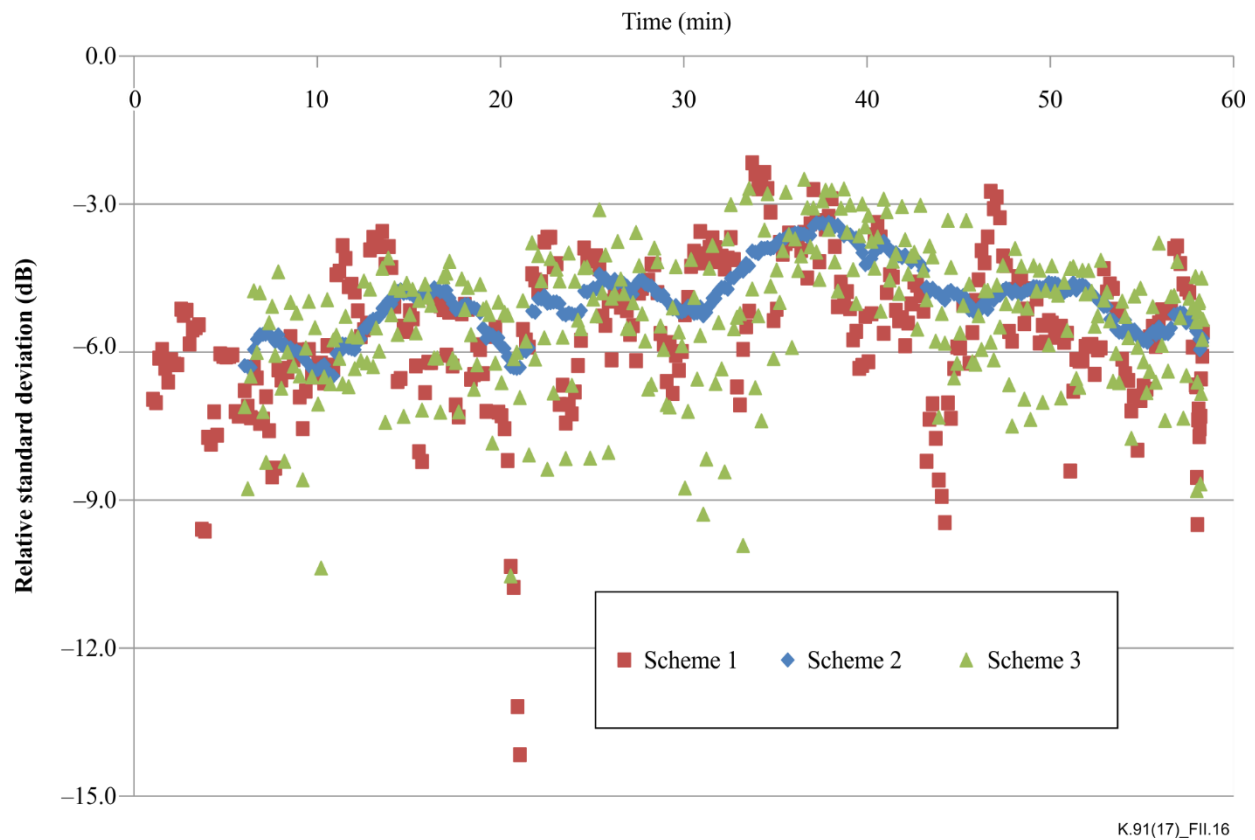


Figure II.16 – Fluctuation of relative standard deviation of a WCDMA base station *E*-field

Appendix III

Examples of RF EMF levels in areas accessible to the general public

(This appendix does not form an integral part of this Recommendation.)

Measurements were carried out in the Republic of Korea in areas where people live or spend substantial time, including hospitals, commercial areas (business areas), residential areas (densely built-up areas), schools and apartment complexes. A total of 50 places were selected: 27 in school areas, 11 in apartment complexes, 5 in commercial areas, 4 in hospitals and 3 in residential areas. Measurement positions in the LOS with the nearest CDMA BS were chosen. The measurement was performed at heights of 1.1 m, 1.5 m, and 1.7 m; using these field values, the spatially averaged value was then calculated. At each point, the field values were averaged over 1 or 6 min. The measurements used an isotropic antenna to ensure all field components (x, y, z in cartesian coordinates) were evaluated correctly considering the variation in field polarization at some distance from the BS antenna. The measurement equipment used was a Narda SRM-3000 all-in-one spectrum analyser and an isotropic probe. This setup makes it possible to perform frequency-selective measurements. When performing the measurement, the detection mode was maintained as the r.m.s.

The spatially averaged field levels, which were calculated using time-averaged values at several places of concern, are shown in Figures III.1 and III.2. Since the reference levels (field strength) are frequency dependent (i.e., mobile communication is operated in different frequency bands, and due to differences in absorbing conditions different reference levels were set up), the exposure levels are characterized by exposure ratio, which is calculated from the square of the ratio of the measured electric field strength, and the reference level, which is expressed as a fraction of the related limit. In Figure III.1, the spatially averaged values are expressed as the median of the specific classification (e.g., hospital, school, rural, single-location) and the horizontal bars denote the highest and lowest spatially averaged field values recorded at each classification, respectively.

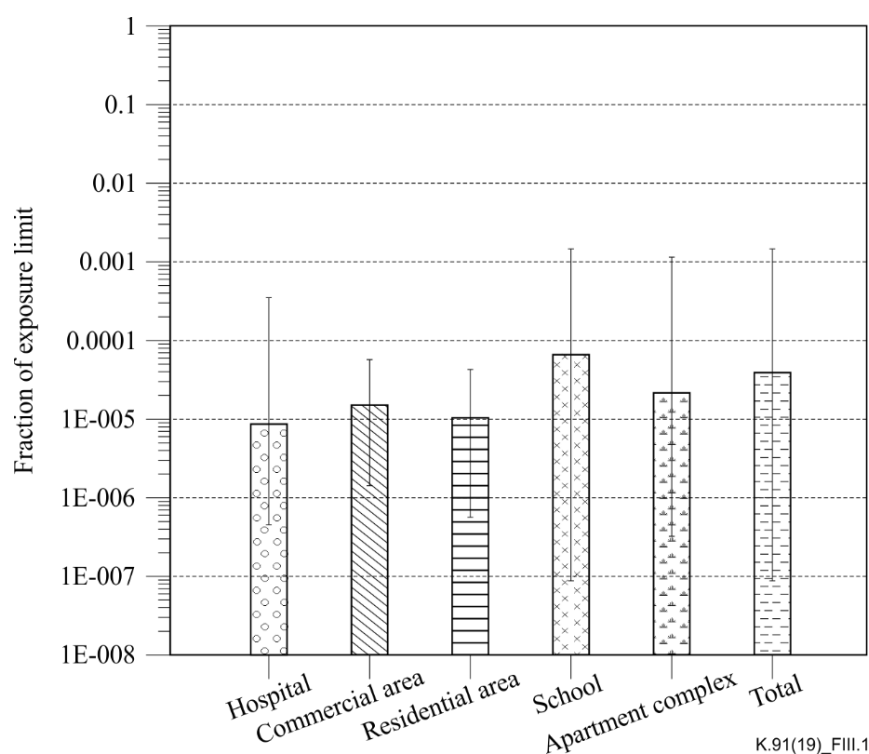
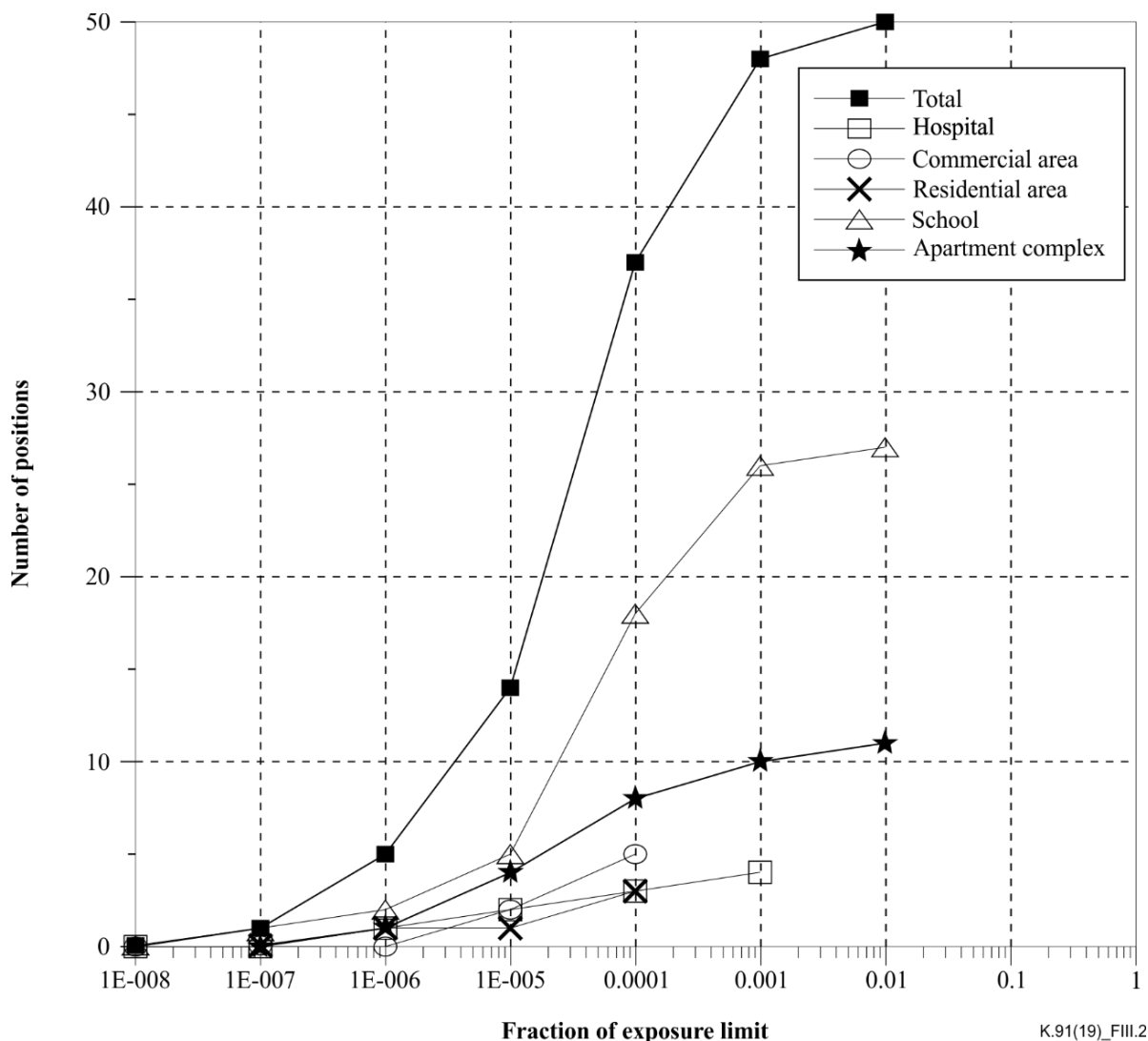


Figure III.1 – Exposure levels at a selection of densely populated areas

At all places measured, the spatially averaged RF EMF levels (median value) were below 0.01% (exposure ratio: 0.000 1) of the limit in the ICNIRP 1998 guidelines in Figure III.1. The highest exposure ratio (highest horizontal bar) occurred at a school, recording a value of 1.5×10^{-3} , which corresponds to a value of 1.5 V/m by the RSS method (CDMA800: 0.1 V/m, CDMA1800: 1.5 V/m).

Figure III.2 shows roughly the cumulative number of positions less than a specific (total) exposure ratio. For instance, in the case of the total line in the graph, the maximum number of positions is 50, meaning that all the spatially averaged field values measured do not exceed 0.01. Almost all of the positions (48 out of 50; 96%) had no more than 0.1% of the limit (exposure ratio: 0.001).



NOTE – Field levels are depicted as a fraction of the exposure limits.

Figure III.2 – Cumulative numbers of positions at a selection of densely populated areas

For the 13 services served and 1 250 positions across the Republic of Korea, the electric field strength by population density is presented in Table III.1. Exposure levels at several places are classified by population density (rural: 123, urban: 517, suburbs: 610) are shown in Table III.1 and Figure III.3. Urban areas have an exposure level that is high. Figure III.3 also shows that the more densely populated the area is, the larger the exposure levels become.

Table III.1 – Average electric field strength in volts per metre of various services by population density

	Frequency band (MHz)	Rural	Urban	Suburbs
TV (VHF)	54 – 174	0.024	0.023	0.025
FM	88 – 108	0.016	0.030	0.029
TV (VHF, T-DMB)	174 – 216	0.015	0.032	0.026
LBS	322 – 328.6	0.006	0.009	0.007
TRS (low)	390 – 400	0.004	0.004	0.004
TV (UHF, D-TV)	470 – 806	0.024	0.030	0.034
TRS (high)	851 – 867	0.008	0.025	0.018
Cellular	869 – 894	0.023	0.100	0.075
Wireless data	938 – 940	0.001	0.008	0.008
PCS	1840 – 1870	0.062	0.152	0.122
IMT-2000	2130 – 2170	0.054	0.151	0.120
S-DMB	2605 – 2655	0.021	0.071	0.051
WiBro	2300 – 2358.5	0.016	0.054	0.029

LBS: Location-based service.

TRS: Trucked radio system.

PCS: Personnel communication system.

IMT-2000: International Mobile Telecommunication-2000.

S-DMB: Satellite-digital multimedia broadcasting.

WiBro: Wireless broadband Internet.

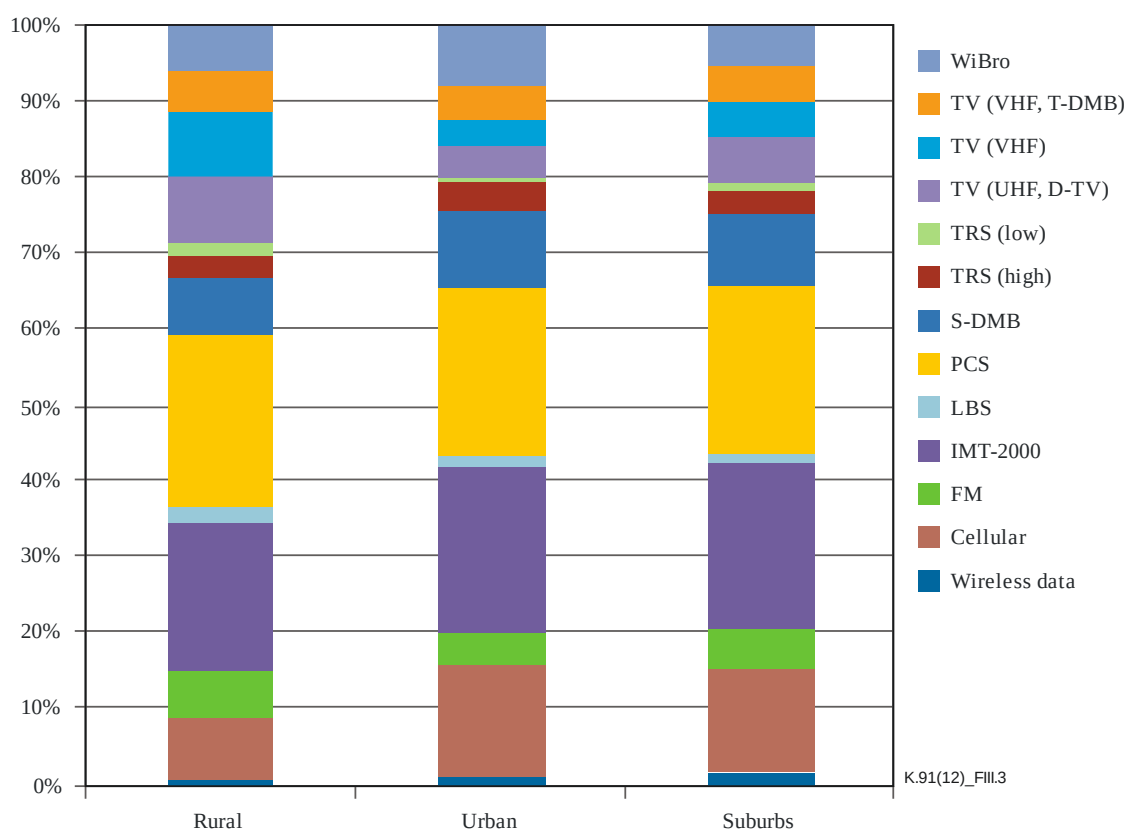


Figure III.3 – Possession rate for various services by population density

During measurement, the following was assumed: the spatial average of three heights: 1.1 m, 1.5 m, and 1.7 m; the number of basic measurement points: three (one dimension); and the number of precise measurements: nine (two dimension). In case of precise measurement, the area is within 0.4 m (horizontal) by 0.6 m (vertical) and the points are placed 0.2 m apart at each height. In addition, to consider incident fields from any direction and polarization, an isotropic probe is used in the measurement system. Moreover, to separate each EMF source under the multi-frequency environment, a spectrum analyser connected to the probe is used for a narrowband measurement and the detection mode of the spectrum analyser is set to r.m.s.

Appendix IV

Watt guard software

(This appendix does not form an integral part of this Recommendation.)

Within the framework of a study commissioned by the Bundesnetzagentur [German Federal Network Agency], a software module has been developed at Karlsruhe University enabling a simple and fast assessment of amateur stations. The software module runs on a wide variety of operating systems.

Features of *Watt guard*

The calculation tool has a graphic user interface at which the relevant data can be entered or the relevant parameters selected. The user enters the parameters: frequency; antenna type and antenna gain; transmit power; his/her location; height of the antenna; antenna orientation; and environment. Figure IV.1 shows the welcome screen of the software.

The depth of electromagnetic wave penetration into the soil and of the reflections is taken account of by means of correction factors for soil conductance. The height dependence correction factor takes account of effects on the field strength from objects in the vicinity. The required safety distances and protection areas are derived from the fields. The safety factor compensates for exceeded field strength limits that may result from reflections from metallic or dielectric objects or from buildings, for instance. The various correction factors are already contained in *Watt guard* so that the user does not have to enter them.

When the required parameters have been entered, the process of calculating the necessary safety distances is started and displayed graphically.

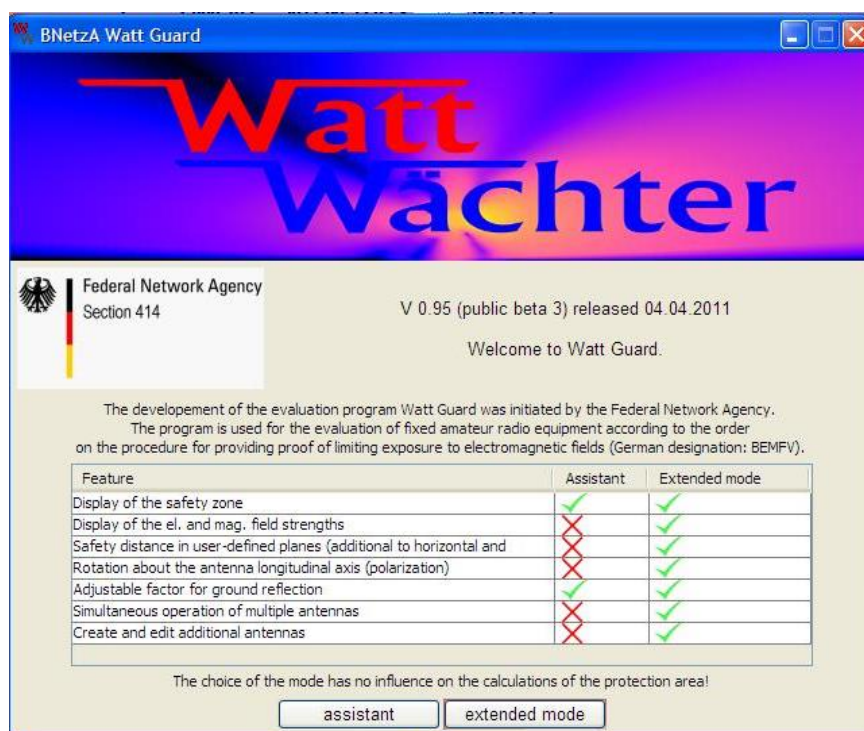


Figure IV.1 – Welcome screen of the *Watt guard* software

Appendix V

Uncertainty calculator software

(This appendix does not form an integral part of this Recommendation.)

V.1 Introduction

This software has been developed to help in the calculation of the uncertainty.

V.2 Brief description of the software

This software can calculate in three steps: type A uncertainty calculation, type B uncertainty calculation and expanded uncertainty calculation. The software references the method of calculating in clause 9 of [IEC 62232].

The main user interface (see Figure V.1) allows the user to perform the following operations:

- 1) select unit (strength or power density);
- 2) call another interface to calculate type A uncertainty, type B uncertainty and expanded uncertainty;
- 3) show results; and
- 4) export results.

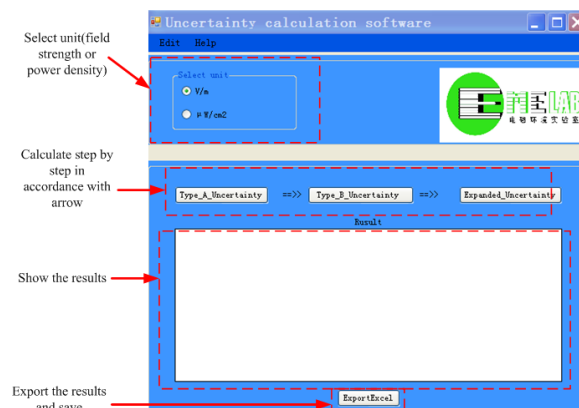


Figure V.1 – Main user interface

– Step 1: Type A uncertainty calculation

Sometimes the probability distribution of an influence quantity can be determined by a simple repeated measure, n independent observations x_k will be obtained under the same conditions of RF field measurement. For convenience, the experimental SD of the mean $S(\bar{x})$ is sometimes called a type A standard uncertainty.

Press the Type_A_Uncertainty key in the main interface (the user can call up the type A uncertainty calculation interface).

In this window (Figure V.2), the user can perform the following operations:

- 1) input the measurement data (the user can generate and fill a form or import Excel data);
- 2) calculate type A uncertainty.

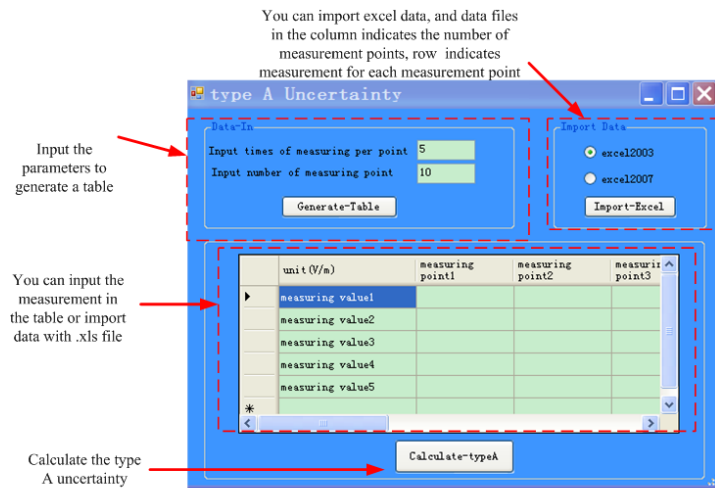


Figure V.2 – Type A uncertainty

NOTE – If the measurement at each point appears only one time, type A uncertainty equals 0.

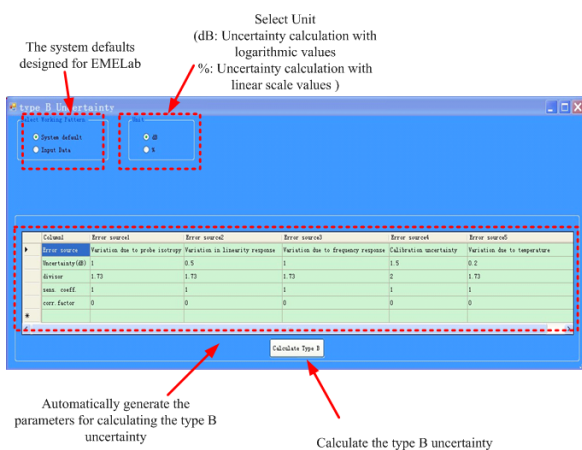
– Step 2: Type B uncertainty calculation

Type B evaluation is based on the following information: data associated with authoritative published quantity values; data associated with the quantity values of certified reference materials; data obtained from a calibration certificate about drift; data obtained from the accuracy class of a verified measuring instrument; or data obtained from limits deduced through personal experience.

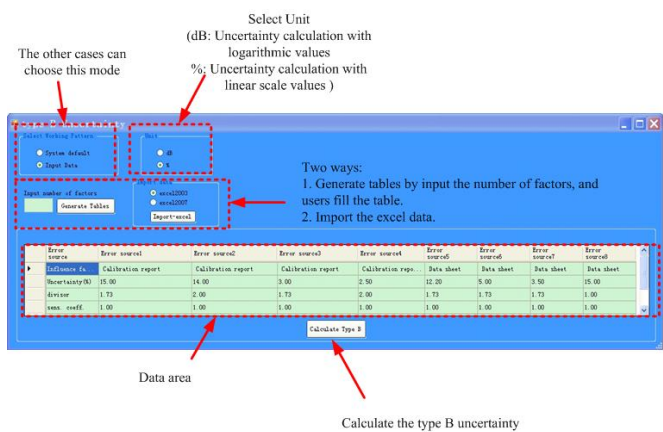
Press the Type_B_Uncertainty key in the main interface (the user can call the type B uncertainty calculation interface).

In this window (Figure V.3), the user can perform the following operations:

- 1) input the error sources and data (the software supports three ways to calculate type B uncertainty: default value, generate tables and import Excel);
- 2) select unit (dB: uncertainty calculation with logarithmic values; %: uncertainty calculation with linear scale values);
- 3) calculate type B uncertainty.



(a) System default



(b) Input data

Figure V.3 – Type B uncertainty

– Step 3: Expanded uncertainty

The additional measure of uncertainty that meets the requirement of providing an interval is termed expanded uncertainty and is denoted by U . The expanded uncertainty U is obtained by multiplying the combined standard uncertainty $u_c(y)$ by a coverage factor k :

$$U = k u_c(y)$$

Usually (it frequently occurs in practice) it can be assumed that taking $k = 1.96$ produces a confidence interval of approximately 95%, so in the software $k = 1.96$ is used.

The final result will be shown in the main interface, which can be exported as an Excel file, see Figure V.4.

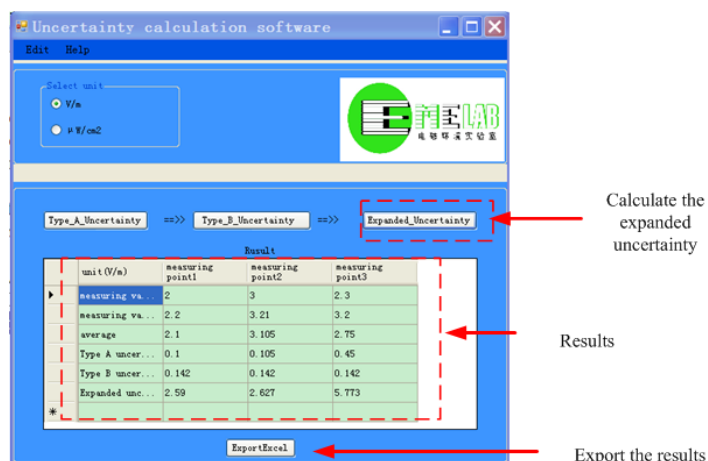


Figure V.4 – Expanded uncertainty

V.3 Examples

Example 1

Clause C2.1 of [b-EN 50413] is an example of a software operation to calculate the uncertainty that only contains type B uncertainty. No repeated measurement is assumed here, which means five points are chosen and just one test is conducted for every point. The measurement results can be found in Table V.1.

Table V.1 – Measurement results for calculation of uncertainty that only contains type B uncertainty

Unit (V/m)	Point 1	Point 2	Point 3	Point 4	Point 5
value 1	0.6	1.3	3.1	2.5	1.9

Operating steps can be seen in Figure V.5, and the type A uncertainty result in the main interface is shown, which is 0 dB and 0%.

(a) Input data

(b) Type A uncertainty results

Figure V.5 – Type A uncertainty, example 1

Type B uncertainty needs to be calculated next. The user needs to choose the unit "dB", because the uncertainty calculation in this example includes logarithmic values. The data are input into tables, like Table V.2, which contain 13 influence factors, and an Excel file is generated.

Table V.1 – Influence factors for type B uncertainty calculation

Error source	Error source 1	Error source 2	Error source 3	Error source 4	Error source 5	Error source 6	Error source 7	Error source 8	Error source 9	Error source 10	Error source 11	Error source 12	Error source 13
Influence factor	Frequency response	Input attenuation	Resolution bandwidths	IF-amplifier	Temperature response	Modulation response	Display reading uncertainty	Calibration uncertainty	Cable attenuation frequency interpolation	Calibration uncertainty	Antenna factor frequency interpolation	Uncertainty caused by reflections	Limited repeatability
Uncertainty (dB)	0.50	0.10	0.05	0.50	1.00	0.50	0.05	0.20	0.10	1.00	0.10	0.80	2.30
divisor	2.00	2.00	2.00	2.00	1.73	1.73	1.73	2.00	1.73	2.00	1.73	1.41	2.00
sens. coeff.	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
corr. factor	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

The operating steps can be seen in Figure V.6 and the type B uncertainty result in the main interface is shown, which is 1.568 dB.

(a) Input data

(b) Type B uncertainty results

Figure V.6 – Type B uncertainty, example 1

In the last step the expanded uncertainty is obtained and the results can be seen in Figure V.7. The result with logarithmic values is 3.073 dB and the result with a linear scale is 42.446%.

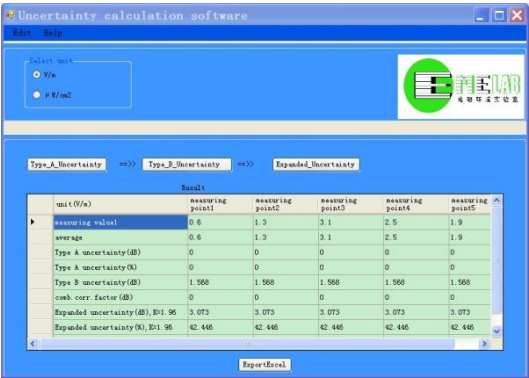


Figure V.7 – Expanded uncertainty, example 1

The results that are calculated by the software are basically in accordance with the results in clause C2.1 of [b-EN 50413]. The error is due to the number of decimal places.

Example 2

The example in clause C2.2 of [b-EN 50413] can also be calculated similarly to Example 1. The calculation steps and assumption of type A uncertainty are the same as in Example 1. However, in the type B uncertainty calculation step, the user needs to choose the unit "%", because of the uncertainty calculation in this example with linear scale, see Figure V.8.

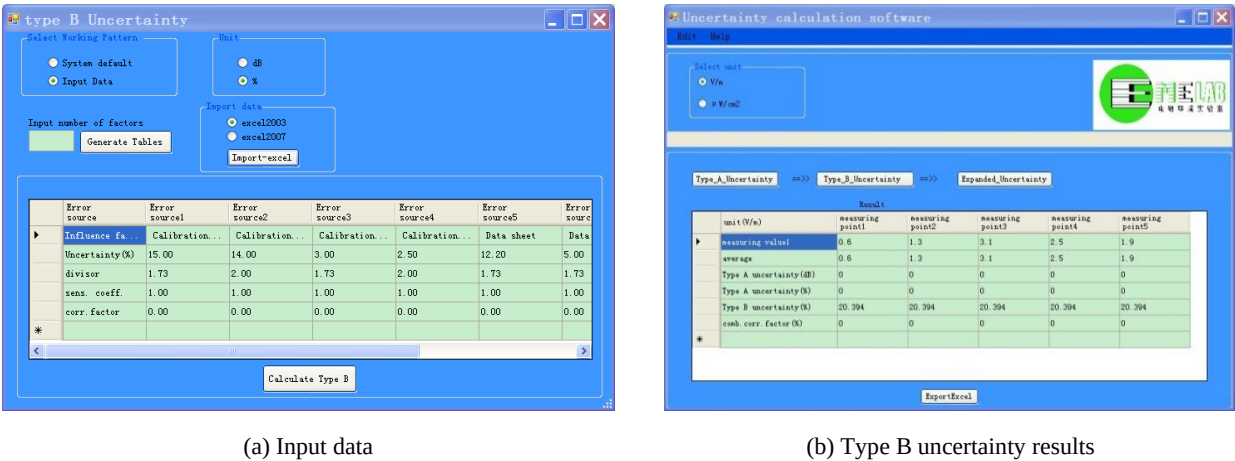


Figure V.8 – Type B uncertainty, example 2

Type B uncertainty is the result in the main interface, which is 20.394%. In the last step, the expanded uncertainty is calculated; the results can be seen in Figure V.9. The result with a linear scale is 39.972%.



Figure V.9 – Expanded uncertainty, example 2

The results calculated by the software are also basically in accordance with the results in clause C2.2 of [b-EN 50413]. The error is due to the number of decimal places.

Appendix VI

Examples for evaluating electromagnetic fields in general public environments with broadband radio signals

(This appendix does not form an integral part of this Recommendation.)

VI.1 Method for evaluating electromagnetic fields in general public environments with broadband signals

When using a spectrum analyser to measure the RF field power density, the RBW has quite a large effect on the measurement results, particularly for broadband signals such as WCDMA and OFDM. Therefore, it should be carefully selected to sufficiently cover the occupied bandwidth of the target radio source for an accurate field power density measurement. On the other hand, in current radio environments, there are various types of radio signal that have different occupied bandwidths in a wide frequency range. Therefore, the RBW should not be sufficiently narrow and as a result, multiple signals are included in the RBW. It increases the uncertainty of the measurement. However, such broadband RBW setting can be applicable with proper consideration of the total power density.

Figure VI.1a) shows an example of the measured frequency spectra of broadband or narrowband RBW settings. Suppose the received signals, $S(f_1)$ and $S(f_2)$, are narrowband signals and $S(f_3)$ and $S(f_4)$ are broadband signals. Figure VI.1b) is a magnified section of the measured frequency spectrum around frequency f_i , which consists of a set of sampled points the number of which is $M(f_i)$. At frequency f_i , the received level, $S(f_i)$, is given by:

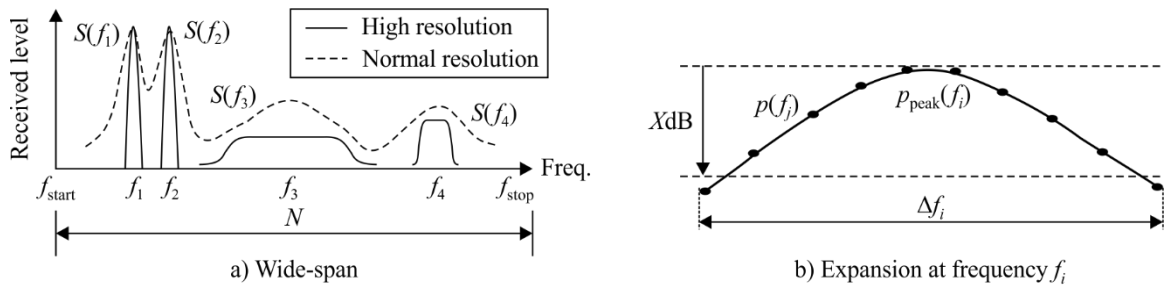
$$S(f_i) = \sum_{j=1}^{M(f_i)} \frac{p(f_i)\Delta f_i}{f_{\text{RBW}}[M(f_i)-1]} \quad (\text{VI-1})$$

When the number of detected radio signals is N , the normalized total power density is calculated by:

$$S_{\text{total}} = \sum_{i=1}^N \left[\frac{S(f_i)}{S_{\text{limit}}(f_i)} \right] \quad (\text{VI-2})$$

where $S_{\text{limit}}(f_i)$ is the frequency dependent limit of the field power density and S_{total} is the total power density normalized by $S_{\text{limit}}(f_i)$.

For instance, a measurement system comprises an isotropic sleeve dipole sensor, an RF switch box, a spectrum analyser, and a laptop computer, as shown in Figure VI.2. The field sensor is based on a sleeve dipole, and the dipole elements have a periodical configuration to reach a wide frequency response in the frequency range from 30 MHz to 3 GHz. Three sleeve dipoles are orthogonally established on a tripod for isotropic response. The laptop computer controls the spectrum analyser and the RF switch box, and calculates the received level, $S(f_i)$, and S_{total} .



K.91(17)_FVI.1

Figure VI.1 – Measured frequency spectrum

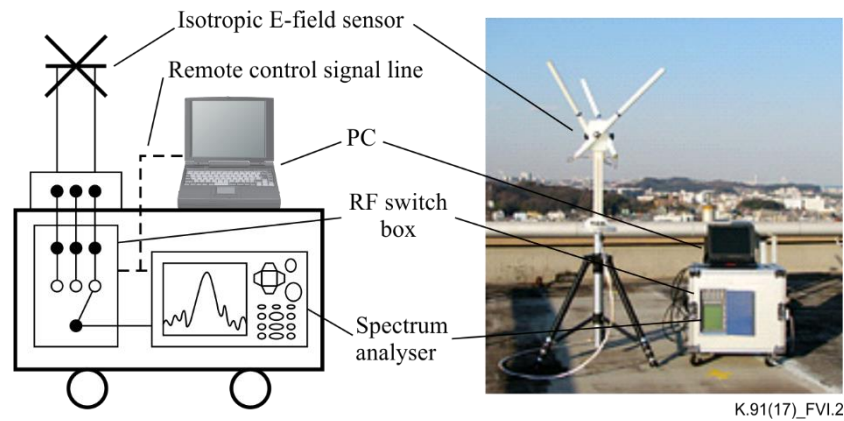


Figure VI.2 – Measurement system

VI.2 Effect of RBWs on measured electromagnetic fields from mobile phone base stations

In order to investigate if RF exposure from mobile phone base stations complies with the exposure limits, EMF measurements are carried out. Although frequency-selective measurements using a spectrum analyser do not provide as much information as signal analysers when measuring WCDMA signal, it is still the method of choice, taking into consideration the simplicity and cost. RBW is one of the important parameters of the frequency-selective equipment and has an influence on the cost of the equipment. Usually equipment with a wide RBW, such as signal analysers, are more expensive. However, for wideband signals such as WCDMA, which has a signal bandwidth of 5 MHz, it is important to confirm whether the use of equipment with narrow RBW can accurately measure these signals and be adopted in compliance testing. In this contribution, data obtained using a Narda SRM-3000 operating in frequency analysis mode are provided. Indoor (i.e., inside an anechoic chamber) and *in-situ* measurements for different RBW values were taken. When spatial averaged *E*-fields were considered, a maximum difference of less than 2.7 dB between values obtained using different RBWs was recorded. For the time-averaged measurements, the maximum difference of less than 1.6 dB was obtained.

Measurements at 800 and 2 000 MHz for indoor and *in-situ* measurements were taken. For indoor measurements, a Narda SRM-3000 is connected to its isotropic probe and the BS simulator (MT8820A) is connected to a transmitting antenna (double ridged horn antenna EMC03115) through an amplifier. This configuration is shown in Figure VI.3. For *in-situ* measurements, some results obtained in two mobile phone base stations are provided. BS1, for the 2000 MHz band, was in an urban area of Tokyo, while BS 2, which was on the outskirts of Tokyo, was chosen for the 800 MHz band. For each of these BSs, three locations were chosen for measurements. In this contribution, the three measurement locations for both BSs are referred to as locations 1, 2 and 3. The distances between these locations and their BSs were between 240 and 270 m.

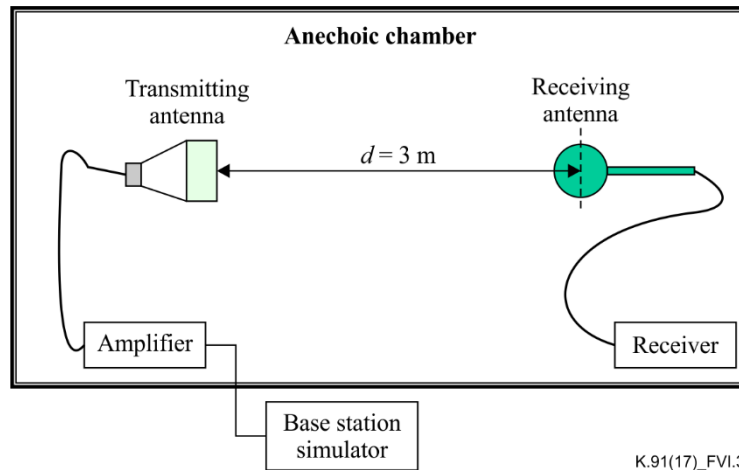


Figure VI.3 – Measurement configuration inside an anechoic chamber

Table VI.1 lists the time-averaged E -field strength. They are normalized to the average E -field obtained for the RBW of 5 MHz. In this case, the measurements were taken for 6 min and the average values were calculated from the measured data points obtained. Because the time between measurements was automatically set, the number of data points was different for each RBW. During this measurement, the antenna was positioned at the height of 1.5 m above the ground. For the *in situ* measurements, results from BS 1 are presented. The maximum difference of 1.6 dB is obtained in the indoor measurements at 800 MHz.

Table VI.1 – Time-averaged E -fields normalized to the 5 MHz value: *in-situ* measurements were obtained at location 1 of each base station

	$f = 882.5 \text{ MHz}$		$f = 2\,137.6 \text{ MHz}$	
RBW	Indoor	<i>In situ</i>	Indoor	<i>In situ</i>
50 kHz	1.0	−0.1	0.3	−0.1
500 kHz	1.6	−0.0	1.1	−0.2
5 MHz	0	0	0	0
NOTE – Results are indicated in decibels.				

Figure VI.4 shows the height pattern of the E -field measured at BS 1. The E -fields were measured up to the height of 200 cm, at an interval of 10 cm for RBWs of 50, 500 kHz and 5 MHz. The difference in the spatial average E -fields obtained using the Japanese Notice No. 300 scheme, which uses 20 points, and 3 points average are summarized in Table VI.2. The difference is referenced to the values obtained with an RBW of 5 MHz. Although, for a given height, large differences in the measured E -fields for different RBWs of up to 6 dB were recorded (as shown in Figure VI.4) the difference in the spatial average E -fields obtained using different RBW is less than 2.7 dB.

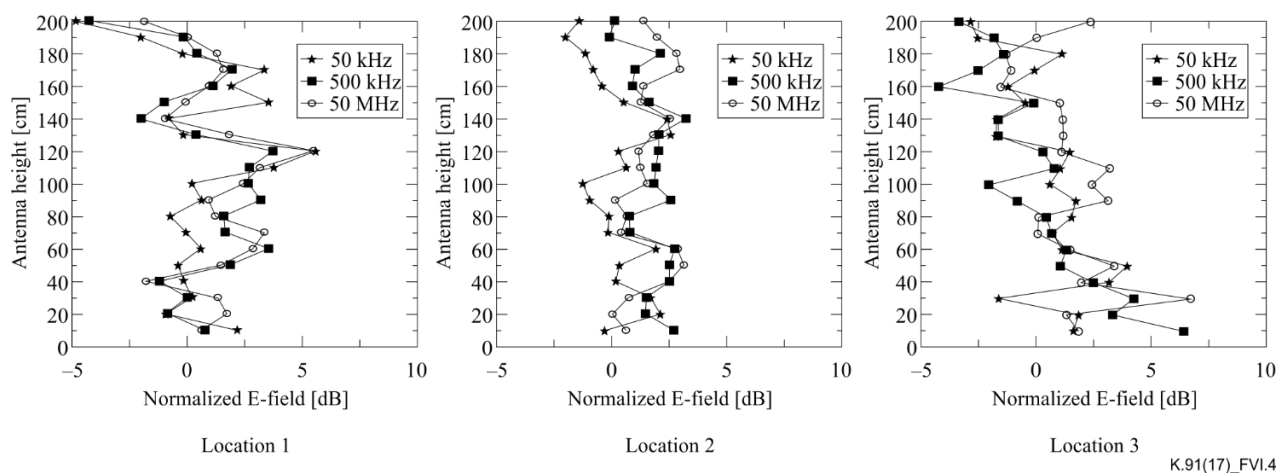


Figure VI.4 – Height pattern of the measured E -fields at 800 MHz band

Table VI.2 – Difference in spatial average E -fields referenced to the E -field obtained with an RBW of 5 MHz for BS1 and BS2

(a) LOCATION 1

	$f = 882.5 \text{ MHz}$		$f = 2\,137.6 \text{ MHz}$	
RBW	20 points	3 points	20 points	3 points
50 kHz	0.9	2.2	-2.5	1.2
500 kHz	0.3	2.7	-0.9	2.5
5 MHz	0	0	0	0

(b) LOCATION 2

	$f = 882.5 \text{ MHz}$		$f = 2\,137.6 \text{ MHz}$	
RBW	20 points	3 points	20 points	3 points
50 kHz	1.4	1.7	1.6	1.1
500 kHz	1.6	1.4	0.9	0.1
5 MHz	0	0	0	0

(c) LOCATION 3

	$f = 882.5 \text{ MHz}$		$f = 2\,137.6 \text{ MHz}$	
RBW	20 points	3 points	20 points	3 points
50 kHz	-1.1	-0.6	-1.8	2.5
500 kHz	0.6	1.0	-0.7	0.8
5 MHz	0	0	0	0

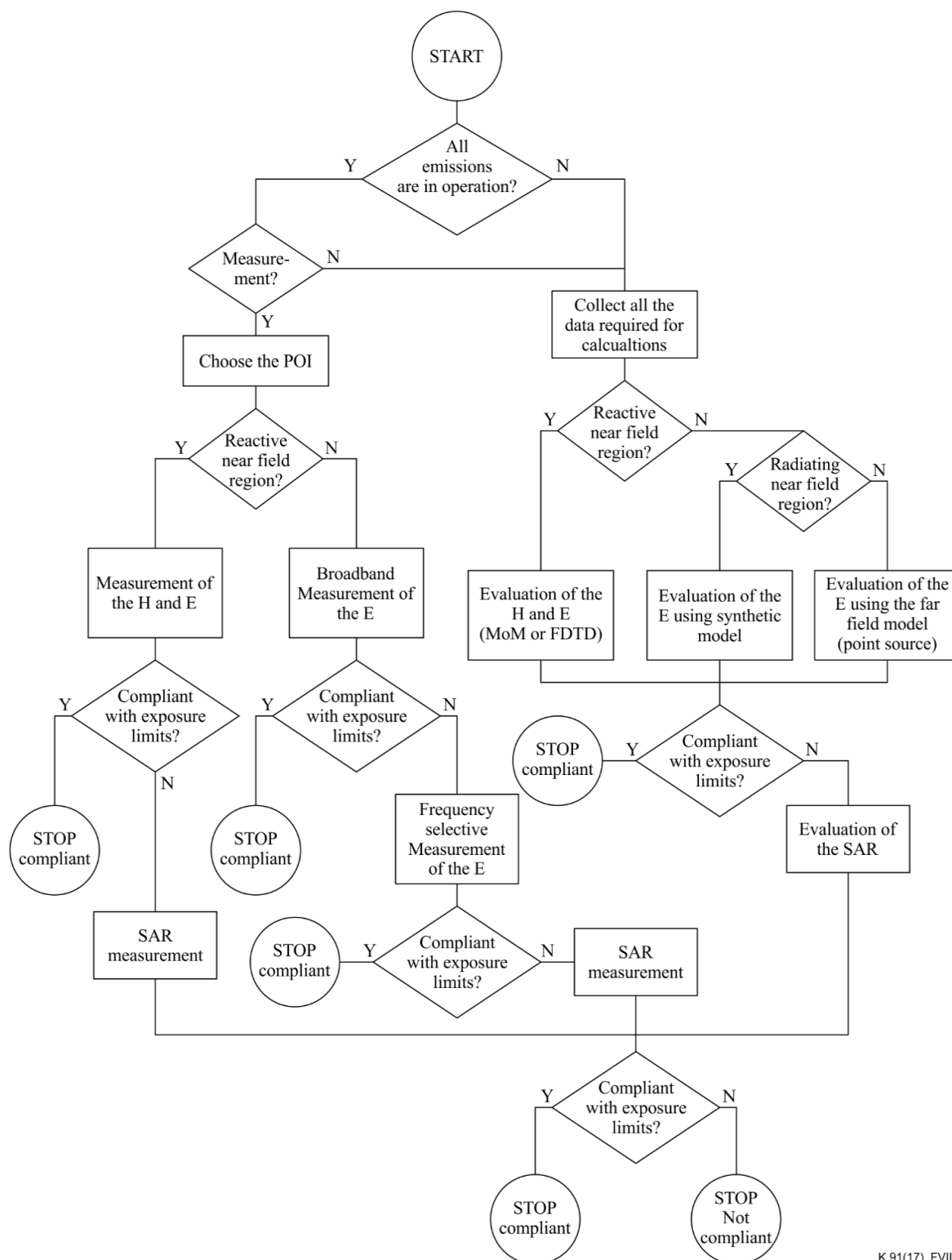
NOTE – Results are indicated in decibels.

Appendix VII

Example of block diagram with possible activities during exposure assessment

(This appendix does not form an integral part of this Recommendation.)

Figure VII.1 is a block diagram of the possible activities during exposure assessment. For a more general block diagram and detailed information, see [IEC 62232].



K.91(17)_FVII.1

Figure VII.1 – Block diagram of the exposure assessment procedure

Appendix VIII

"EMF exposure" mobile app

(This appendix does not form an integral part of this Recommendation.)

VIII.1 Introduction

"EMF exposure" is a mobile app available for Android users. The user of this application should have knowledge about how to conduct the RF exposure assessment using the parameters specified in [ITU-T K.52]. It allows the user to add BSs on a map and to calculate the approximate total exposure ratio due to the emissions from them. The user must have access to the BS input data. The app was developed with the support of the Spectrum, Orbit and Broadcasting Office of Anatel, the Brazilian Regulatory Telecommunications Agency. The source code is available from [b-GitHub].

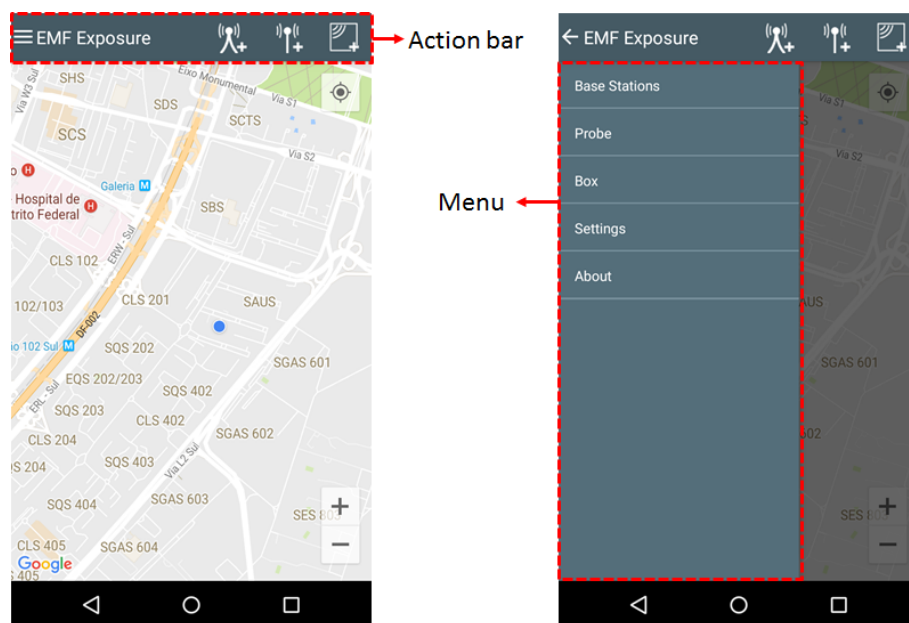
Disclaimer

This open-source implementation tool is provided "as is" with no warranties, express or implied, including, but not limited to, the warranties of merchantability, fitness for a particular purpose and non-infringement of intellectual property rights and neither the developer (or its affiliates) nor ITU shall be held liable in any event for any damages whatsoever (including, without limitation, damages for loss of profits, business interruption, loss of information or any other pecuniary loss) arising out of or related to the use of or inability to use this tool.

The evaluations are based on conservative models and may overestimate actual exposure.

VIII.2 Brief description of the main features of "EMF exposure"

The app provides a simple user interface that allows the user to manually add BSs and calculate the total exposure ratio due to them. Figure VIII.1 shows its main screen.

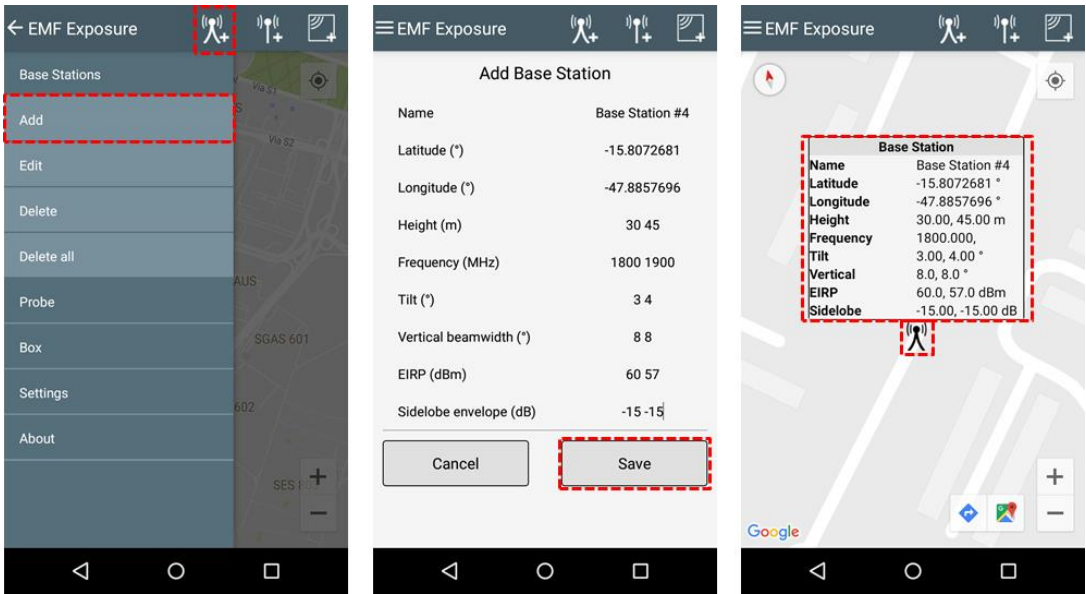


NOTE – "EMF exposure" has three icons that allow the user to access the most common operations (add base station, add probe and add box); (Right) The lateral menu is highlighted. It allows the user to access all the functions of the app.

**Figure VIII.1 – Main user interface
in the left-hand image, the action bar is highlighted**

To add BSs, users can tap on the "Add base station" icon on the action bar or they can use the lateral menu, as shown in Figure VIII.2. In that case, the user taps on the map to add a BS. "EMF exposure"

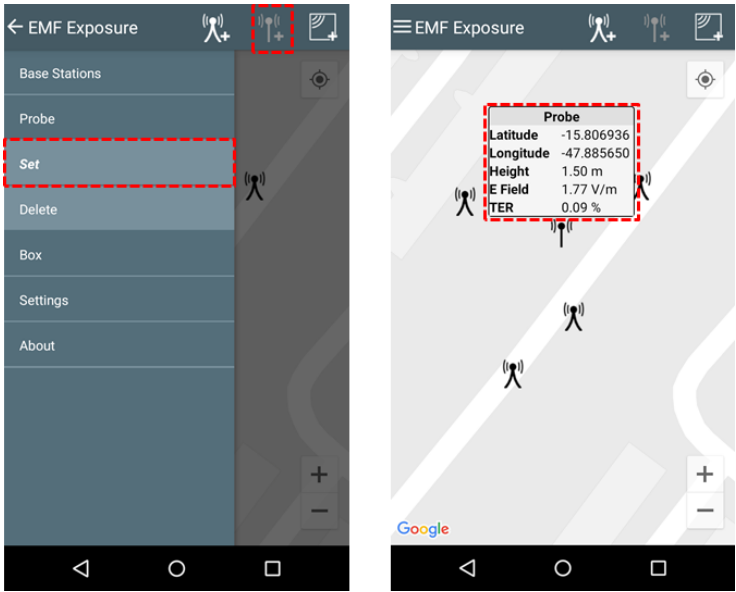
will show a screen for the user to enter the BS information. Note that one BS can have multiple radio sources. In this case, the data should be provided, separated by an empty space. Once a BS has been added, a BS icon is shown on the map. If the user taps on it, the BS properties are shown.



NOTE – (Left) the possibilities of adding a base station; centre) add base station screen; (right) base station shown on the map and its information.

Figure VIII.2 – Procedure for adding base stations

When the BSs are configured, the user can check the total exposure level using the probe function. To add a probe, the user can tap on the "Add probe" icon on the action bar or use the lateral menu as shown in Figure VIII.3. The user can then tap on the map to add a probe. The app will show electric field and the total exposure rate (TER) at that location.



NOTE – (Left) Possibilities to add a probe; (Right) Probe shown on the map and its information.

Figure VIII.3 – Procedure to add a probe

In Figure VIII.3, the information is shown for a probe height of 1.5 m. This height can be changed using the settings options (Figure VIII.4).

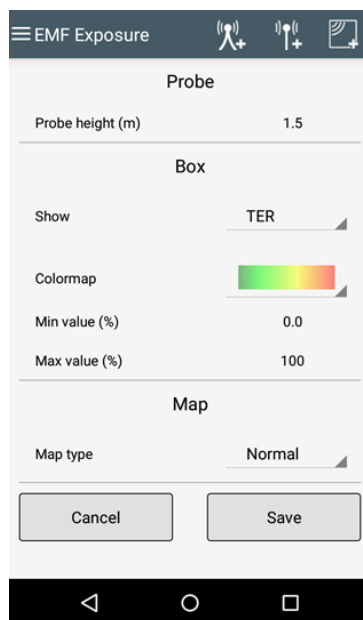


Figure VIII.4 – Changing settings

VIII.3 Brief description of the calculations done by "EMF exposure"

"EMF exposure" calculates the TER as specified in clause 9.6 of [ITU K.100]. The transmitting antennas of the BSs are modelled as omnidirectional in the horizontal pattern (assumes equal energy in all horizontal directions, which is not the case for all mobile network antennas) and as a "Directivity type-2 antenna" (specified in clause IV.2.2 of [ITU K.52]) in the vertical pattern. The propagation model is defined by Equation 7-6 of this Recommendation. A detailed description of "EMF exposure" can be found in [b-CF 2017a] and the app can also be downloaded from [b-CF 2017b].

Bibliography

- [b-ITU-T K-Sup.1] ITU-T K-series Recommendations – Supplement 1 (2020), *ITU-T K.91 – Guide on electromagnetic fields and health*.
- [b-ITU-T K-Sup.4] ITU-T K-series Recommendations – Supplement 4 (2018), *ITU-T K.91 – Electromagnetic field considerations in smart sustainable cities*.
- [b-ITU-T K-Sup.9] ITU-T K-series of Recommendations – Supplement 9 (2019), *5G technology and human exposure to radiofrequency electromagnetic fields*.
- [b-ITU-T K-Sup.13] ITU-T K-series of Recommendations – Supplement 13 (2021), *Radiofrequency electromagnetic field (RF-EMF) exposure levels from mobile and portable devices during different conditions of use*.
- [b-ITU-T K-Sup.14] ITU-T K-series of Recommendations – Supplement 14 (2019), *The impact of RF-EMF exposure limits stricter than the ICNIRP or IEEE guidelines on 4G and 5G mobile network deployment*.
- [b-ITU-T K-Sup.16] ITU-T K-series of Recommendations – Supplement 16 (2022), *Electromagnetic field compliance assessments for 5G wireless networks*.
- [b-ITU-T K-Sup.19] ITU-T K-series of Recommendations – Supplement 19 (2019), *Electromagnetic field (EMF) strength inside underground railway trains*.
- [b-ITU-T K-Sup.20] ITU-T K-series of Recommendations – Supplement 20 (2021), *Radiofrequency exposure evaluation around underground base stations*.
- [b-ITU-T K-Sup.29] ITU-T K-series of Recommendations – Supplement 29 (2022), *Electromagnetic field strength inside and outside of electric vehicles using wireless power transfer technology*.
- [b-ITU-T K-Sup.32] ITU-T K-series of Recommendations – Supplement 32 (2023), *Case studies of radio frequency-electromagnetic field (RF-EMF) assessment*.
- [b-ITU-T EMF LA] ITU-T Report, *Monitoring of electromagnetic field levels in Latin America – Implementation of Recommendation ITU-T K.83*. Geneva: ITU. 38 pp.
https://www.itu.int/dms_pub/itu-t/oth/0b/11/T0B110000283301PDFE.pdf
- [b-ITU-D Q 7/2] ITU-D Question 7/2 (2021), *Policies, guidelines, regulations and assessments of human exposure to radio-frequency electromagnetic fields*.
https://www.itu.int/dms_pub/itu-d/opb/stg/d-stg-sq02.07.2-2021-pdf-e.pdf
- [b-ITU-R SM.2452] Report ITU-R SM.2452 (2022), *Electromagnetic field measurements to assess human exposure*.
<https://www.itu.int/pub/R-REP-SM.2452-2022>
- [b-ITU-R SpecMon] ITU-R (2011). *Handbook: Spectrum Monitoring*.
<http://www.itu.int/pub/R-HDB-23-2011>
- [b-IEC TR 63170] IEC TR 63170:2018, *Measurement procedure for the evaluation of power density related to human exposure to radio frequency fields from wireless communication devices operating between 6 GHz and 100 GHz*.
- [b-ISO/IEC 17025] ISO/IEC 17025:2017, *General requirements for the competence of testing and calibration laboratories*.
- [b-ICNIRP 2010] ICNIRP (2010). Guidelines for limiting exposure to time-varying electric and magnetic fields (1 Hz TO 100 kHz), Health Physics, Vol. 99, No. 6; pp. 818-836.
<https://www.icnirp.org/cms/upload/publications/ICNIRPLFgdl.pdf>

- [b-ICNIRP 1998] Guidelines for limiting exposure to time-varying electric, magnetic, and electromagnetic fields (up to 300 GHz), *Health Physics*, Vol. 74, no. 4, pp. 494-522.
<http://www.icnirp.org/cms/upload/publications/ICNIRPemfgdl.pdf>
- [b-ICNIRP 2020] ICNIRP guidelines for limiting exposure to electromagnetic fields (100 kHz to 300 GHz). *Health Physics*, 118(5), 2020, pp. 483-524.
<https://www.icnirp.org/cms/upload/publications/ICNIRPfgdl2020.pdf>
- [b-IEC 62209-1] IEC 62209-1:2016, *Measurement procedure for the assessment of specific absorption rate of human exposure to radio frequency fields from hand-held and body-mounted wireless communication devices – Part 1: Devices used next to the ear (Frequency range of 300 MHz to 6 GHz)*.
- [b-IEC 62209-3] IEC 62209-3:2019, *Measurement procedure for the assessment of specific absorption rate of human exposure to radio frequency fields from hand-held and body-mounted wireless communication devices – Part 3: Vector measurement-based systems (Frequency range of 600 MHz to 6 GHz)*.
- [b-IEC 62369-1] IEC 62369-1:2008, *Evaluation of human exposure to electromagnetic fields from short range devices (SRDs) in various applications over the frequency range 0 GHz to 300 GHz – Part 1: Fields produced by devices used for electronic article surveillance, radio frequency identification and similar systems*.
- [b-IEC PAS 63184] IEC PAS 63184:2021, *Assessment methods of the human exposure to electric and magnetic fields from wireless power transfer systems – Models, instrumentation, measurement and numerical methods and procedures (frequency range of 1 kHz to 30 MHz)*.
- [b-IEC TR 62630] IEC Technical Report 62630:2010, *Guidance for evaluating exposure from multiple electromagnetic sources*.
- [b-IEC TR 62669] IEC TR 62669:2019, *Case studies supporting IEC 62232 – Determination of RF field strength, power density and SAR in the vicinity of radiocommunication base stations for the purpose of evaluating human exposure*.
- [b-IEC/IEEE 62704-1] IEC/IEEE 62704-1:2017, *Determining the peak spatial-average specific absorption rate (SAR) in the human body from wireless communications devices, 30 MHz to 6 GHz – Part 1: General requirements for using the finite difference time-domain (FDTD) method for SAR calculations*.
- [b-IEC/IEEE 62704-2] IEC/IEEE 62704-2:2017, *Determining the peak spatial-average specific absorption rate (SAR) in the human body from wireless communications devices, 30 MHz to 6 GHz – Part 2: Specific requirements for finite difference time domain (FDTD) modelling of exposure from vehicle mounted antennas*.
- [b-IEC/IEEE 62704-3] IEC/IEEE 62704-3:2017, *Determining the peak spatial-average specific absorption rate (SAR) in the human body from wireless communications devices, 30 MHz to 6 GHz – Part 3: Specific requirements for using the finite difference time domain (FDTD) method for SAR calculations of mobile phones*.
- [b-IEEE C95.1] IEEE C95.1-2019, *IEEE standard for safety levels with respect to human exposure to radio frequency electromagnetic fields, 3 kHz to 300 GHz*.
<https://emfguide.itu.int/pdfs/c95.1-2005.pdf>

- [b-IEEE C95.7] IEEE C95.7-2014, *IEEE recommended practice for radio frequency safety programs, 3 kHz to 300 GHz*.
- [b-IEEE 802.15.1] IEEE 802.15.1-2005, *IEEE Standard for information technology – Local and metropolitan area networks – Specific requirements – Part 15.1a: Wireless medium access control (MAC) and physical layer (PHY) specifications for wireless personal area networks (WPAN)*.
- [b-WHO 2014] WHO (2014). Electromagnetic fields and public health: Mobile phones. Geneva: World Health Organization.
<https://www.who.int/en/news-room/fact-sheets/detail/electromagnetic-fields-and-public-health-mobile-phones>
- [b-WHO Internet] World Health Organization (Internet). Electromagnetic fields (EMF): Existence of national standards, 2017. Geneva: World Health Organization.: <http://www.who.int/gho/phe/emf/legislation/en/>
- [b-EN 50413] CENELEC EN 50413:2008, *Basic standard on measurement and calculation procedures for human exposure to electric, magnetic and electromagnetic fields (0 Hz-300 GHz)*.
- [b-CF 2017a] Carísio Fernandes, L. (2017a). The design and development of an app to compute exposure to electromagnetic fields, *IEEE Antennas and Propagation Magazine*, **59**(2), pp. 149-160. DOI: 10.1109/MAP.2017.2655527.
- [b-CF 2017b] Carísio Fernandes, L. (2017b). *EMF exposure*, Amazon.
Available [viewed 2019-12-21] at: <https://www.amazon.com/Leandro-Car%C3%ADsio-Fernandes-EMF-Exposure/dp/B01JF59LBO/>
- [b-GitHub] GitHub (2019). *EMF exposure*.
Available [viewed 2019-12-20] at: <https://github.com/carisio/emf-exposure>
- [b-JCGM 100] JCGM 100:2008, *Evaluation of measurement data – Guide to the expression of uncertainty in measurement*.
- [b-Mazar 2016] Radio Spectrum Management: Policies, Regulations and Techniques
https://mazar.atwebpages.com/Downloads/WileyChapter9RF-EMF_HumanHazards_MazarJanuary2021.pdf

SERIES OF ITU-T RECOMMENDATIONS

Series A	Organization of the work of ITU-T
Series D	Tariff and accounting principles and international telecommunication/ICT economic and policy issues
Series E	Overall network operation, telephone service, service operation and human factors
Series F	Non-telephone telecommunication services
Series G	Transmission systems and media, digital systems and networks
Series H	Audiovisual and multimedia systems
Series I	Integrated services digital network
Series J	Cable networks and transmission of television, sound programme and other multimedia signals
Series K	Protection against interference
Series L	Environment and ICTs, climate change, e-waste, energy efficiency; construction, installation and protection of cables and other elements of outside plant
Series M	Telecommunication management, including TMN and network maintenance
Series N	Maintenance: international sound programme and television transmission circuits
Series O	Specifications of measuring equipment
Series P	Telephone transmission quality, telephone installations, local line networks
Series Q	Switching and signalling, and associated measurements and tests
Series R	Telegraph transmission
Series S	Telegraph services terminal equipment
Series T	Terminals for telematic services
Series U	Telegraph switching
Series V	Data communication over the telephone network
Series X	Data networks, open system communications and security
Series Y	Global information infrastructure, Internet protocol aspects, next-generation networks, Internet of Things and smart cities
Series Z	Languages and general software aspects for telecommunication systems